1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

Thu Apr 17 17:11:44 2014

REV	ECN	DESCRIPTION OF REVISION	CK APPD
7	0002727241	ENGINEERING RELEASED	2014-04-18

N61 CARRIER BUILD

PDF PAGE CONTENTS

SCH 051-9903
BRD 820-3486
MCO 056-6825

BOM 639-4237 (16GB,BETTER)
BOM 639-5838 (32GB,BEST)
BOM 639-5839 (64GB,ULTRA)

BOM 639-00025(128GB,SUPREME,TLC)

BOM 639-00208 (16GB,BETTER,DTD)
BOM 639-00209 (32GB,BEST,DTD)
BOM 639-00210 (64GB,ULTRA,DTD)

BOM 639-00212(128GB,SUPREME,TLC,DTD)

NAND BOM OPTIONS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S0998	1	NAND,19NM,16GX8,MLC,PPN1.5	U0604	CRITICAL	NAND_16G
335S0993	1	NAND,19NM,32GX8,MLC,PPN1.5	U0604	CRITICAL	NAND_32G
335S0994	1	NAND,19NM,64GX8,MLC,PPN1.5	U0604	CRITICAL	NAND_64G
335S00010	1	NAND,19NM,128GX8,TLC,PPN1.5	U0604	CRITICAL	NAND_128G
138S0867	1	CAP,X5R,10UF,20%,6.3V,0.65MM,HRTZ,0402	C0610,C0611,C0614,C0634	CRITICAL	NAND_16G
138S0867	1	CAP,X5R,10UF,20%,6.3V,0.65MM,HRTZ,0402	C0613,C0633,C0610,C0611,C0614,C0634	CRITICAL	NAND_32G & NAND_64G
138S00003	1	CAP,X5R,15UF,20%,6.3V,0.65MM,HRTZ,0402	C0613,C0633,C0610,C0611,C0614,C0634	CRITICAL	NAND_128G

ALTERNATE NAND BOM OPTIONS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S0992	335S0998	ALTERNATE	U0604	TOSHIBA,NAND,16GB
335S1038	335S0998	ALTERNATE	U0604	HYNIX,NAND,16GB
335S1040	335S0994	ALTERNATE	U0604	HYNIX,NAND,64GB
335S00014	335S0994	ALTERNATE	U0604	TOSHIBA,NAND,64GB
335S00015	335S00010	ALTERNATE	U0604	TOSHIBA,NAND128GB
335S00009	335S0994	ALTERNATE	U0604	SANDISK,NAND,64GB,TLC

SHIELD BOM OPTIONS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
604-00241	1	SUBASSY, SHIELD, UPPER FRONT, N61	SH2501	CRITICAL	COMMON
604-00242	1	SUBASSY, SHIELD, LOWER FRONT, N61	SH2502	CRITICAL	COMMON
604-00243	1	SUBASSY, SHIELD, LOWER BACK, N61	SH2504	CRITICAL	COMMON
604-00244	1	SUBASSY, SA SHIELD, N61	SH2506	CRITICAL	COMMON

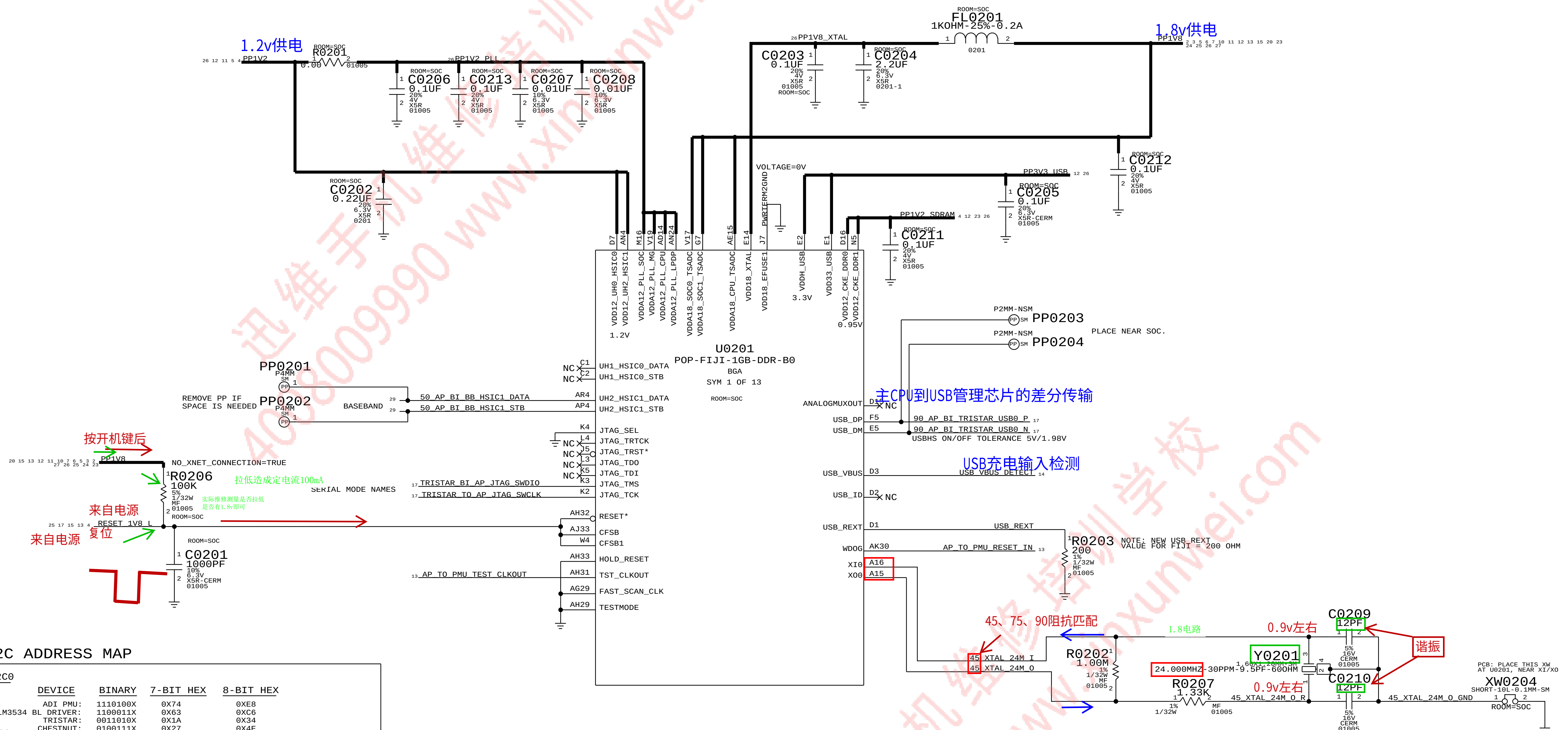
N61 BOM CALLOUTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9903	1	SCH, MLB, N61	SCH	CRITICAL	?
820-3486	1	PCBF, MLB, N61	PCB	CRITICAL	?
825-6838	1	EEEE FOR 639-4237 16GB	EEEE_G16T	CRITICAL	EEEE_16G
825-6838	1	EEEE FOR 639-5838 32GB	EEEE_G16R	CRITICAL	EEEE_32G
825-6838	1	EEEE FOR 639-5839 64GB	EEEE_G16Q	CRITICAL	EEEE_64G
825-6838	1	EEEE FOR 639-00025 128GB	EEEE_G16N	CRITICAL	EEEE_128G
825-6838	1	EEEE FOR 639-00208 16GB	EEEE_F98F	CRITICAL	EEEE_16G_TDDLTE
825-6838	1	EEEE FOR 639-00209 32GB	EEEE_FQK0	CRITICAL	EEEE_32G_TDDLTE
825-6838	1	EEEE FOR 639-00210 64GB	EEEE_FQJY	CRITICAL	EEEE_64G_TDDLTE
825-6838	1	EEEE FOR 639-00212 128GB	EEEE_FY9W	CRITICAL	EEEE_128G_TLC_TDDLTE

ALTERNATE BOM OPTIONS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S1844	152S1836	ALTERNATE	L1604	TY ALT INDUCTOR
152S1842	152S1849	ALTERNATE	L1519	TY ALT INDUCTOR
197S0392	197S0369	ALTERNATE	Y1200	ESPON ALT XTAL
197S0399	197S0369	ALTERNATE	Y1200	NDK ALT XTAL
338S1285	338S1202	ALTERNATE	U1601	L21 SPKAMP
152S2034	152S2033	ALTERNATE	L1209,L1211, L1213	1.2MM 1.0UH, CYNTEC
152S00004	152S2049	ALTERNATE	L1210,L1212, L1214	1.2MM 0.47UH, CYNTEC
339S00005	339S0246	ALTERNATE	U0201	FIJI, B0, SAMSUNG
339S0247	339S0246	ALTERNATE	U0201	FIJI, B0, HYNIX
339S00006	339S0246	ALTERNATE	U0201	FIJI, B1, E
339S00007	339S0246	ALTERNATE	U0201	FIJI, B1, H
339S00008	339S0246	ALTERNATE	U0201	FIJI, B1, S
155S0773	155S0453	ALTERNATE		TY 120OHM FERRITE
118S0764	118S0717	ALTERNATE	R1309	3.92KOHM, 01005
343S0688	343S0638	ALTERNATE	U2401	CUMULUS C1, FAB4
138S00005	138S00003	ALTERNATE	C1290	15UF,0402,HRTZL CAP
155S00011	155S00008	ALTERNATE	L1135	CMC,90OHM,MURATA
377S0168	377S0140	ALTERNATE	DZ1113	SUPPR,TRANS,VARISTOR,AMOTECH
155S0885	155S0610	ALTERNATE	FL1802,FL1803	FERR BD,150OHM,200MA,01005
138S0648	138S0652	ALTERNATE	C1018	CAP,4.7UF,20%,6.3V,0402,H=0.65MM
138S0657	138S0702	ALTERNATE	C1106	CAP,4.3UF,20%,4V,0610
338S00028	338S00017	ALTERNATE	U2203	CARBON, BOSCH, BMI162BC
338S00029	338S00017	ALTERNATE	U2203	CARBON, ST, AP6DS2AA
335S00013	335S0894	ALTERNATE	U0301	ST 8K EEPROM

FIJI: JTAG,USB,HSIC,XTAL

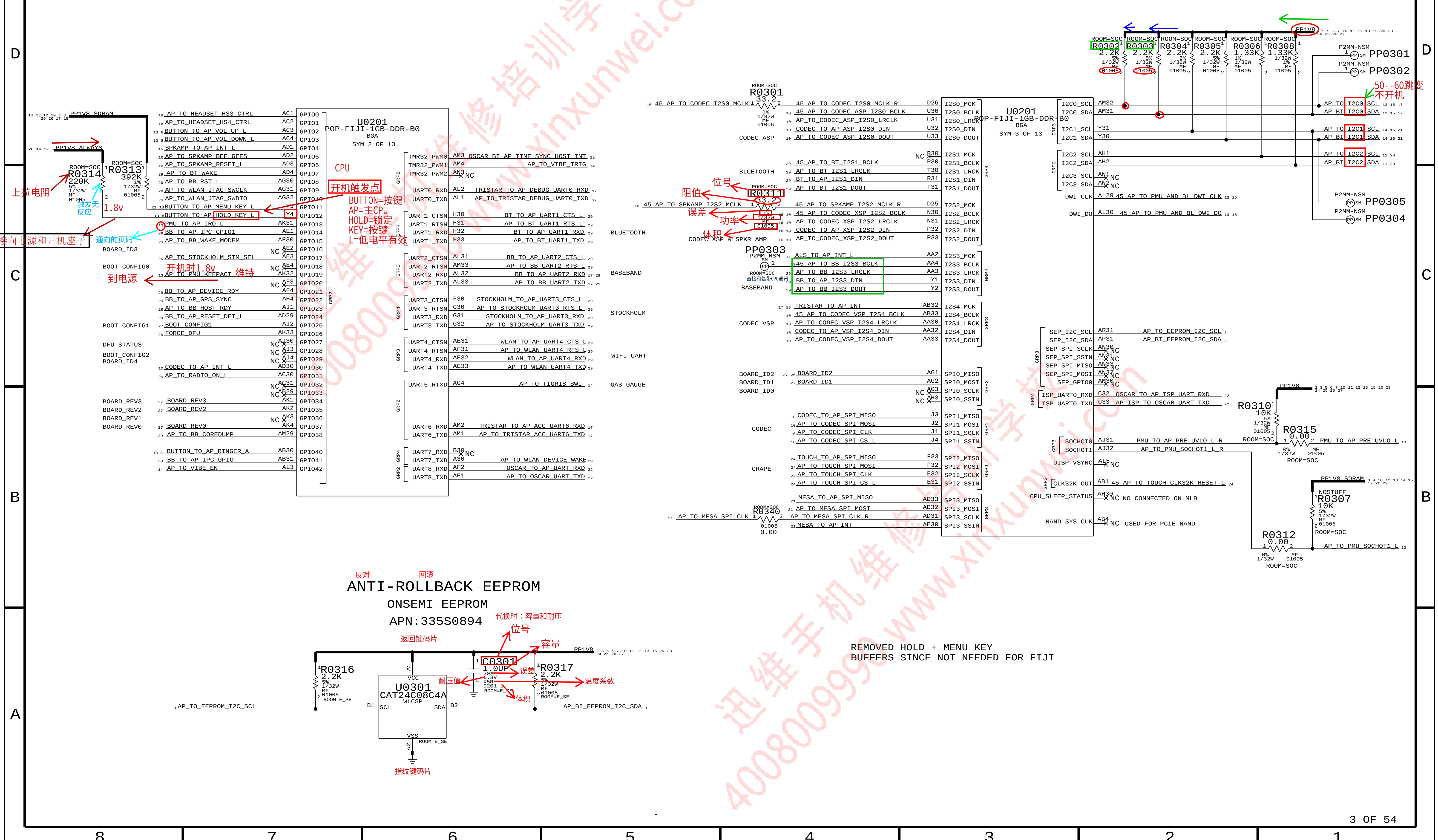


I2C ADDRESS MAP

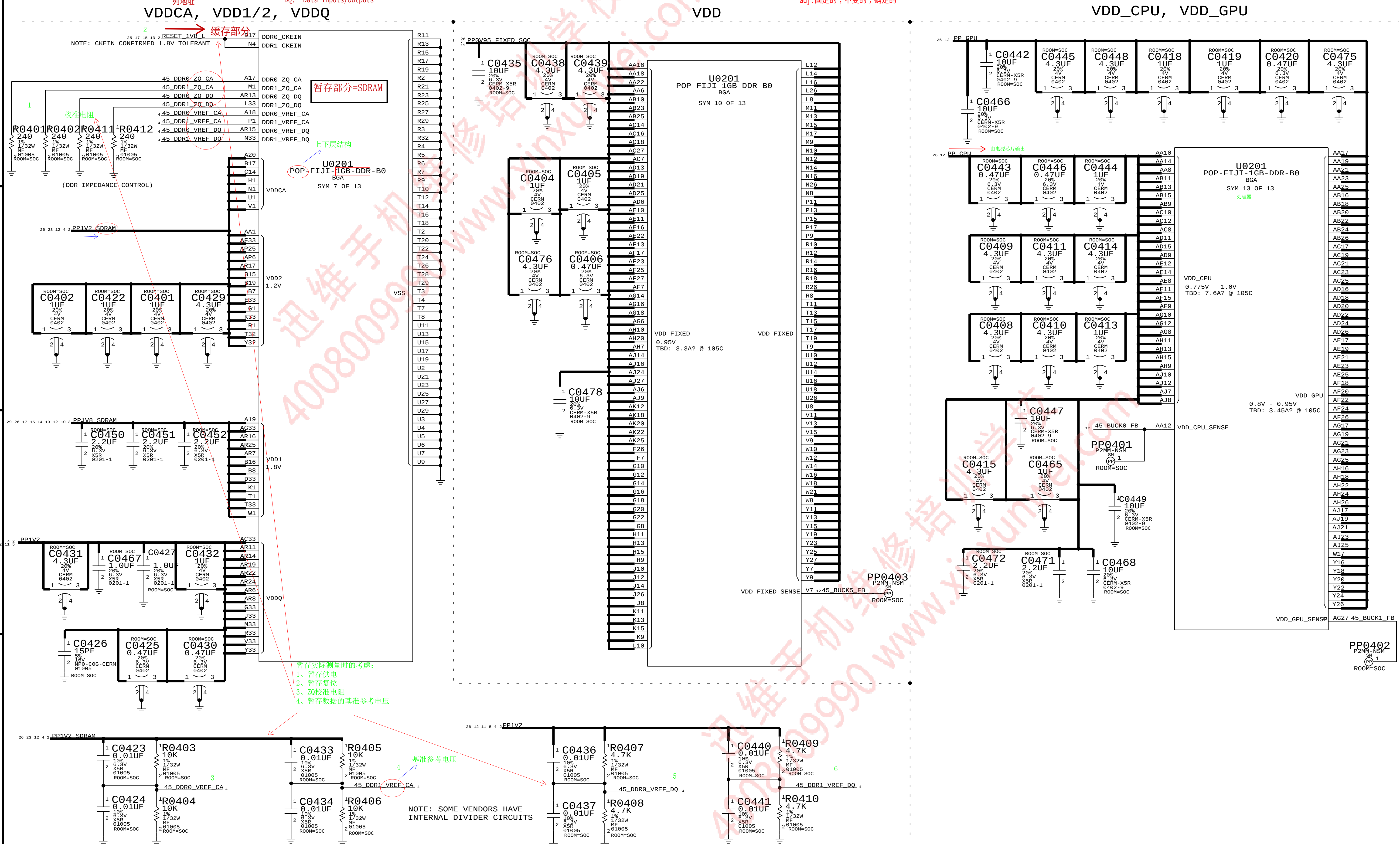
DEVICE	BINARY	7-BIT HEX	8-BIT HEX
I2C0			
ADI PMU:	1110100X	0X74	0XE8
LM3534 BL DRIVER:	1100011X	0X63	0XC6
TRISTAR:	0011010X	0X1A	0X34
CHESTNUT:	0100111X	0X27	0X4E
I2C1			
TIGRIS CHARGER:	1110101X	0X75	0XEA
LINEAR VIBE:	1011010X	0X5A	0XB4
CS35L19B AMP:	1000000X	0X40	0X80
MESA EEPROM (MEMORY):	1010110X	0X56	0XAC
MESA EEPROM (ID):	1011110X	0X5E	0XBC
I2C2			
CT814 ALS:	0101001X	0X29	0X52
DISPLAY EEPROM:	1010001X	0X51	0XA2
RCAM I2C			
OPEL STROBE DRIVER:	1100011X	0X63	0XC6
REAR FACING CAM:	0010000X	0X10	0X20
VCM AF DRIVER:	0001100X	0X0C	0X18
FCAM I2C			
FRONT FACING CAM:	0010000X	0X10	0X20

NOTE: ACCEL, GYRO, COMPASS ALL USING SPI (VIA OSCAR) FOR AP COMMUNICATION.

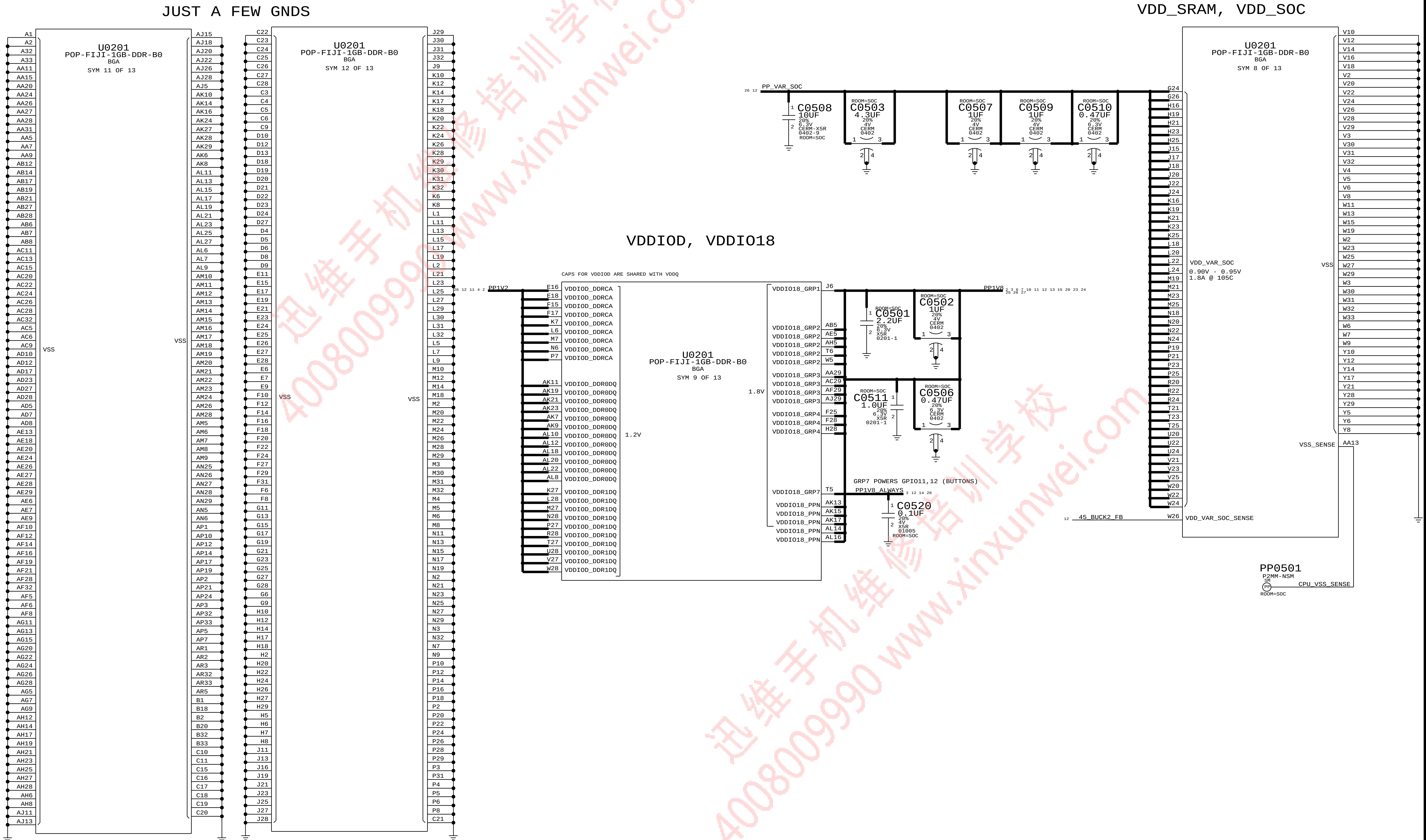
FIJI: DIGITAL I/O,BOOTSTRAPPING



FIJI: VDDCA,VDD1/2,VDDQ,VDD,VDD_FIXED,VDD_CPU,VDD_GPU

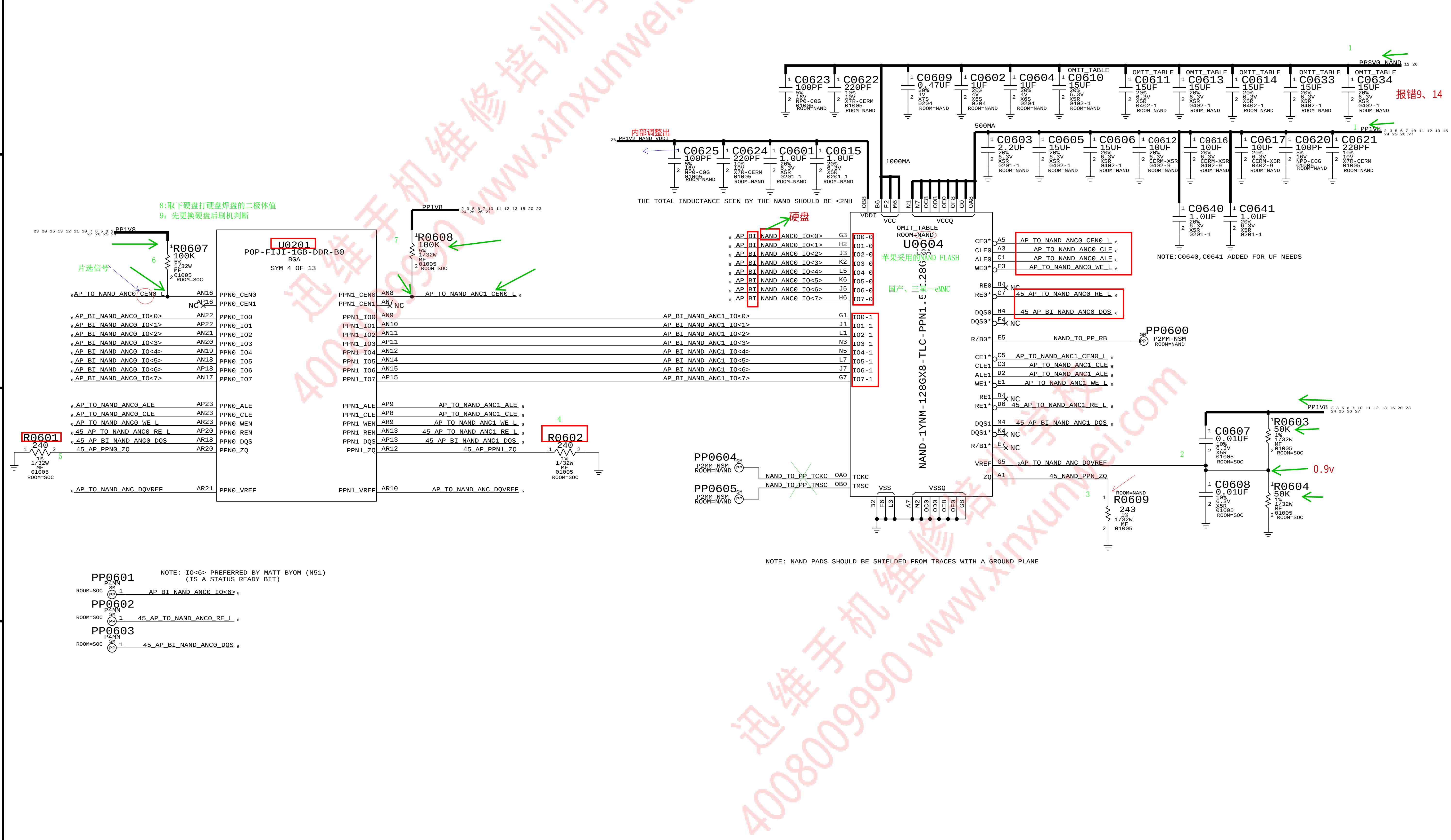


FIJI: VDDIOD,VDDIO18,VDD_VAR_SOC

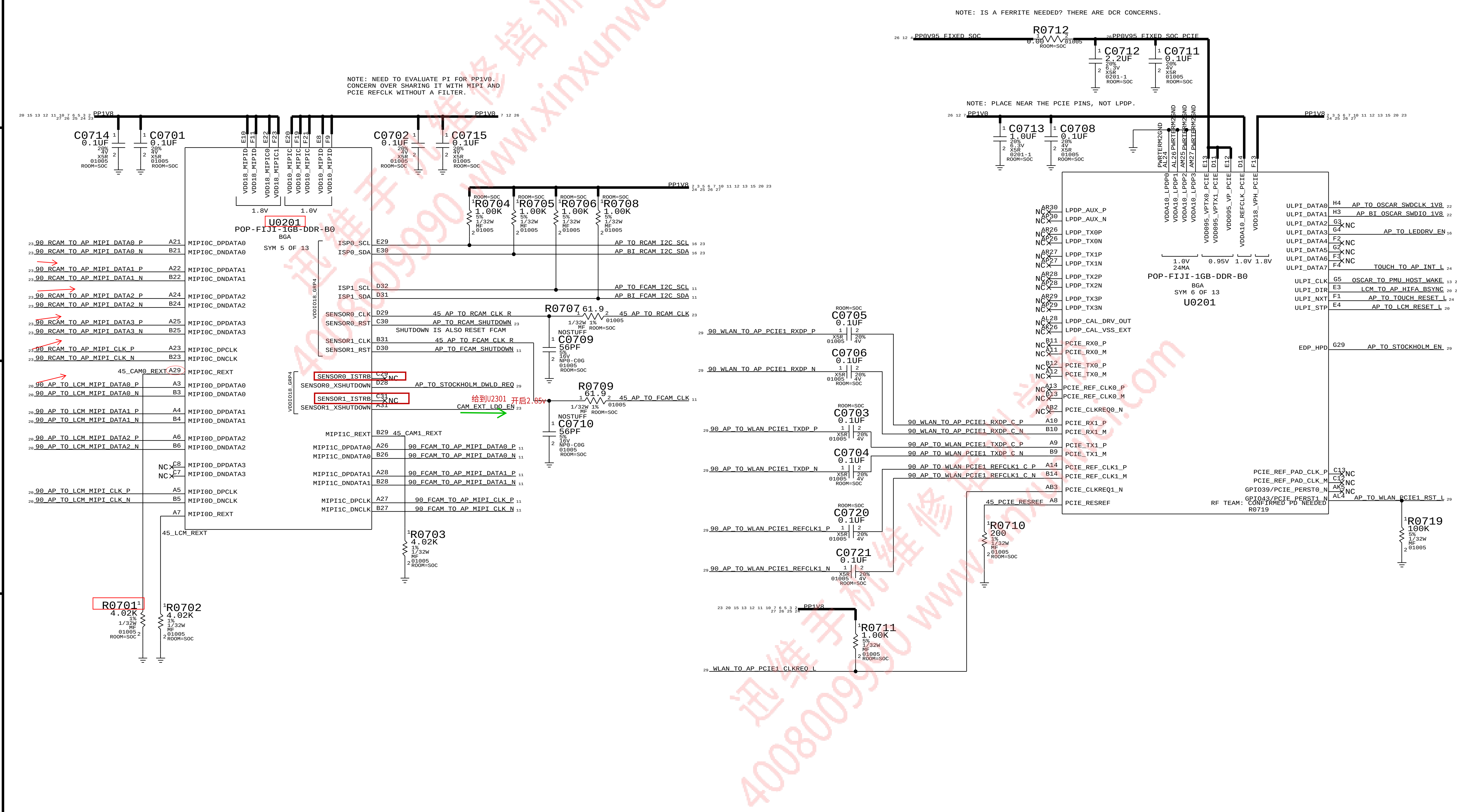


FIJI: NAND + 12X17 NAND PKG

SUPPORT FOR PPN1.5 (1.8V IO) ONLY

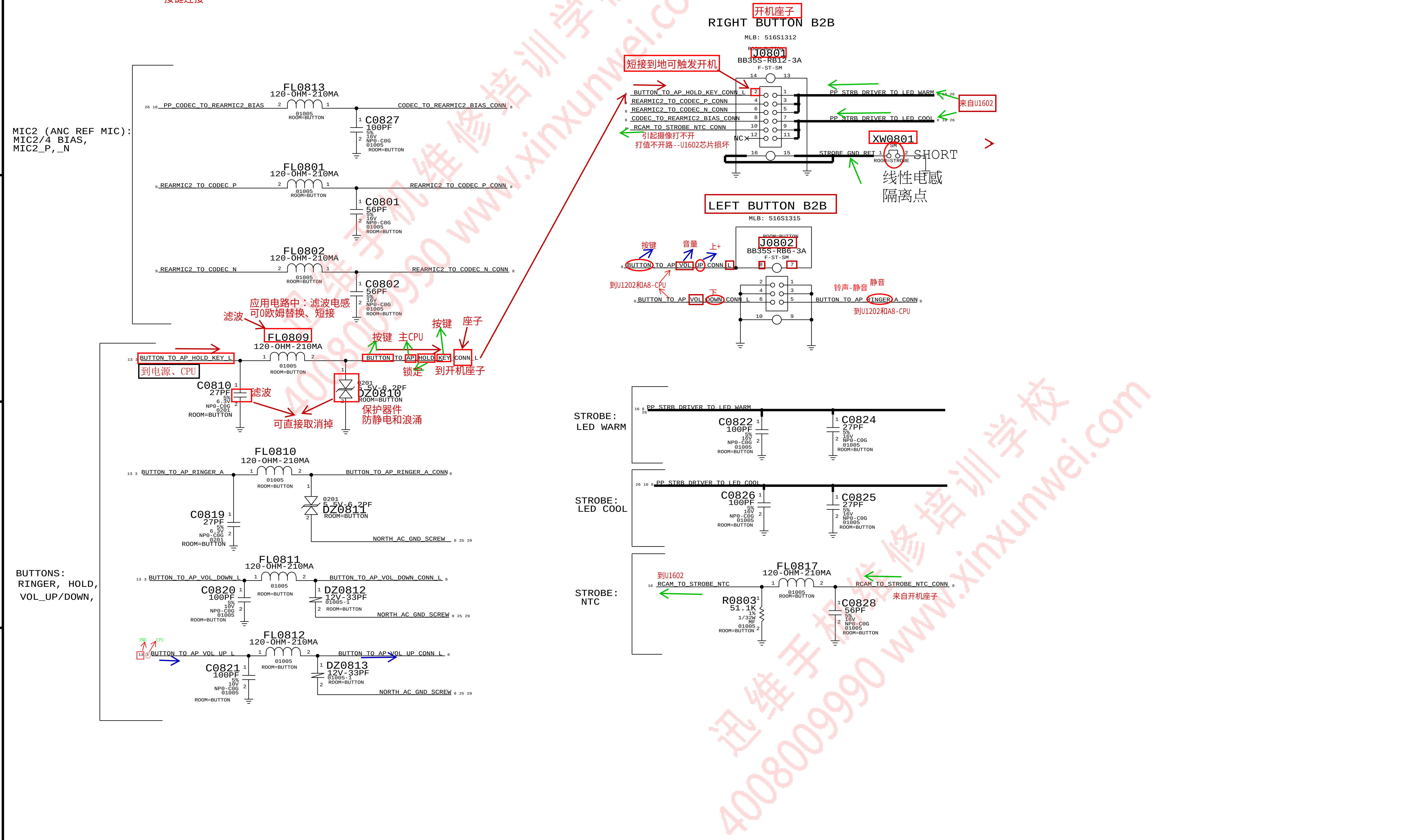


FIJI: HIGH SPEED DIG (CAM,LCD,LPDP,PCIE)



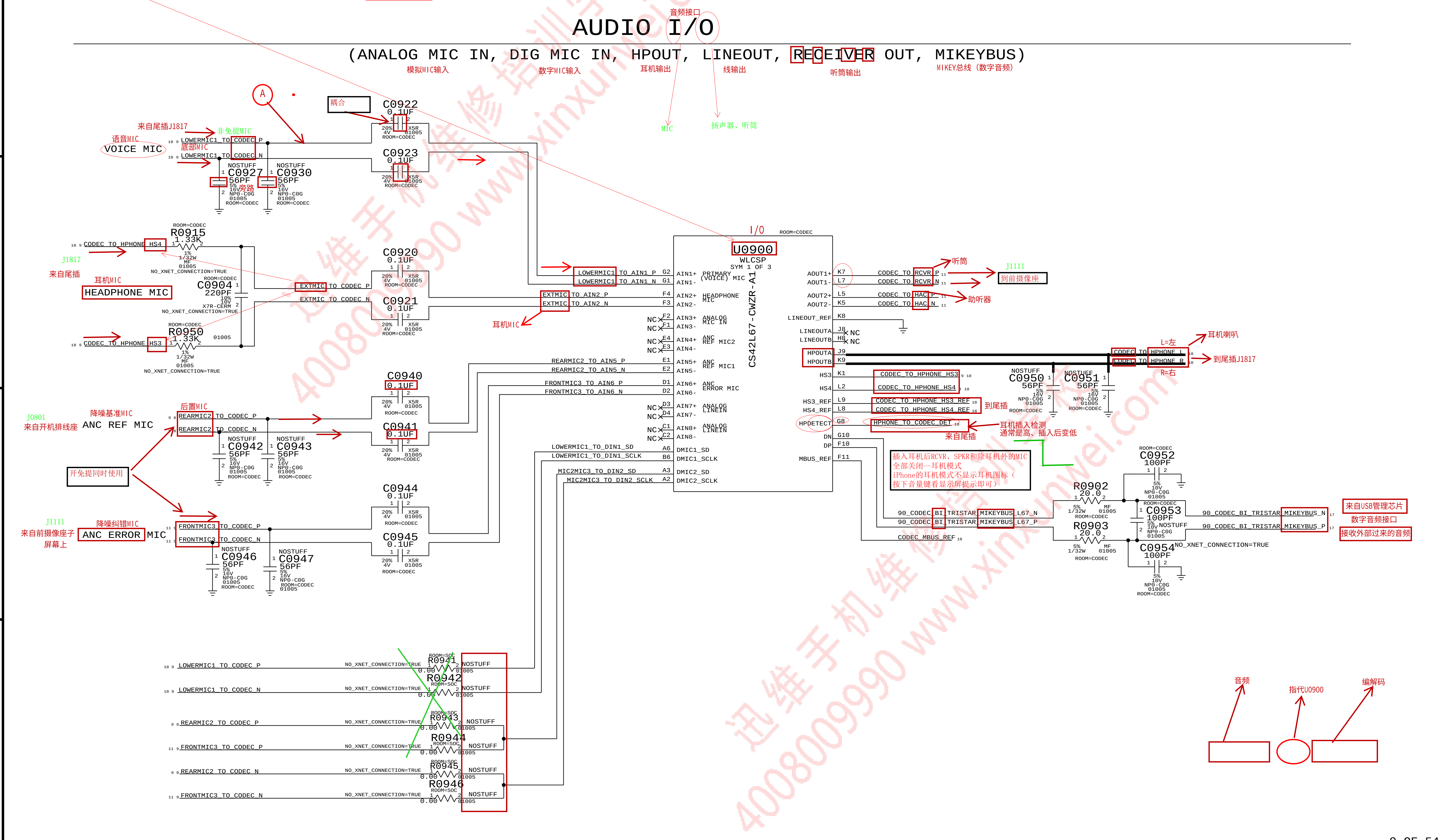
BUTTON FLEX (BUTTONS, ANC REF MIC, STROBE, STROBE_NTC, WIFI FLEX PAC)

按键连接



L67 AUDIO CODEC

音频编解码 abbr.信号编解码器(=decoder coder)



L67 AUDIO CODEC

音频 编解码器

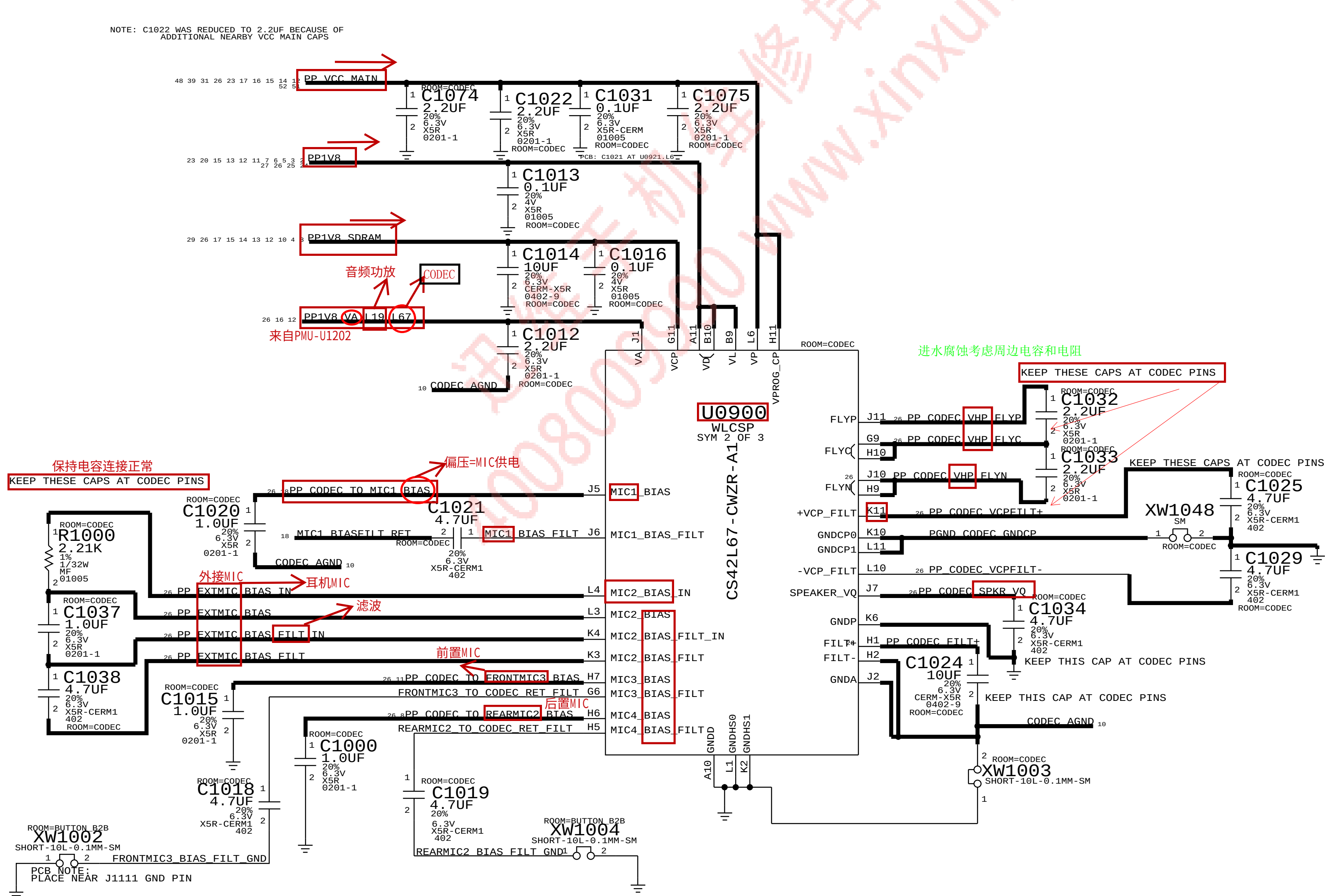
U0900

MIC--模拟输入--放大--A/D转换（音频编码）--D/A---放大--模拟输出--喇叭（听筒）

供电 MIC供电

POWER, MICBIAS

NOTE: C1022 WAS REDUCED TO 2.2UF BECAUSE OF ADDITIONAL NEARBY VCC MAIN CAPS

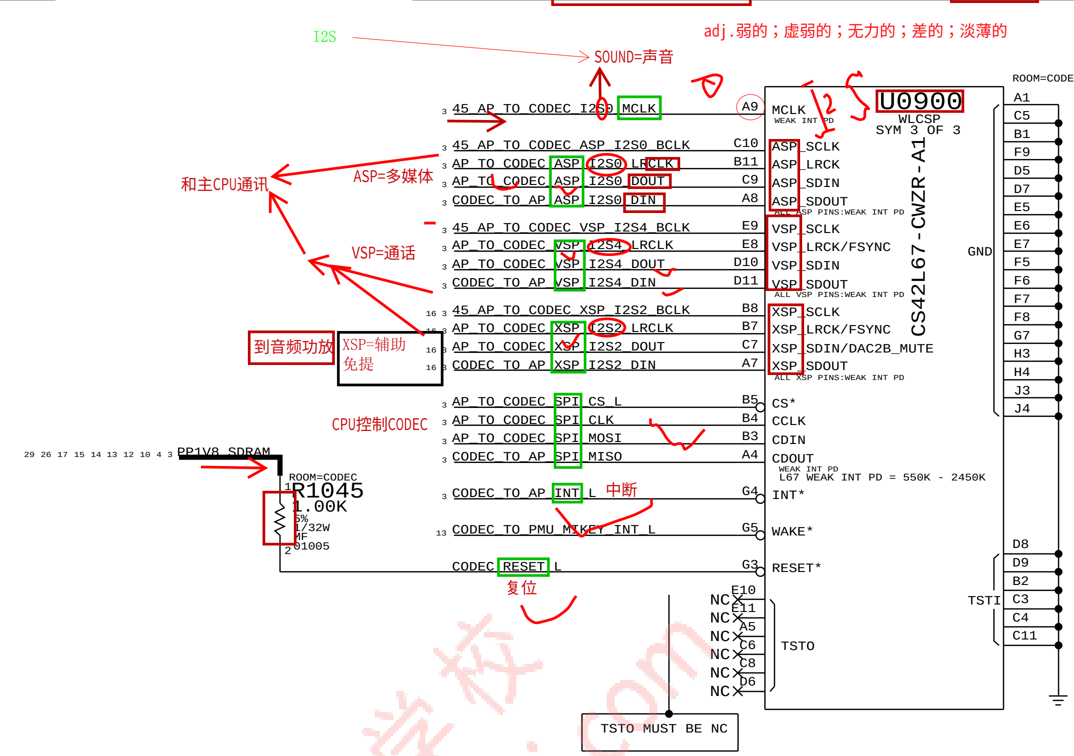


数字信号传输接口

DIGITAL SYSTEM I/O

I2S SOUND=声音

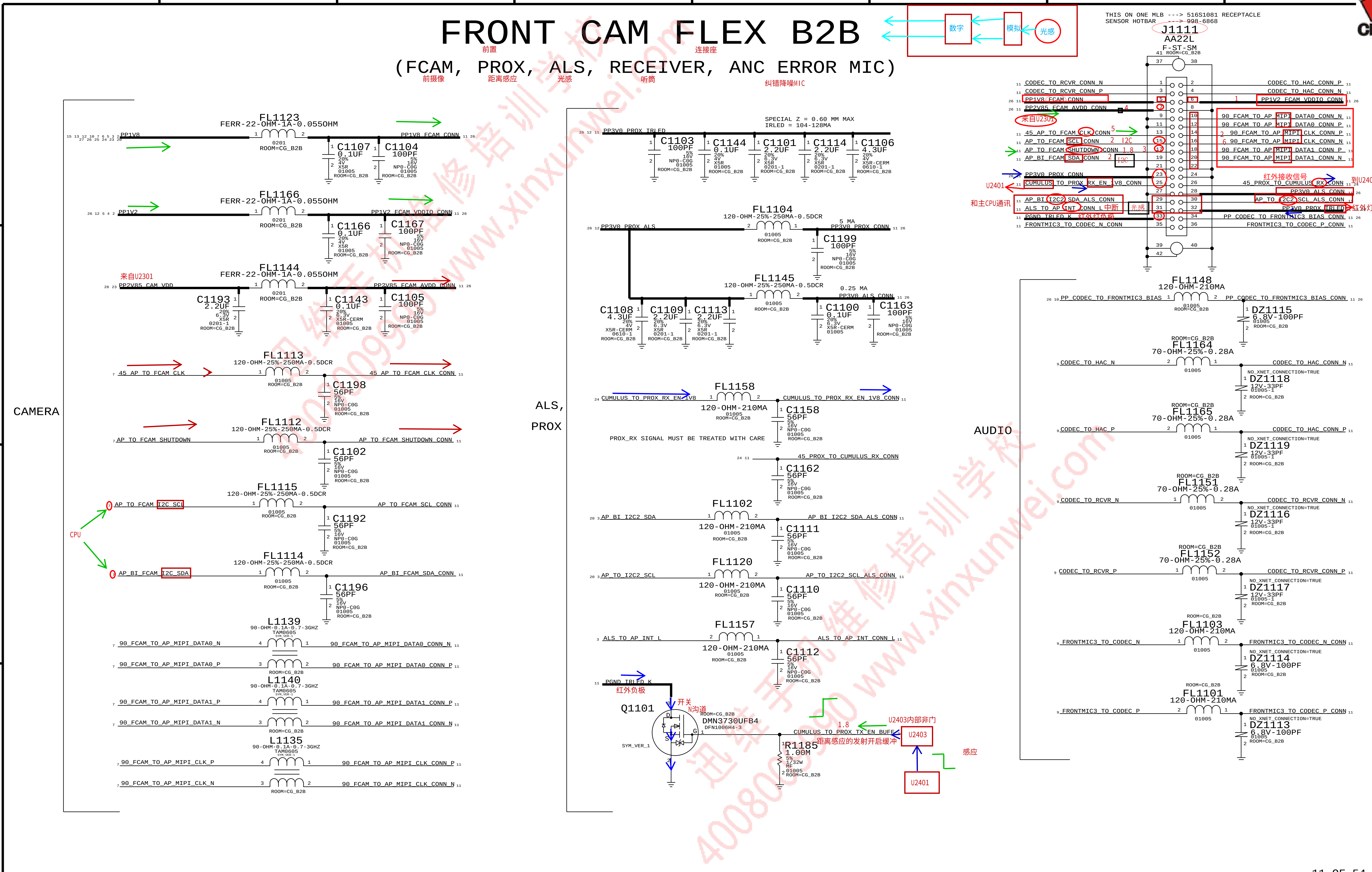
adj.弱的；虚弱的；无力的；差的；淡薄的

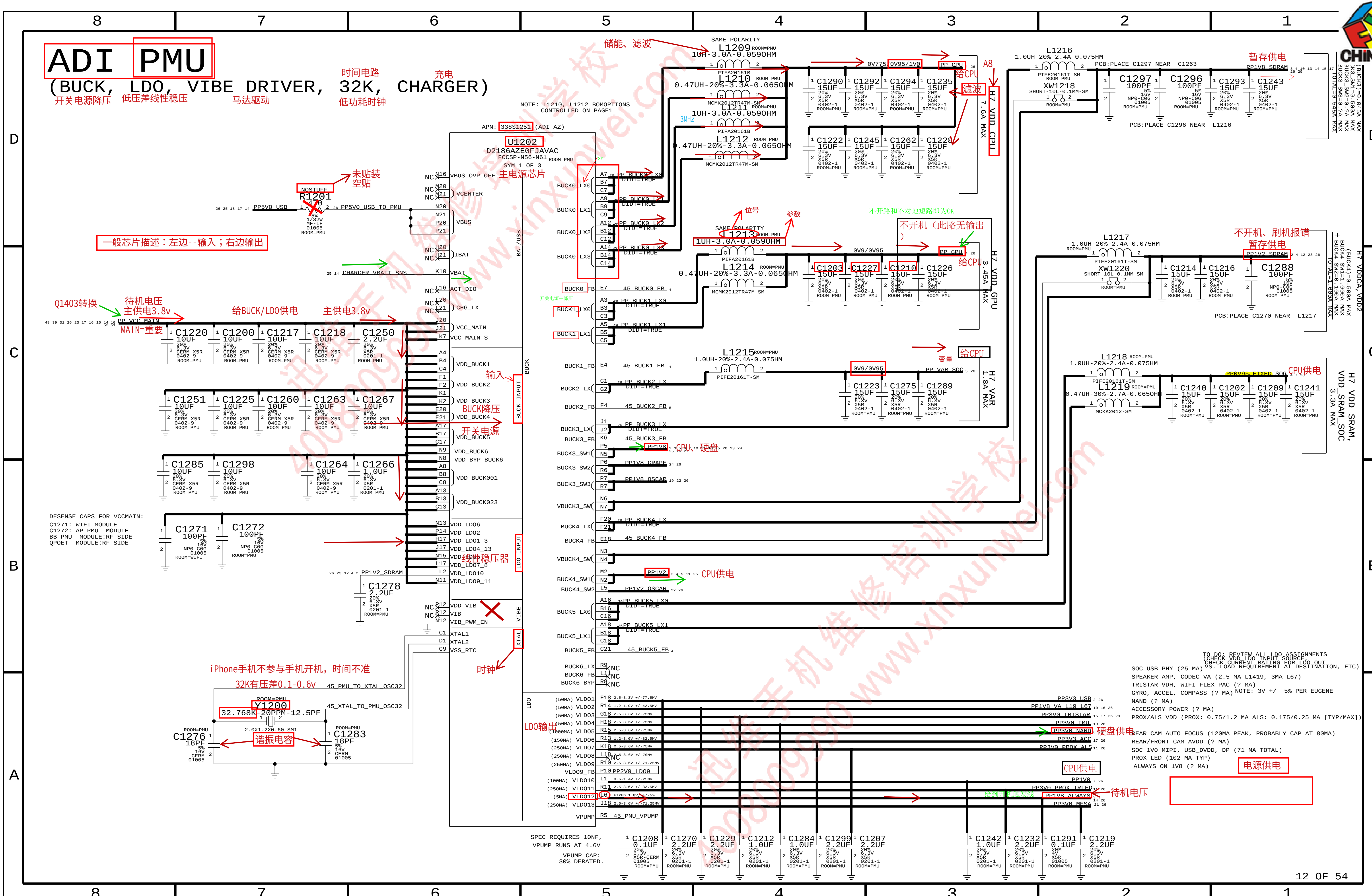


FRONT CAM FLEX B2B

(FCAM, PROX, ALS, RECEIVER, ANC ERROR MIC)

前置 连接座 前摄像 距离感应 光感 听筒 纠错降噪MIC





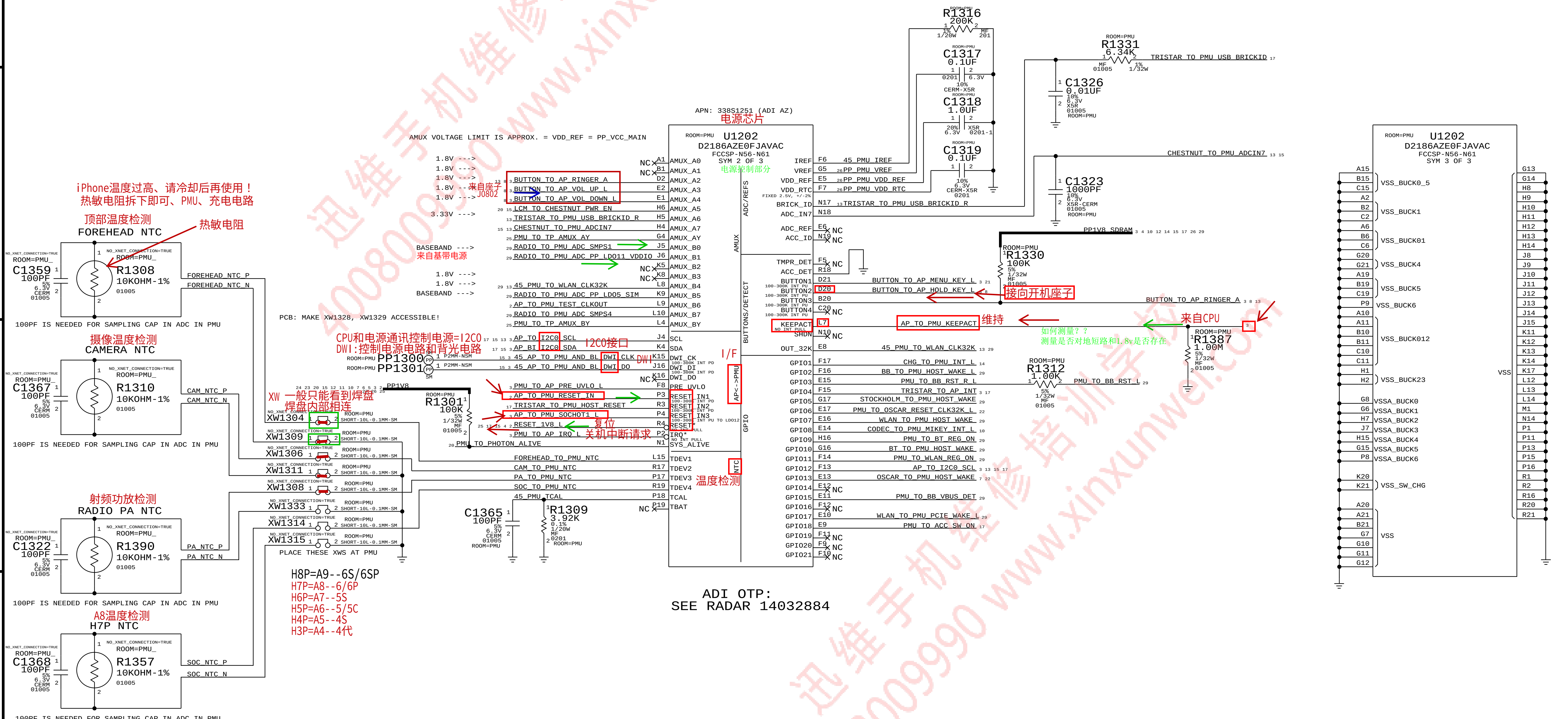
ADI PMU
(BUCK, LDO, VIBE DRIVER, 32K, CHARGER)
开关电源降压
低压差线性稳压
马达驱动
时间电路
低功耗时钟
充电
储能、滤波
NOTE: L1210, L1212 BOMOPTIONS CONTROLLED ON PAGE1
APN: 338S1251 (ADI AZ)
U1202
D2186AZE0FJAVAC
FCCSP-N56-N61
SYM 1 OF 3
主电源芯片
未贴装
空贴
NOSTUFF
R1201
PP5V0_USB
PP5V0_USB_TO_PMU
一般芯片描述：左边--输入；右边输出
CHARGER_VBATT_SNS
Q1403转换
待机电压
主供电3.8v
给BUCK/LDO供电
主供电3.8v
PP_VCC_MAIN
MAIN=重要
输入
BUCK降压
开关电源
线性稳压器
DESENSE CAPS FOR VCCMAIN:
C1271: WIFI MODULE
C1272: AP PMU MODULE
BB PMU MODULE:RF SIDE
QPOET MODULE:RF SIDE
PP1V2_SDRAM
iPhone手机不参与手机开机，时间不准
32K有压差0.1-0.6v
45_PMU_TO_XTAL_OSC32
45_XTAL_TO_PMU_OSC32
Y1200
32.768K-20PPM-12.5PF
谐振电容
时钟
LDO输出
SAME POLARITY
L1209
L1210
L1211
L1212
L1213
L1214
L1215
L1216
L1217
L1218
L1219
XW1218
XW1220
PP_CPU
给CPU
滤波
A8
H7 VDD_CPU
7.6A MAX
PP_GPU
H7 VDD_GPU
3.45A MAX
PP_VAR_SOC
H7 VAR
1.8A MAX
0V775/0V95/1V0
0V9/0V95
0V9/0V95
位号
参数
不开路和不对地短路即为OK
不开机（此路无输出）
变量
不开机、刷机报错
暂存供电
PP1V8_SDRAM
PP1V2_SDRAM
PCB:PLACE C1297 NEAR C1263
PCB:PLACE C1296 NEAR L1216
PCB:PLACE C1270 NEAR L1217
PP0V95_FIXED_SOC
CPU供电
H7 VDD_SRAM, VDD_SRAM_SOC
3.3A MAX
PP1V8
CPU、硬盘
PP1V8_GRAPE
PP1V8_OSCAR
PP1V2
CPU供电
PP1V2_OSCAR
PP3V3_USB
PP1V8_VA_L19_L67
PP3V0_TRISTAR
PP3V0_IMU
PP3V0_NAND
硬盘供电
PP3V3_ACC
PP3V0_PROX_ALS
CPU供电
PP1V0
PP3V0_PROX_IRLED
PP1V8_ALWAYS
待机电压
PP3V0_MESA
电源供电
TO DO: REVIEW ALL LDO ASSIGNMENTS
CHECK VDD_LDO INPUT SOURCE
CHECK CURRENT RATING FOR LDO OUT VS. LOAD REQUIREMENT AT DESTINATION, ETC)
SOC USB PHY (25 MA)
SPEAKER AMP, CODEC VA (2.5 MA L1419, 3MA L67)
TRISTAR VDH, WIFI_FLEX PAC (? MA)
GYRO, ACCEL, COMPASS (? MA)
NAND (? MA)
ACCESSORY POWER (? MA)
PROX/ALS VDD (PROX: 0.75/1.2 MA ALS: 0.175/0.25 MA [TYP/MAX])
REAR CAM AUTO FOCUS (120MA PEAK, PROBABLY CAP AT 80MA)
REAR/FRONT CAM AVDD (? MA)
SOC 1V0 MIPI, USB_DVDD, DP (71 MA TOTAL)
PROX LED (102 MA TYP)
ALWAYS ON 1V8 (? MA)
NOTE: 3V +/- 5% PER EUGENE
SPEC REQUIRES 10NF, VPUMP RUNS AT 4.6V
VPUMP CAP: 30% DERATED.
45_PMU_VPUMP
CHINAFX

ADI PMU

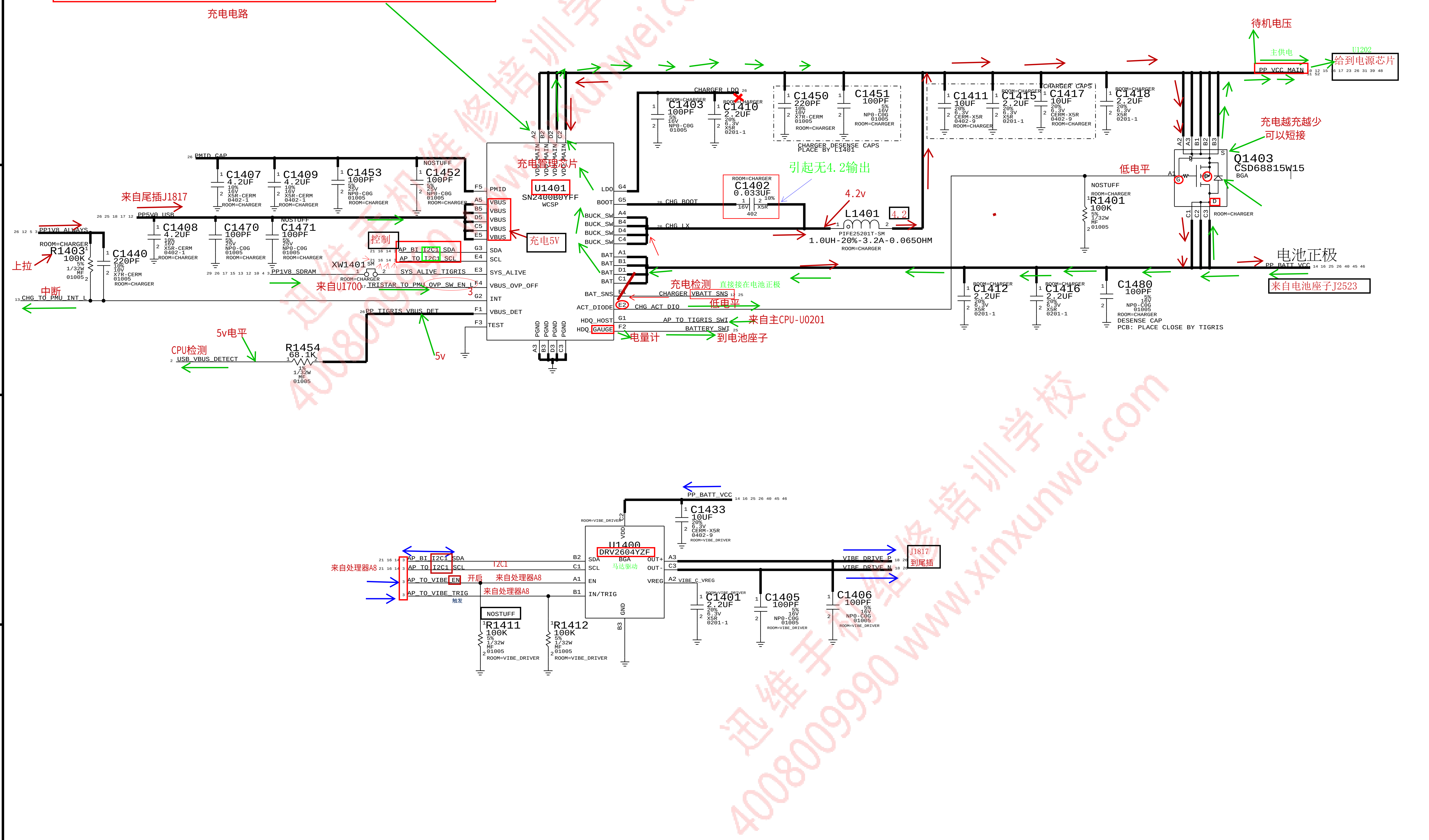
按键 数模转换 温度检测电路 系统接口 地

(AMUX, GPIO, BUTTONS, ADC, THERMISTORS, SYSTEM I/F, GND)

模拟多路复合开关 通用型输入输出接口 NTC 负极



TIGRIS CHARGER & VIBE DRIVER

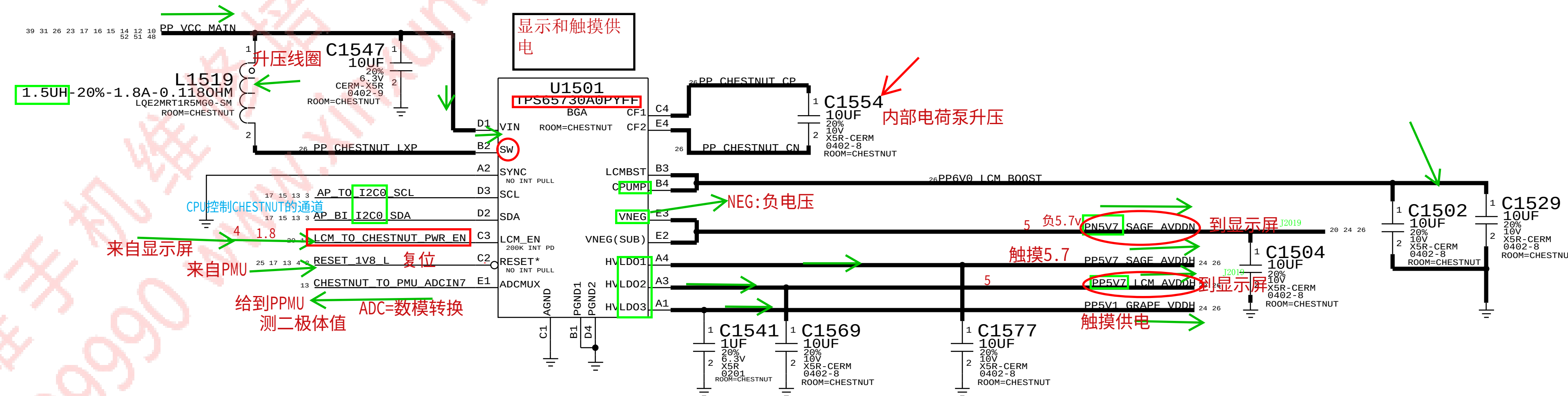


CHESTNUT, BACKLIGHT DRIVER, MESA BOOST

显示电源芯片 背光驱动 指纹升压

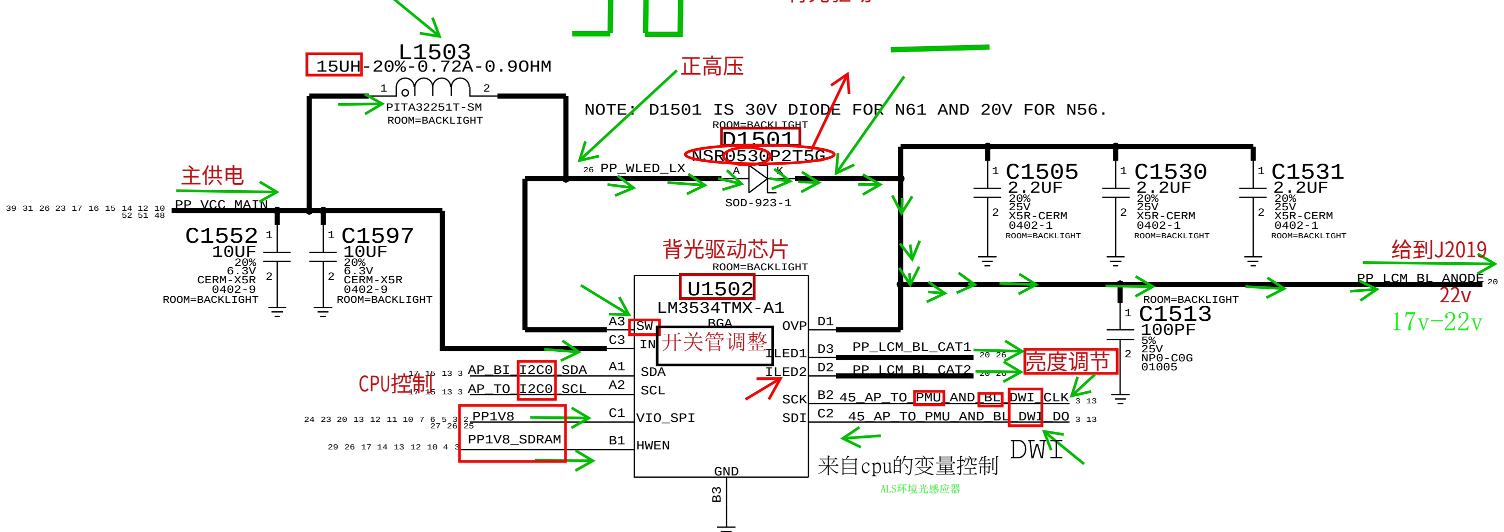
D500 DISPLAY PMU (TI CHESTNUT, 338S1149)

显示电源

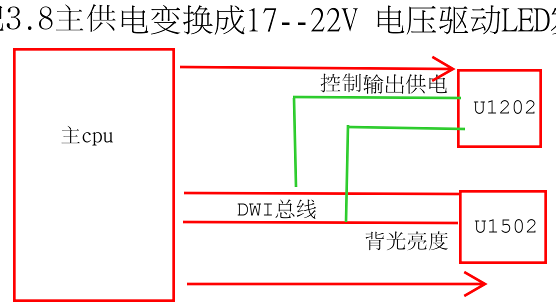


D500 BACKLIGHT DRIVER

背光驱动



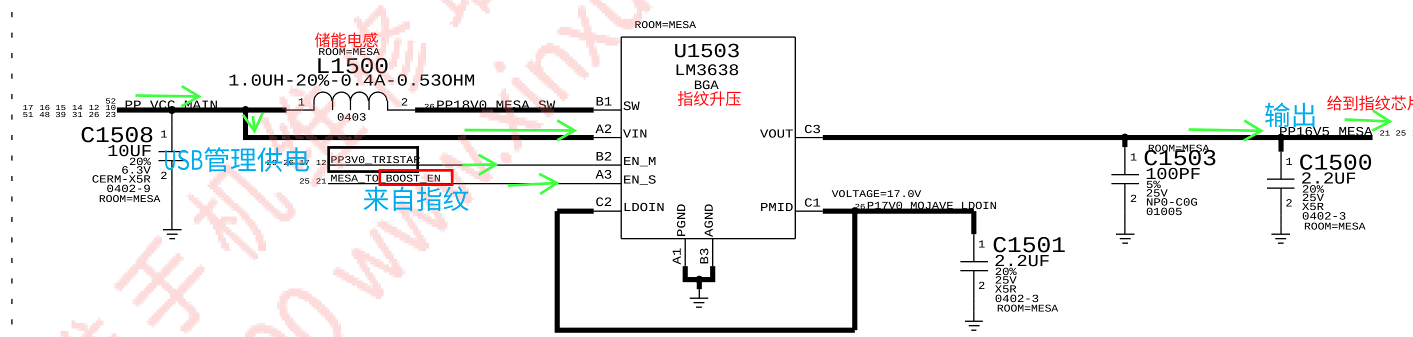
把3.8主供电变换成17--22V 电压驱动LED发光（屏背光部分）



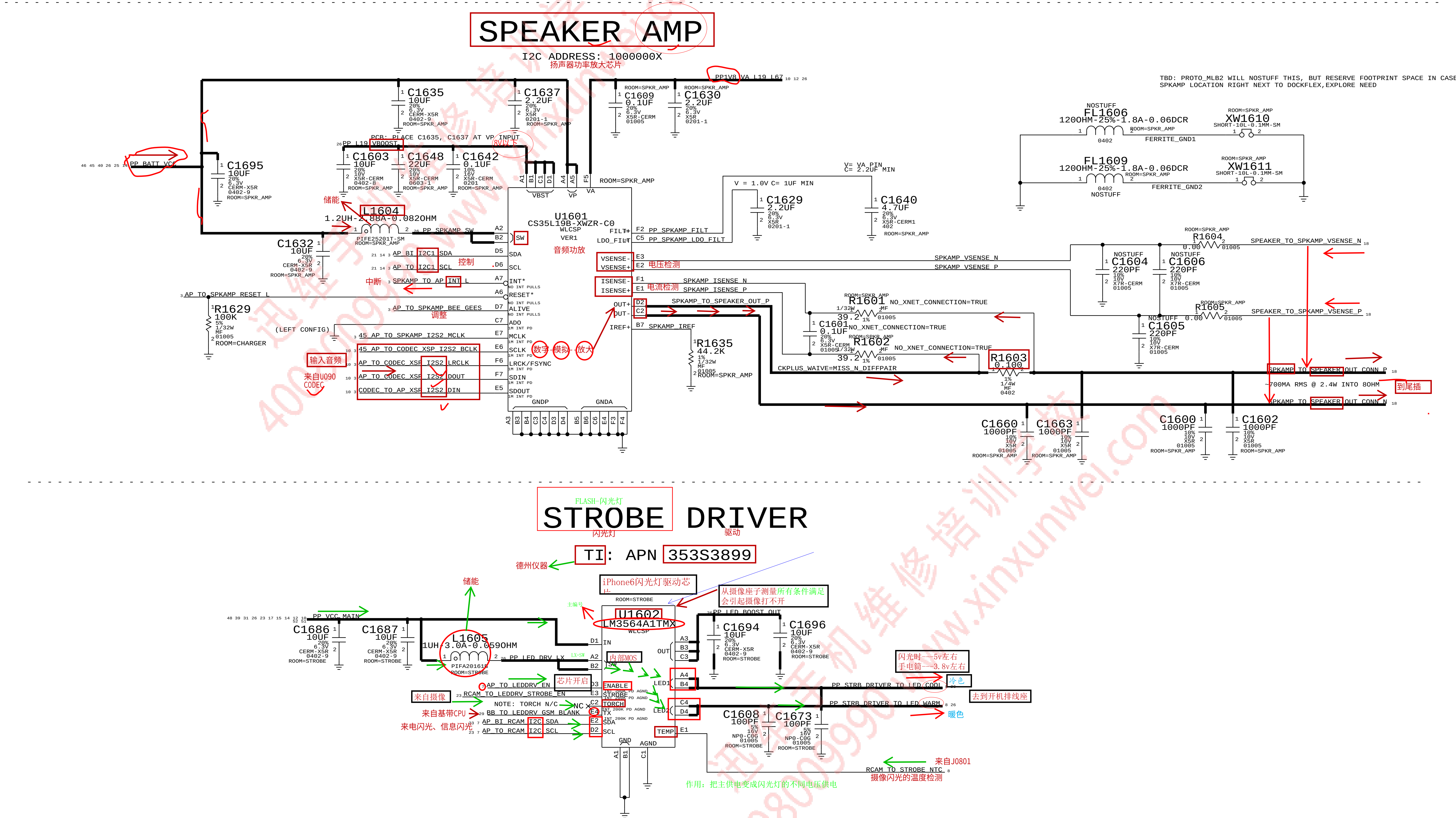
MESA BOOST A0

指纹升压

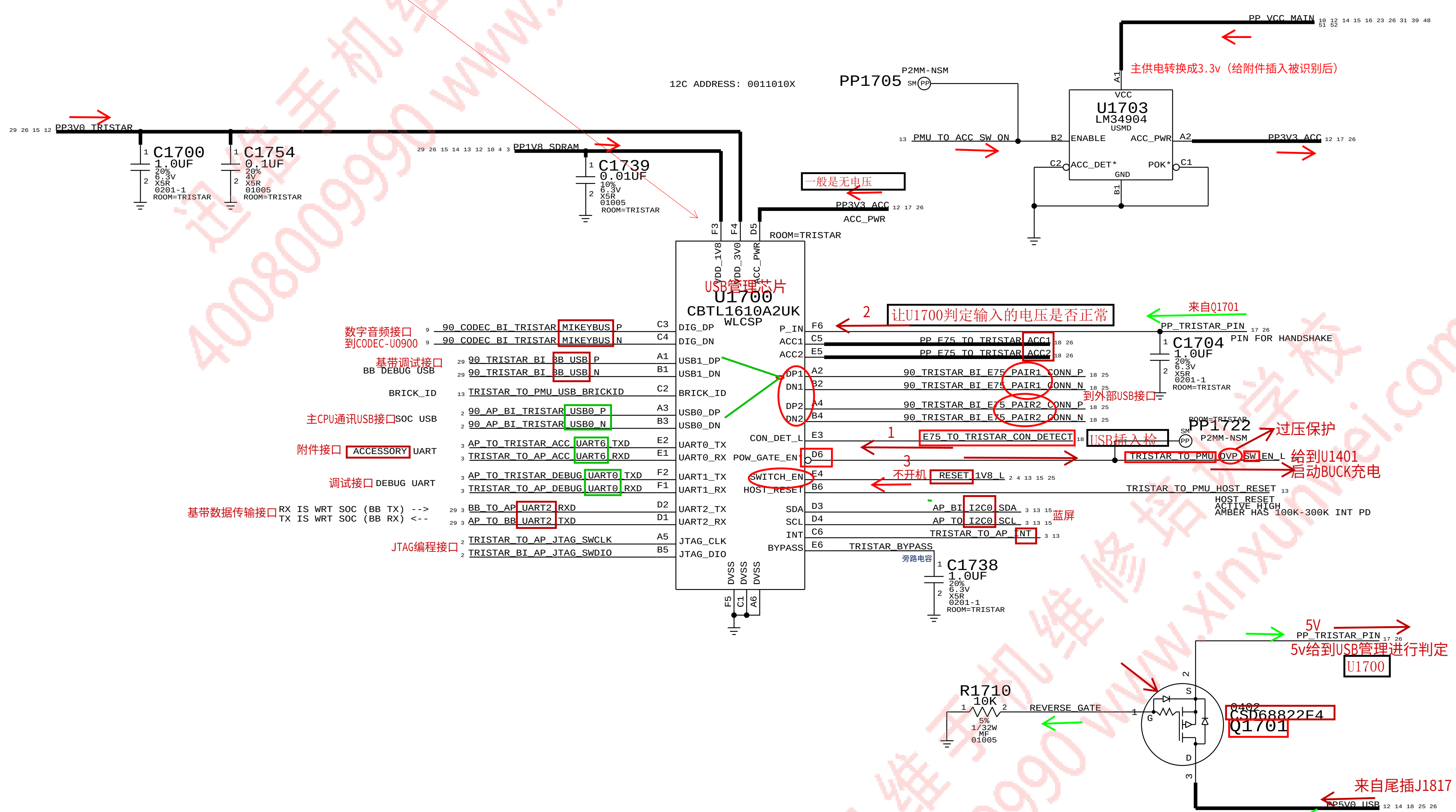
APN: 353S3978



SPEAKER AMP, LED DRIVER

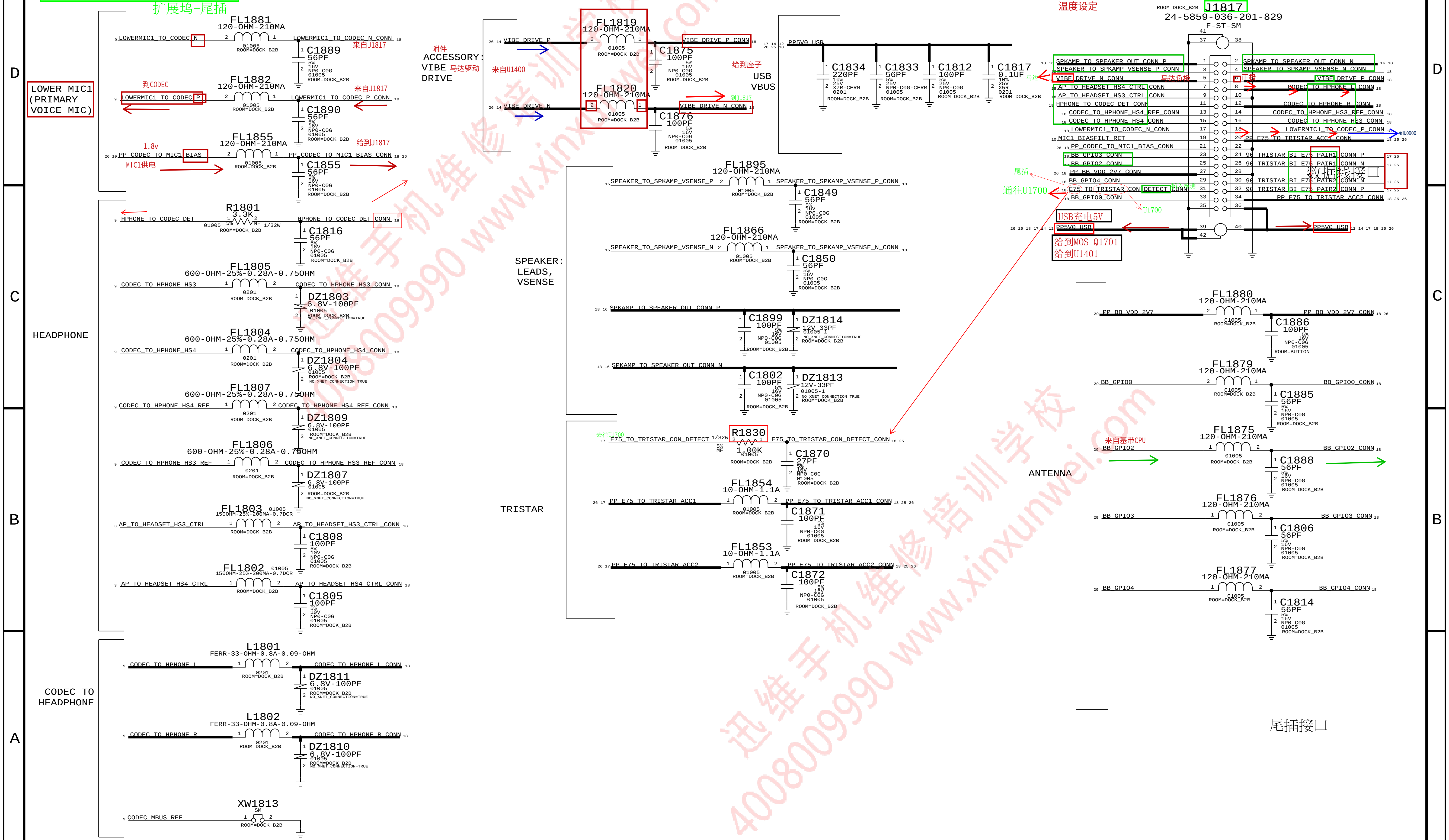


TRISTAR2



I2C ADDRESS: 0011010X
PP1705
P2MM-NSM
PP_VCC_MAIN
主供电转换成3.3v（给附件插入被识别后）
U1703
LM34904
VCC
ENABLE
ACC_PWR
ACC_DET*
POK*
GND
PMU_TO_ACC_SW_ON
PP3V3_ACC
PP3V0_TRISTAR
PP1V8_SDRAM
C1700
1.0UF
C1754
0.1UF
C1739
0.01UF
ROOM=TRISTAR
一般是无电压
ACC_PWR
USB管理芯片
U1700
CBTL1610A2UK
WLCSP
VDD_1V8
VDD_3V0
ACC_PWR
数字音频接口
到CODEC-U0900
90_CODEC_BI_TRISTAR_MIKEYBUS_P
90_CODEC_BI_TRISTAR_MIKEYBUS_N
基带调试接口
BB DEBUG USB
90_TRISTAR_BI_BB_USB_P
90_TRISTAR_BI_BB_USB_N
BRICK_ID
TRISTAR_TO_PMU_USB_BRICKID
主CPU通讯USB接口
SOC USB
90_AP_BI_TRISTAR_USB0_P
90_AP_BI_TRISTAR_USB0_N
附件接口
ACCESSORY UART
AP_TO_TRISTAR_ACC_UART6_TXD
TRISTAR_TO_AP_ACC_UART6_RXD
调试接口
DEBUG UART
AP_TO_TRISTAR_DEBUG_UART0_TXD
TRISTAR_TO_AP_DEBUG_UART0_RXD
基带数据传输接口
RX IS WRT SOC (BB TX) -->
TX IS WRT SOC (BB RX) <--
BB_TO_AP_UART2_RXD
AP_TO_BB_UART2_TXD
JTAG编程接口
TRISTAR_TO_AP_JTAG_SWCLK
TRISTAR_BI_AP_JTAG_SWDIO
DIG_DP
DIG_DN
USB1_DP
USB1_DN
USB0_DP
USB0_DN
UART0_TX
UART0_RX
UART1_TX
UART1_RX
UART2_TX
UART2_RX
JTAG_CLK
JTAG_DIO
DVSS
P_IN
ACC1
ACC2
DP1
DN1
DP2
DN2
CON_DET_L
POW_GATE_EN
SWITCH_EN
HOST_RESET
SDA
SCL
INT
BYPASS
2 让U1700判定输入的电压是否正常
来自Q1701
PP_TRISTAR_PIN
PIN FOR HANDSHAKE
C1704
1.0UF
PP_E75_TO_TRISTAR_ACC1
PP_E75_TO_TRISTAR_ACC2
90_TRISTAR_BI_E75_PAIR1_CONN_P
90_TRISTAR_BI_E75_PAIR1_CONN_N
90_TRISTAR_BI_E75_PAIR2_CONN_P
90_TRISTAR_BI_E75_PAIR2_CONN_N
到外部USB接口
1
E75_TO_TRISTAR_CON_DETECT
USB插入检
PP1722
过压保护
TRISTAR_TO_PMU_OVP_SW_EN_L
给到U1401
启动BUCK充电
3
不开机
RESET_1V8_L
TRISTAR_TO_PMU_HOST_RESET
HOST RESET
ACTIVE HIGH
AMBER HAS 100K-300K INT PD
AP_BI_I2C0_SDA
AP_TO_I2C0_SCL
蓝屏
TRISTAR_TO_AP_INT
TRISTAR_BYPASS
旁路电容
C1738
1.0UF
5V
PP_TRISTAR_PIN
5v给到USB管理进行判定
U1700
R1710
10K
REVERSE_GATE
CSD68822F4
Q1701
来自尾插J1817
PP5V0_USB

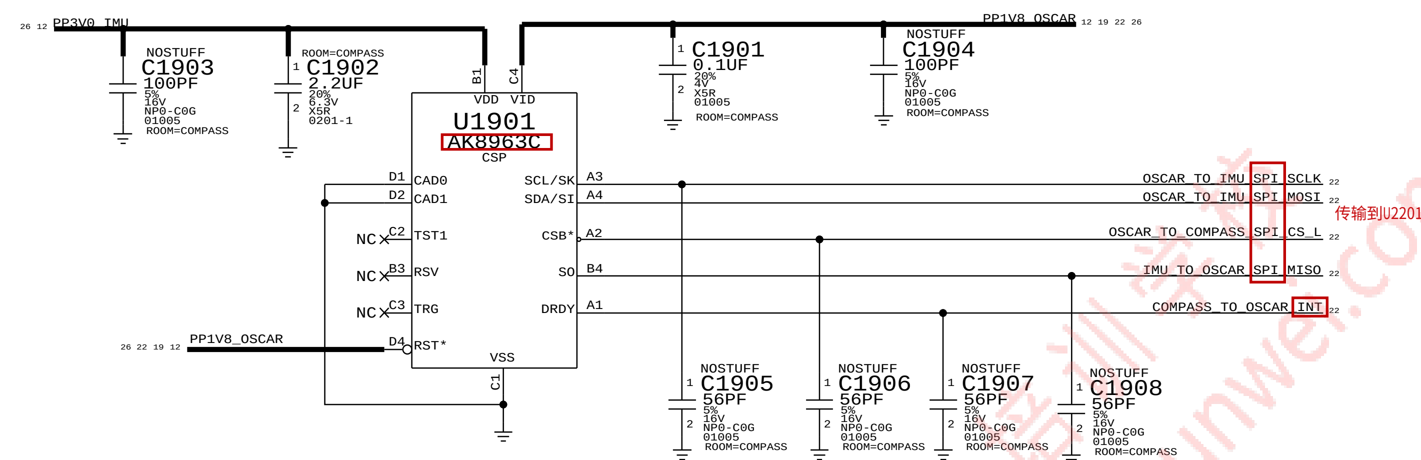
DOCKFLEX B2B (USB VBUS, SPEAKER,ANTENNA LAT SW CTRL, MIC1 (PRIMARY MIC), ACC DET/ID/PWR, E75 DIFFPAIRS)

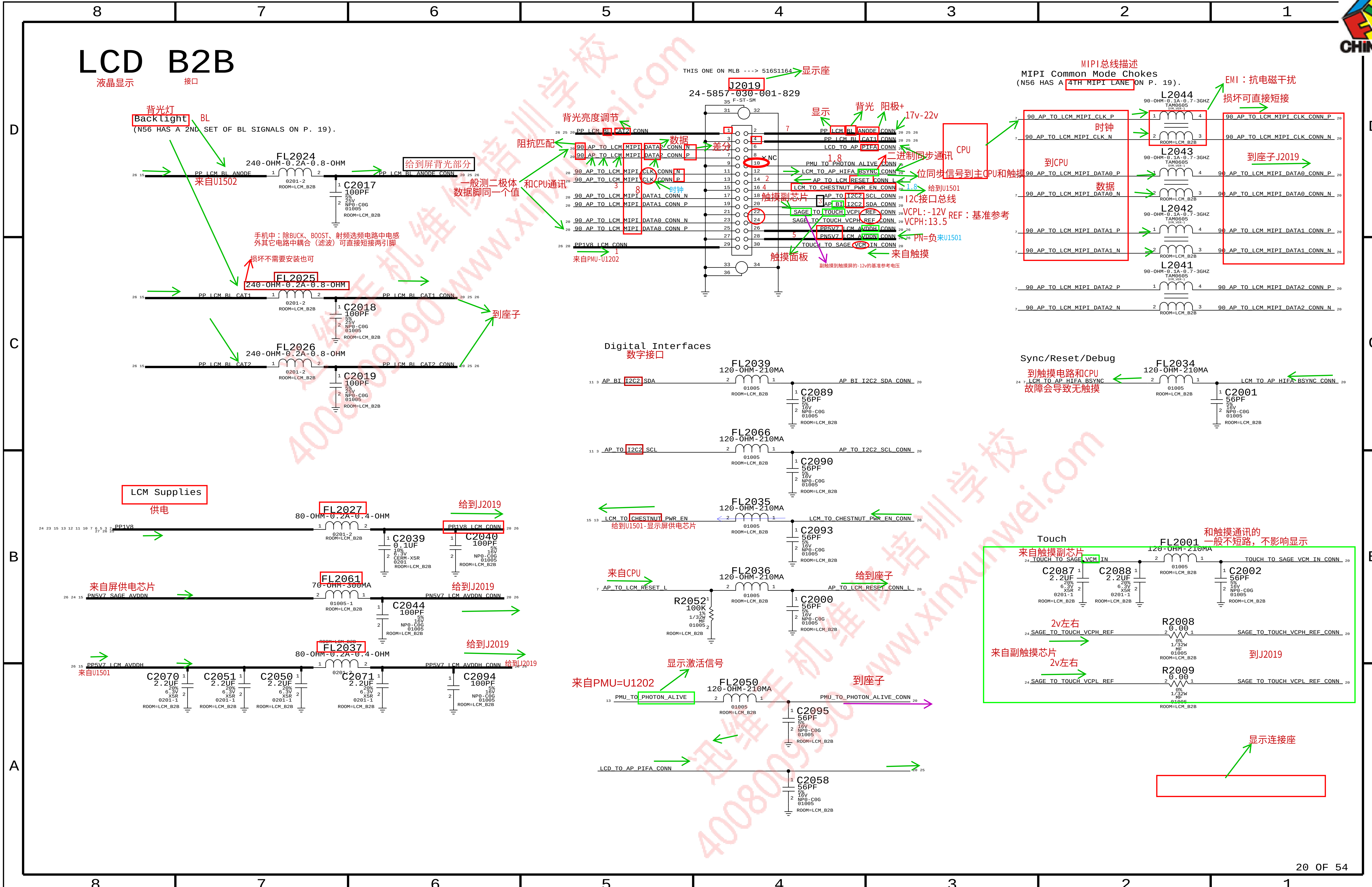


COMPASS - AKM COMPASS IN POR LOCATION

COMPASS CSP: 338S1014

指南针

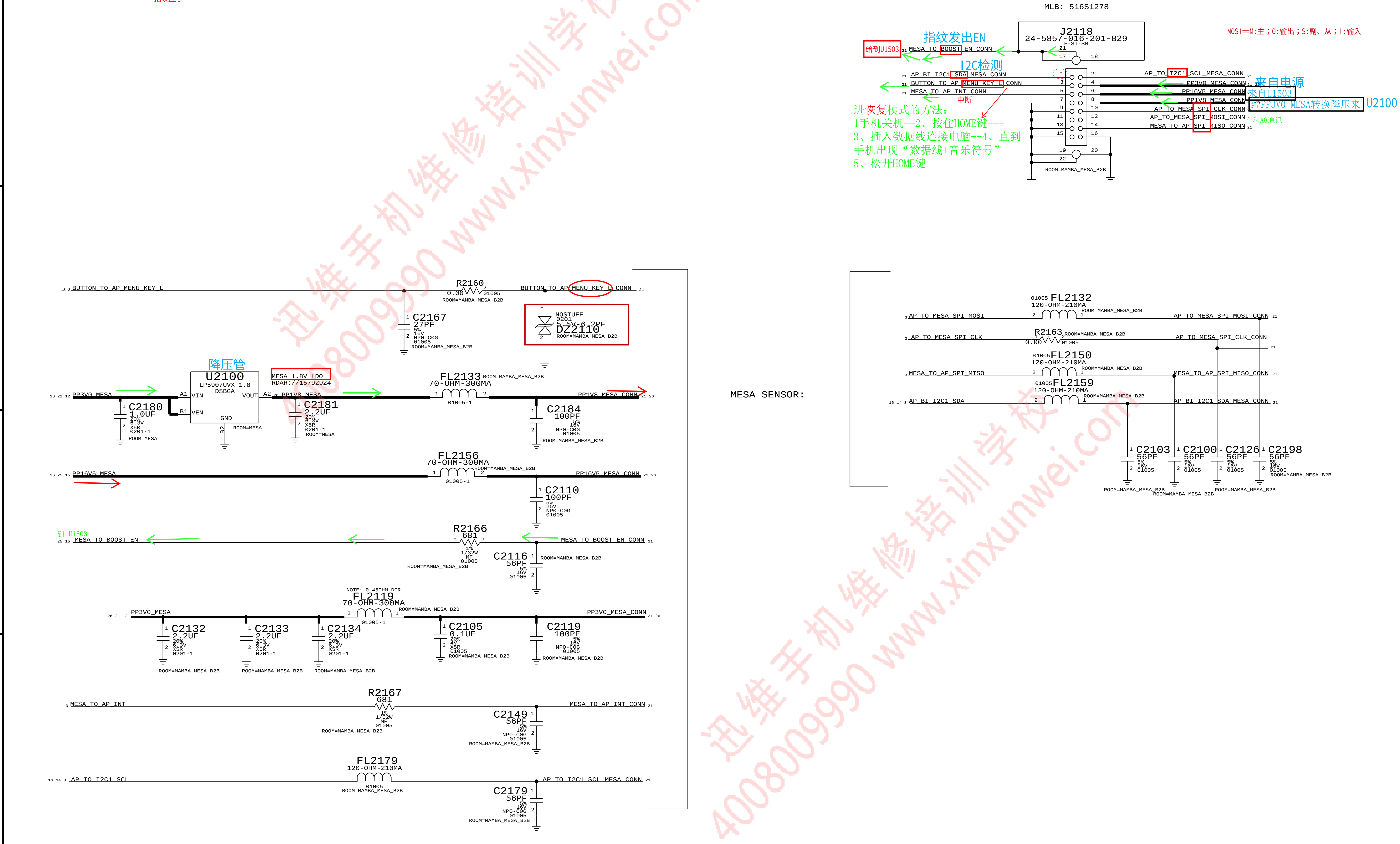


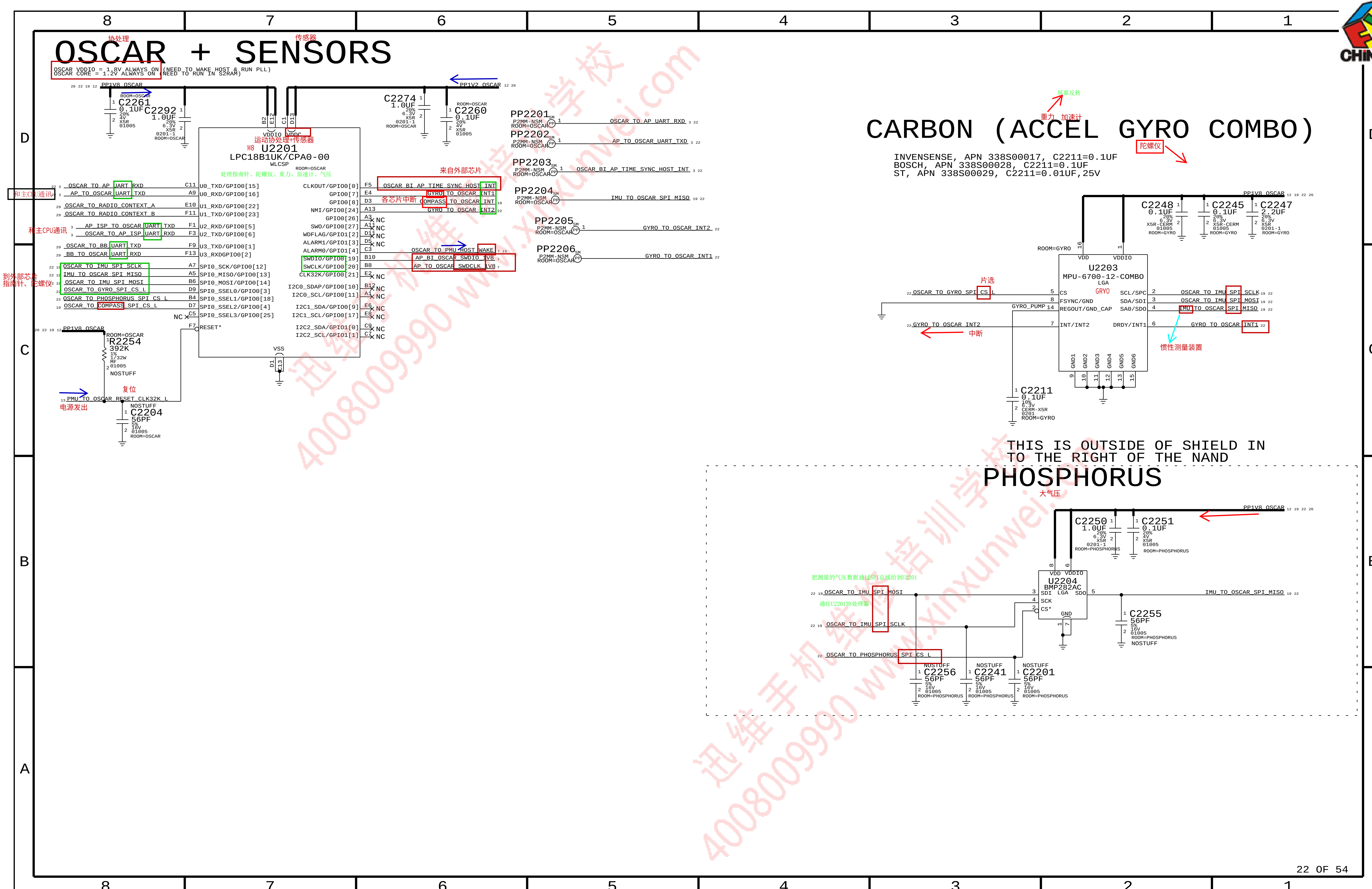


LCD B2B
液晶显示
接口
Backlight
背光灯
BL
(N56 HAS A 2ND SET OF BL SIGNALS ON P. 19).
FL2024
240-OHM-0.2A-0.8-OHM
PP_LCM_BL_ANODE
来自U1502
PP_LCM_BL_ANODE_CONN
给到屏背光部分
C2017
100PF
一般测二极体数据脚同一个值
手机中：除BUCK、BOOST、射频选频电路中电感外其它电路中耦合（滤波）可直接短接两引脚
损坏不需要安装也可
FL2025
240-OHM-0.2A-0.8-OHM
PP_LCM_BL_CAT1
PP_LCM_BL_CAT1_CONN
C2018
100PF
到座子
FL2026
240-OHM-0.2A-0.8-OHM
PP_LCM_BL_CAT2
PP_LCM_BL_CAT2_CONN
C2019
100PF
THIS ONE ON MLB ---> 516S1164
显示座
J2019
24-5857-030-001-829
F-ST-SM
背光亮度调节
阻抗匹配
数据
差分
和CPU通讯
时钟
PP_LCM_BL_CAT2_CONN
90_AP_TO_LCM_MIPI_DATA2_CONN_N
90_AP_TO_LCM_MIPI_DATA2_CONN_P
90_AP_TO_LCM_MIPI_CLK_CONN_N
90_AP_TO_LCM_MIPI_CLK_CONN_P
90_AP_TO_LCM_MIPI_DATA1_CONN_N
90_AP_TO_LCM_MIPI_DATA1_CONN_P
90_AP_TO_LCM_MIPI_DATA0_CONN_N
90_AP_TO_LCM_MIPI_DATA0_CONN_P
PP1V8_LCM_CONN
来自PMU-U1202
显示
背光
阳极+
17v-22v
PP_LCM_BL_ANODE_CONN
PP_LCM_BL_CAT1_CONN
LCD_TO_AP_PIFA_CONN
PMU_TO_PHOTON_ALIVE_CONN
二进制同步通讯
LCM_TO_AP_HIFA_BSYNC_CONN
AP_TO_LCM_RESET_CONN_L
位同步信号到主CPU和触摸
LCM_TO_CHESTNUT_PWR_EN_CONN
给到U1501
触摸副芯片
AP_TO_I2C2_SCL_CONN
AP_BI_I2C2_SDA_CONN
I2C接口总线
SAGE_TO_TOUCH_VCPL_REF_CONN
SAGE_TO_TOUCH_VCPH_REF_CONN
VCPL:-12V
VCPH:13.5
REF：基准参考
PP5V7_LCM_AVDDH_CONN
PN5V7_LCM_AVDDN_CONN
PN=负
TOUCH_TO_SAGE_VCM_IN_CONN
触摸面板
来自触摸
副触摸到触摸屏的-12v的基准参考电压
CPU
MIPI总线描述
MIPI Common Mode Chokes
(N56 HAS A 4TH MIPI LANE ON P. 19).
EMI：抗电磁干扰
损坏可直接短接
L2044
90-OHM-0.1A-0.7-3GHZ
TAM0605
90_AP_TO_LCM_MIPI_CLK_P
90_AP_TO_LCM_MIPI_CLK_N
时钟
90_AP_TO_LCM_MIPI_CLK_CONN_P
90_AP_TO_LCM_MIPI_CLK_CONN_N
L2043
到CPU
到座子J2019
90_AP_TO_LCM_MIPI_DATA0_P
90_AP_TO_LCM_MIPI_DATA0_N
数据
90_AP_TO_LCM_MIPI_DATA0_CONN_P
90_AP_TO_LCM_MIPI_DATA0_CONN_N
L2042
90_AP_TO_LCM_MIPI_DATA1_P
90_AP_TO_LCM_MIPI_DATA1_N
90_AP_TO_LCM_MIPI_DATA1_CONN_P
90_AP_TO_LCM_MIPI_DATA1_CONN_N
L2041
90_AP_TO_LCM_MIPI_DATA2_P
90_AP_TO_LCM_MIPI_DATA2_N
90_AP_TO_LCM_MIPI_DATA2_CONN_P
90_AP_TO_LCM_MIPI_DATA2_CONN_N
Digital Interfaces
数字接口
FL2039
120-OHM-210MA
AP_BI_I2C2_SDA
AP_BI_I2C2_SDA_CONN
C2089
56PF
FL2066
120-OHM-210MA
AP_TO_I2C2_SCL
AP_TO_I2C2_SCL_CONN
C2090
56PF
FL2035
120-OHM-210MA
LCM_TO_CHESTNUT_PWR_EN
给到U1501-显示屏供电芯片
LCM_TO_CHESTNUT_PWR_EN_CONN
C2093
56PF
来自CPU
FL2036
120-OHM-210MA
AP_TO_LCM_RESET_L
给到座子
AP_TO_LCM_RESET_CONN_L
R2052
100K
C2000
56PF
显示激活信号
来自PMU=U1202
FL2050
120-OHM-210MA
PMU_TO_PHOTON_ALIVE
PMU_TO_PHOTON_ALIVE_CONN
到座子
C2095
56PF
LCD_TO_AP_PIFA_CONN
C2058
56PF
Sync/Reset/Debug
FL2034
120-OHM-210MA
到触摸电路和CPU
故障会导致无触摸
LCM_TO_AP_HIFA_BSYNC
LCM_TO_AP_HIFA_BSYNC_CONN
C2001
56PF
Touch
和触摸通讯的
一般不短路，不影响显示
来自触摸副芯片
FL2001
120-OHM-210MA
TOUCH_TO_SAGE_VCM_IN
TOUCH_TO_SAGE_VCM_IN_CONN
C2087
2.2UF
C2088
2.2UF
C2002
56PF
2v左右
R2008
0.00
SAGE_TO_TOUCH_VCPH_REF
SAGE_TO_TOUCH_VCPH_REF_CONN
来自副触摸芯片
到J2019
2v左右
R2009
0.00
SAGE_TO_TOUCH_VCPL_REF
SAGE_TO_TOUCH_VCPL_REF_CONN
显示连接座
LCM Supplies
供电
FL2027
80-OHM-0.2A-0.4-OHM
PP1V8
给到J2019
PP1V8_LCM_CONN
C2039
0.1UF
C2040
100PF
来自屏供电芯片
FL2061
70-OHM-300MA
PN5V7_SAGE_AVDDN
给到J2019
PN5V7_LCM_AVDDN_CONN
C2044
100PF
FL2037
80-OHM-0.2A-0.4-OHM
PP5V7_LCM_AVDDH
来自U1501
给到J2019
PP5V7_LCM_AVDDH_CONN
C2070
2.2UF
C2051
2.2UF
C2050
2.2UF
C2071
2.2UF
C2094
100PF

MESA CONNECTOR

指纹座子

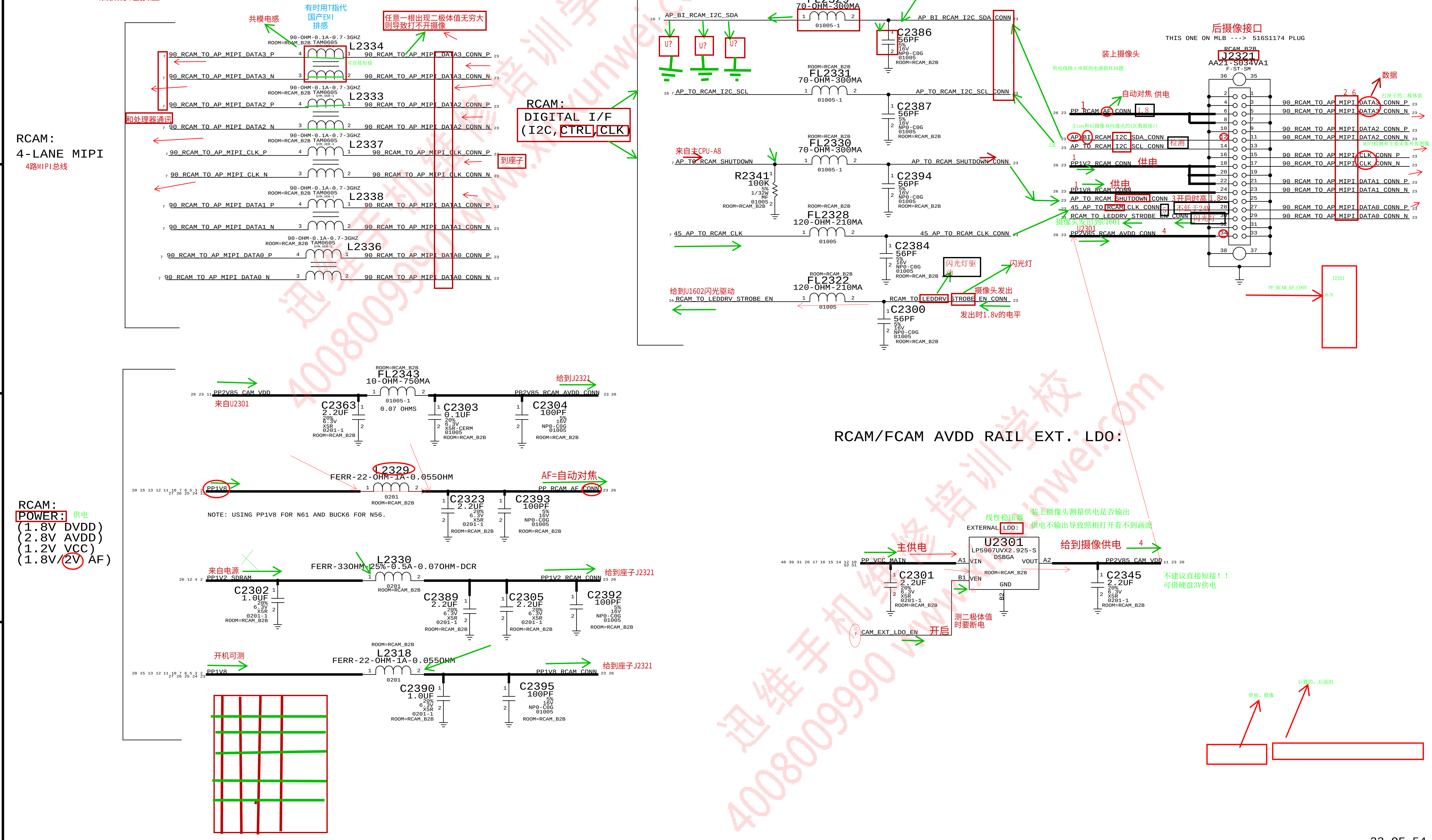


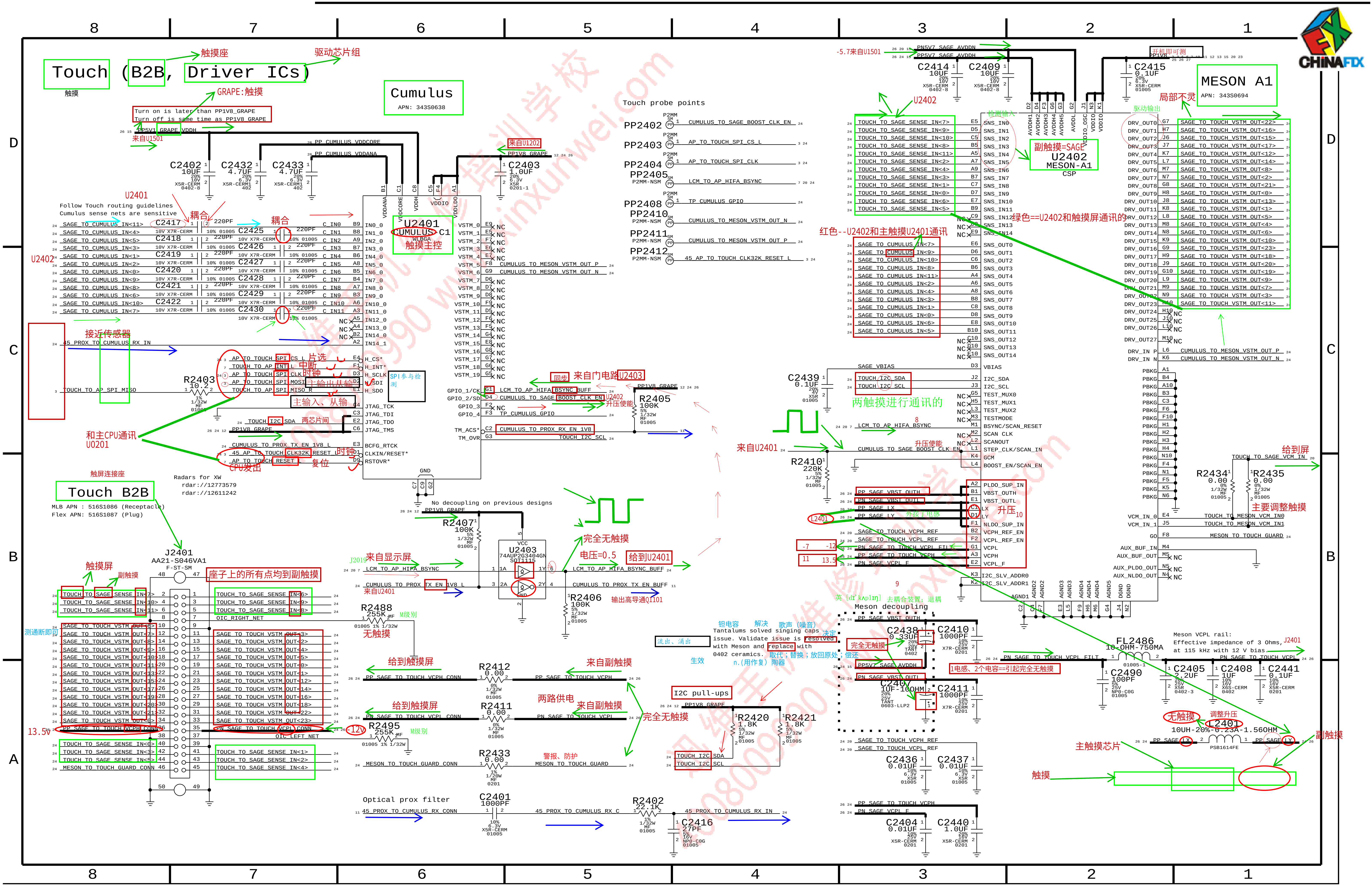


OSCAR + SENSORS
协处理
传感器
OSCAR VDDIO = 1.8V ALWAYS ON (NEED TO WAKE HOST & RUN PLL)
OSCAR CORE = 1.2V ALWAYS ON (NEED TO RUN IN S2RAM)
PP1V8_OSCAR
PP1V2_OSCAR
C2261 0.1UF 20% 4V X5R 01005
C2292 1.0UF 20% 6.3V X5R 0201-1 ROOM=OSCAR
C2274 1.0UF 20% 6.3V X5R 0201-1 ROOM=OSCAR
C2260 0.1UF 20% 4V X5R 01005 ROOM=OSCAR
运动协处理+传感器
U2201
LPC18B1UK/CPA0-00
WLCSP
处理指南针、陀螺仪、重力、加速计、气压
和主CPU通讯
OSCAR_TO_AP_UART_RXD
AP_TO_OSCAR_UART_TXD
OSCAR_TO_RADIO_CONTEXT_A
OSCAR_TO_RADIO_CONTEXT_B
AP_ISP_TO_OSCAR_UART_TXD
OSCAR_TO_AP_ISP_UART_RXD
OSCAR_TO_BB_UART_TXD
BB_TO_OSCAR_UART_RXD
到外部芯片 指南针、陀螺仪
OSCAR_TO_IMU_SPI_SCLK
IMU_TO_OSCAR_SPI_MISO
OSCAR_TO_IMU_SPI_MOSI
OSCAR_TO_GYRO_SPI_CS_L
OSCAR_TO_PHOSPHORUS_SPI_CS_L
OSCAR_TO_COMPASS_SPI_CS_L
U0_TXD/GPIO0[15]
U0_RXD/GPIO0[16]
U1_RXD/GPIO0[22]
U1_TXD/GPIO0[23]
U2_RXD/GPIO0[5]
U2_TXD/GPIO0[6]
U3_TXD/GPIO0[1]
U3_RXDGPIO0[2]
SPI0_SCK/GPIO0[12]
SPI0_MISO/GPIO0[13]
SPI0_MOSI/GPIO0[14]
SPI0_SSEL0/GPIO0[3]
SPI0_SSEL1/GPIO0[18]
SPI0_SSEL2/GPIO0[4]
SPI0_SSEL3/GPIO0[25]
RESET*
CLKOUT/GPIO0[0]
GPIO0[7]
GPIO0[8]
NMI/GPIO0[24]
GPIO0[26]
SWO/GPIO0[27]
WDFLAG/GPIO1[2]
ALARM1/GPIO1[3]
ALARM0/GPIO1[4]
SWDIO/GPIO0[19]
SWCLK/GPIO0[20]
CLK32K/GPIO0[21]
I2C0_SDAP/GPIO0[10]
I2C0_SCL/GPIO0[11]
I2C1_SDA/GPIO0[9]
I2C1_SCL/GPIO0[17]
I2C2_SDA/GPIO1[0]
I2C2_SCL/GPIO1[1]
来自外部芯片
各芯片中断
OSCAR_BI_AP_TIME_SYNC_HOST_INT
GYRO_TO_OSCAR_INT1
COMPASS_TO_OSCAR_INT
GYRO_TO_OSCAR_INT2
OSCAR_TO_PMU_HOST_WAKE
AP_BI_OSCAR_SWDIO_1V8
AP_TO_OSCAR_SWDCLK_1V8
R2254 392K 1% 1/32W MF 01005 NOSTUFF
复位
PMU_TO_OSCAR_RESET_CLK32K_L
电源发出
C2204 56PF 5% 16V 01005 ROOM=OSCAR NOSTUFF
PP2201 OSCAR_TO_AP_UART_RXD
PP2202 AP_TO_OSCAR_UART_TXD
PP2203 OSCAR_BI_AP_TIME_SYNC_HOST_INT
PP2204 IMU_TO_OSCAR_SPI_MISO
PP2205 GYRO_TO_OSCAR_INT2
PP2206 GYRO_TO_OSCAR_INT1
CARBON (ACCEL GYRO COMBO)
重力 加速计
陀螺仪
INVENSENSE, APN 338S00017, C2211=0.1UF
BOSCH, APN 338S00028, C2211=0.1UF
ST, APN 338S00029, C2211=0.01UF,25V
C2248 0.1UF 20% 6.3V X5R-CERM 01005 ROOM=GYRO
C2245 0.1UF 20% 6.3V X5R-CERM 01005 ROOM=GYRO
C2247 2.2UF 20% 6.3V X5R 0201-1 ROOM=GYRO
U2203
MPU-6700-12-COMBO
LGA
片选
OSCAR_TO_GYRO_SPI_CS_L
GYRO_PUMP
GYRO_TO_OSCAR_INT2
中断
OSCAR_TO_IMU_SPI_SCLK
OSCAR_TO_IMU_SPI_MOSI
IMU_TO_OSCAR_SPI_MISO
GYRO_TO_OSCAR_INT1
惯性测量装置
C2211 0.1UF 10% 6.3V CERM-X5R 0201 ROOM=GYRO
THIS IS OUTSIDE OF SHIELD IN TO THE RIGHT OF THE NAND
PHOSPHORUS
大气压
C2250 1.0UF 20% 6.3V X5R 0201-1 ROOM=PHOSPHORUS
C2251 0.1UF 20% 4V X5R 01005 ROOM=PHOSPHORUS
U2204
BMP282AC
把测量的气压数据通过SPI总线给到U2201
通往U2201协处理器
OSCAR_TO_IMU_SPI_MOSI
OSCAR_TO_IMU_SPI_SCLK
OSCAR_TO_PHOSPHORUS_SPI_CS_L
IMU_TO_OSCAR_SPI_MISO
C2255 56PF 5% 16V 01005 ROOM=PHOSPHORUS NOSTUFF
C2256 56PF 5% 16V 01005 ROOM=PHOSPHORUS NOSTUFF
C2241 56PF 5% 16V 01005 ROOM=PHOSPHORUS NOSTUFF
C2201 56PF 5% 16V 01005 ROOM=PHOSPHORUS NOSTUFF

RCAM B2B (REAR CAMERA CONNECTOR)

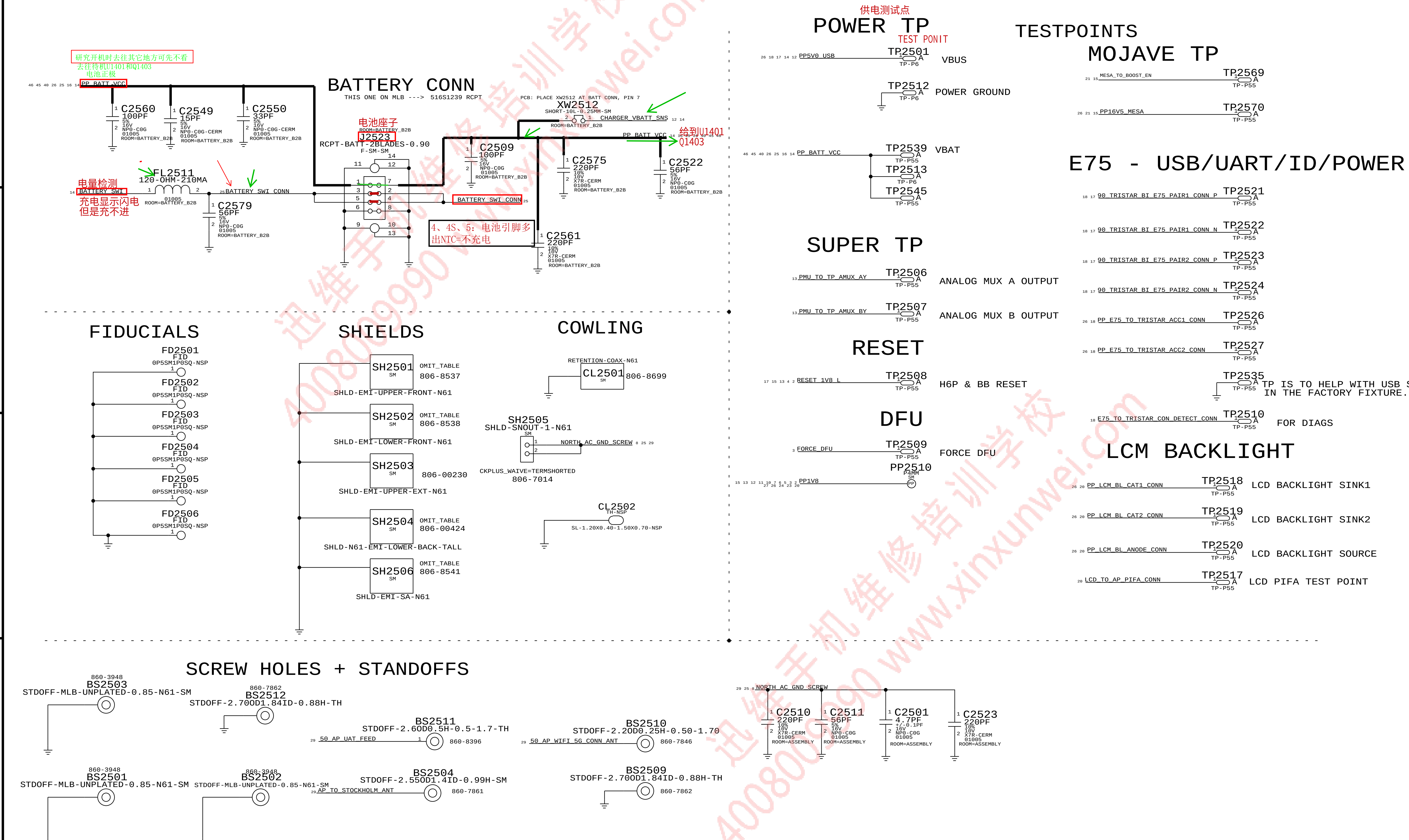
后摄像连接座 后置 摄像 座子





Touch (B2B, Driver ICs)
触摸
触摸座
驱动芯片组
GRAPE:触摸
Turn on is later than PP1V8_GRAPE
Turn off is same time as PP1V8_GRAPE
PP5V1_GRAPE_VDDH
来自U1501
Cumulus
APN: 343S0638
PP_CUMULUS_VDDCORE
PP_CUMULUS_VDDANA
来自U1202
PP1V8_GRAPE
C2402 10UF
C2432 4.7UF
C2433 4.7UF
C2403 1.0UF
U2401
Follow Touch routing guidelines
Cumulus sense nets are sensitive
耦合
U2402
U2401 CUMULUS
WLBGA
触摸主控
C2417 C2418 C2419 C2420 C2421 C2422 C2425 C2426 C2427 C2428 C2429 C2430
220PF
接近传感器
45_PROX_TO_CUMULUS_RX_IN
TOUCH_TO_AP_SPI_MISO
R2403 10.2
片选
中断
时钟
主输出从输入
主输入、从输出
SPI参与检测
和主CPU通讯
U0201
两芯片间
CUMULUS_TO_PROX_TX_EN_1V8_L
45_AP_TO_TOUCH_CLK32K_RESET_L
AP_TO_TOUCH_RESET
CPU发出
复位
触屏连接座
Radars for XW
rdar://12773579
rdar://12611242
Touch B2B
MLB APN : 516S1086 (Receptacle)
Flex APN: 516S1087 (Plug)
J2401
AA21-S046VA1
F-ST-SM
触摸屏
副触摸
座子上的所有点均到副触摸
测通断即可
13.5V
-12V
OIC_RIGHT_NET
OIC_LEFT_NET
MESON_TO_TOUCH_GUARD_CONN
Touch probe points
PP2402 PP2403 PP2404 PP2405 PP2408 PP2410 PP2411 PP2412
CUMULUS_TO_SAGE_BOOST_CLK_EN
AP_TO_TOUCH_SPI_CS_L
AP_TO_TOUCH_SPI_CLK
LCM_TO_AP_HIFA_BSYNC
TP_CUMULUS_GPIO
CUMULUS_TO_MESON_VSTM_OUT_N
CUMULUS_TO_MESON_VSTM_OUT_P
45_AP_TO_TOUCH_CLK32K_RESET_L
来自门电路U2403
同步
升压使能
R2405 100K
GND
No decoupling on previous designs
R2407 100K
U2403
74AUP2G3404GN
SOT1115
完全无触摸
电压=0.5
给到U2401
来自显示屏
LCM_TO_AP_HIFA_BSYNC
LCM_TO_AP_HIFA_BSYNC_BUFF
CUMULUS_TO_PROX_TX_EN_BUFF
来自U2401
R2406 100K
输出高导通Q1101
R2488 255K
无触摸
给到触摸屏
来自副触摸
R2412 0.00
PP_SAGE_TO_TOUCH_VCPH_CONN
PP_SAGE_TO_TOUCH_VCPH
R2411 0.00
两路供电
PN_SAGE_TO_TOUCH_VCPL_CONN
PN_SAGE_TO_TOUCH_VCPL
R2495 255K
R2433 0.00
警报、防护
MESON_TO_TOUCH_GUARD
Optical prox filter
C2401 1000PF
45_PROX_TO_CUMULUS_RX_CONN
45_PROX_TO_CUMULUS_RX_C
R2402 22.1K
C2416 27PF
-5.7来自U1501
PN5V7_SAGE_AVDDN
PP5V7_SAGE_AVDDH
C2414 10UF
C2409 10UF
C2415 0.1UF
开机即可测
MESON A1
APN: 343S0694
局部不灵
U2402
检测输入
驱动输出
副触摸=SAGE
U2402
MESON-A1
CSP
绿色==U2402和触摸屏通讯的
红色--U2402和主触摸U2401通讯
C2439 0.1UF
SAGE_VBIAS
TOUCH_I2C_SDA
TOUCH_I2C_SCL
两触摸进行通讯的
LCM_TO_AP_HIFA_BSYNC
升压使能
来自U2401
R2410 220K
CUMULUS_TO_SAGE_BOOST_CLK_EN
给到屏
TOUCH_TO_SAGE_VCM_IN
R2434 0.00
R2435 0.00
主要调整触摸
TOUCH_TO_MESON_VCM_IN0
TOUCH_TO_MESON_VCM_IN1
MESON_TO_TOUCH_GUARD
PP_SAGE_VBST_OUTH
PN_SAGE_VBST_OUTL
PP_SAGE_LX
PP_SAGE_LY
升压
PP_SAGE_TO_TOUCH_VCPH_REF
PP_SAGE_TO_TOUCH_VCPL_REF
PN_SAGE_TO_TOUCH_VCPL_FILT
PP_SAGE_TO_TOUCH_VCPH
PN_SAGE_VCPL_F
-7 -12
11 13.5
Meson decoupling
去耦合装置，退耦
钽电容 解决 歌声(噪音)
Tantalums solved singing caps issue. Issue is resolved with Meson replace with 0402 ceramics.
流出、涌出
生效
I2C pull-ups
R2420 1.8K
R2421 1.8K
C2438 0.33UF
C2410 1000PF
完全无触摸
PP5V7_SAGE_AVDDH
PN_SAGE_VBST_OUTL
C2411 1000PF
L2400 1UF-16OHM
1电感、2个电容==引起完全无触摸
FL2486 10-OHM-750MA
Meson VCPL rail:
Effective impedance of 3 Ohms, at 115 kHz with 12 V bias.
J2401
PN_SAGE_TO_TOUCH_VCPL
C2490 100PF
C2405 2.2UF
C2408 1UF
C2441 0.1UF
无触摸
调整升压
L2401
10UH-20%-0.23A-1.560HM
PSB1614FE
主触摸芯片
副触摸
触摸
C2436 0.01UF
C2437 0.01UF
C2404 0.01UF
C2440 1.0UF
CHINAFX

BATT CONN, TPS, STANDOFFS/SHIELDS/FIDUCIALS



VOLTAGE PROPERTIES

电压描述

输入到电池接口的电压

描述

描述电压具体情况

I55	VOLTAGE=3.3V	PP3V3_USB	2 12
I56	VOLTAGE=1.8V	PP1V8_VA_L19_L67	10 12 16
I57	VOLTAGE=3.0V	PP3V0_TRISTAR	12 15 17 29
I58	VOLTAGE=3.0V	PP3V0_IMU	12 19
I60	VOLTAGE=3.0V	PP3V0_NAND	6 12
I59	VOLTAGE=3.0V	PP3V3_ACC	12 17
I61	VOLTAGE=3.0V	PP3V0_PROX_ALS	11 12
I64	VOLTAGE=4.6V	PP_VCC_MAIN	10 12 14 15 16 17 23 31 39 48 51 52
I65	VOLTAGE=1.0V	PP1V0	7 12
I67	VOLTAGE=3.0V	PP3V0_PROX_IRLED	11 12
I66	VOLTAGE=1.8V	PP1V8_ALWAYS	3 5 12 14
I68	VOLTAGE=3.0V	PP3V0_MESA	12 21
I70	VOLTAGE=1.1V	PP_CPU	4 12
I69	VOLTAGE=1.1V	PP_GPU	4 12
I71	VOLTAGE=1.2V	PP1V2_SDRAM	2 4 12 23
I72	VOLTAGE=1.8V	PP1V8_SDRAM	3 4 10 12 13 14 15 17 29
I73	VOLTAGE=1.8V	PP1V8	2 3 5 6 7 10 11 12 13 15 20 23 24 25 27
I77	VOLTAGE=1.8V	PP1V8_GRAPE	12 24
I76	VOLTAGE=1.8V	PP1V8_OSCAR	12 19 22
I75	VOLTAGE=1.2V	PP1V2_NAND_VDDI	6
I74	VOLTAGE=1.8V	PP_EXTMIC_BIAS_FILT_IN	10
I78	VOLTAGE=1.8V	BOARD_ID2	3 27
I79	VOLTAGE=1.2V	PP1V2	2 4 5 11 12
I80	VOLTAGE=5.0V	PP_E75_TO_TRISTAR_ACC1_CONN	18 25
I81	VOLTAGE=5.0V	PP_E75_TO_TRISTAR_ACC1	17 18
I82	VOLTAGE=22.0V	PP_LCM_BL_ANODE	15 20
I83	VOLTAGE=0.2V	PP_LCM_BL_CAT2	15 20
I87	VOLTAGE=0.2V	PP_LCM_BL_CAT1	15 20
I86	VOLTAGE=0.2V	PP_LCM_BL_CAT2_CONN	20 25
I85	VOLTAGE=0.2V	PP_LCM_BL_CAT1_CONN	20 25
I88	VOLTAGE=-5.7V	PN5V7_SAGE_AVDDN	15 20 24
I89	VOLTAGE=1.2V	PP1V2_OSCAR	12 22
I90	VOLTAGE=3.0V	PP3V0_MESA_CONN	21
I91	VOLTAGE=6V	PP6V0_LCM_BOOST	15
I92	VOLTAGE=5.0V	PP_STRB_DRIVER_TO_LED_WARM	8 16
I96	VOLTAGE=5.0V	PP_STRB_DRIVER_TO_LED_COOL	8 16
I93	VOLTAGE=1.8V	PP_CODEC_TO_MIC1_BIAS	10 18
I97	VOLTAGE=1.8V	PP_EXTMIC_BIAS_IN	10
I98	VOLTAGE=1.8V	PP_EXTMIC_BIAS_FILT	10
I99	VOLTAGE=1.8V	PP_CODEC_TO_FRONTMIC3_BIAS	10 11
I100	VOLTAGE=1.8V	PP_CODEC_TO_REARMIC2_BIAS	8 10
I101	VOLTAGE=1.8V	PP_CODEC_FILT+	10
I103	VOLTAGE=2.2V	PP_CODEC_SPKR_VQ	10
I140	VOLTAGE=2.5V	PP_CODEC_VCPFILT-	10
I104	VOLTAGE=2.5V	PP_CODEC_VCPFILT+	10
I106	VOLTAGE=2.5V	PP_CODEC_VHP_FLYN	10
I105	VOLTAGE=0.2V	PP_CODEC_VHP_FLYC	10
I107	VOLTAGE=2.5V	PP_CODEC_VHP_FLYP	10
I108	VOLTAGE=1.8V	PP1V8_FCAM_CONN	11
I109	VOLTAGE=3.0V	PP2V85_FCAM_AVDD_CONN	11
I110	VOLTAGE=1.8V	PP_CODEC_TO_FRONTMIC3_BIAS_CONN	11
I111	VOLTAGE=3.0V	PP3V0_ALS_CONN	11
I113	VOLTAGE=1.2V	PP1V2_FCAM_VDDIO_CONN	11
I112	VOLTAGE=5.0V	PP5V0_USB	12 14 17 18 25
I114	VOLTAGE=5.0V	PP5V0_USB_TO_PMU	12
I116	VOLTAGE=4.6V	PP_BUCK5_LX0	12
I115	VOLTAGE=4.6V	PP_BUCK3_LX	12
I117	VOLTAGE=4.6V	PP_BUCK4_LX	12
I118	VOLTAGE=4.6V	PP_BUCK2_LX	12
I120	VOLTAGE=4.6V	PP_BUCK1_LX1	12
I119	VOLTAGE=4.6V	PP_BUCK1_LX0	12
I121	VOLTAGE=4.6V	PP_BUCK0_LX3	12
I122	VOLTAGE=4.6V	PP_BUCK0_LX2	12
I126	VOLTAGE=4.6V	PP_BUCK0_LX1	12
I124	VOLTAGE=4.6V	PP_BUCK0_LX0	12
I125	VOLTAGE=6.0V	PP_CHESTNUT_LXP	15
I123	VOLTAGE=6.0V	PP_CHESTNUT_CP	15
I127	VOLTAGE=6.0V	PP_CHESTNUT_CN	15
I128	VOLTAGE=5.7V	PP5V7_SAGE_AVDDH	15 24
I129	VOLTAGE=5.7V	PP5V7_LCM_AVDDH	15 20
I130	VOLTAGE=5.1V	PP5V1_GRAPE_VDDH	15 24
I131	VOLTAGE=22.0V	PP_WLED_LX	15
I132	VOLTAGE=18.0V	PP18V0_MESA_SW	15
I133	VOLTAGE=17.0V	P17V0_MOJAVE_LDOIN	15
I134	VOLTAGE=16.5V	PP16V5_MESA	15 21 25
I136	VOLTAGE=8.0V	PP_SPKAMP_SW	16
I135	VOLTAGE=8.0V	PP_L19_VBOOST	16
I137	VOLTAGE=1.8V	PP_SPKAMP_FILT	16
I138	VOLTAGE=1.8V	PP_SPKAMP_LDO_FILT	16

I1	VOLTAGE=5.0V	PP_LED_DRV_LX	16
I2	VOLTAGE=5.0V	PP_LED_BOOST_OUT	16
I4	VOLTAGE=2.9V	PP2V9_LDO9	12
I5	VOLTAGE=1.8V	PP_CODEC_TO_MIC1_BIAS_CONN	18
I7	VOLTAGE=4.6V	PP_E75_TO_TRISTAR_ACC2	17 18
I8	VOLTAGE=4.6V	PP_E75_TO_TRISTAR_ACC2_CONN	18 25
I10	VOLTAGE=1.8V	PP1V8_LCM_CONN	20
I11	VOLTAGE=22.0V	PP_LCM_BL_ANODE_CONN	20 25
I12	VOLTAGE=-5.7V	PN5V7_LCM_AVDDN_CONN	20
I13	VOLTAGE=5.7V	PP5V7_LCM_AVDDH_CONN	20
I16	VOLTAGE=1.8V	PP1V8_MESA	21
I17	VOLTAGE=16.5V	PP16V5_MESA_CONN	21
I143	VOLTAGE=5.0V	PP_TRISTAR_PIN	17
I20	VOLTAGE=1.2V	PP1V2_RCAM_CONN	23
I21	VOLTAGE=1.8V	PP1V8_RCAM_CONN	23
I23	VOLTAGE=3.0V	PP2V85_CAM_VDD	11 23
I24	VOLTAGE=1.8V	PP2V85_RCAM_AVDD_CONN	23
I25	VOLTAGE=1.8V	PP_CUMULUS_VDDCORE	24
I26	VOLTAGE=1.2V	PP_CUMULUS_VDDANA	24
I27	VOLTAGE=13.5V	PP_SAGE_TO_TOUCH_VCPH_CONN	24
I28	VOLTAGE=-12V	PN_SAGE_TO_TOUCH_VCPL_CONN	24
I30	VOLTAGE=13.5V	PP_SAGE_TO_TOUCH_VCPH	24
I29	VOLTAGE=-12V	PN_SAGE_TO_TOUCH_VCPL	24
I33	VOLTAGE=-12V	PN_SAGE_VCPL_F	24
I35	VOLTAGE=5.7V	PP_SAGE_LX	24
I34	VOLTAGE=17.0V	PP_SAGE_LY	24
I38	VOLTAGE=1.8V	PP_PMU_VREF	13
I37	VOLTAGE=14V	PP_SAGE_VBST_OUTH	24
I40	VOLTAGE=5.0V	PP_TIGRIS_VBUS_DET	14
		PP1V8_PLL	
		PP_MIPIOD_VREG	
		BOARD_ID0	
I41	VOLTAGE=2.5V	PP_PMU_VDD_REF	13
I42	VOLTAGE=1.8V	PP_EXTMIC_BIAS	10
I43	VOLTAGE=1.8V	PP1V8_XTAL	2
I44	VOLTAGE=1.8V	PP_PMU_VDD_RTC	13
I46	VOLTAGE=4.6V	PP_BATT_VCC	14 16 25 40 45 46
I48	VOLTAGE=1.8V	PP1V8_MESA_CONN	21
I47	VOLTAGE=3.0V	PP3V0_PROX_CONN	11
I50	VOLTAGE=1.0V	PP0V95_FIXED_SOC	4 7 12
I51	VOLTAGE=1.0V	PP0V95_FIXED_SOC_PCIE	7
I52	VOLTAGE=1.2V	PP1V2_PLL	2
I53	VOLTAGE=1.0V	PP_BUCK5_LX1	12
I54	VOLTAGE=1.0V	PP_VAR_SOC	5 12
I141	VOLTAGE=5.0V	PMID_CAP	14
I147	VOLTAGE=5.0V	CHARGER_LDO	14
I148	VOLTAGE=4.6V	CHG_BOOT	14
I149	VOLTAGE=4.6V	CHG_LX	14
I150	VOLTAGE=3.0V	VIBE_DRIVE_P	14 18
I151	VOLTAGE=3.0V	VIBE_DRIVE_N	14 18
I152	VOLTAGE=1.8V	PP_RCAM_AF_CONN	23
I154	VOLTAGE=-14.0V	PN_SAGE_VBST_OUTL	24
I153	VOLTAGE=-12.0V	PN_SAGE_TO_TOUCH_VCPL_FILT	24
I155	VOLTAGE=2.7V	PP_BB_VDD_2V7_CONN	18

N61 SPECIFIC

BOOTSTRAPPING (BOARD_REV, BOARD_ID, BOOT_CFG)

BOARD_REV[3:0]={GPIO34, GPIO35, GPIO36, GPIO37}
FLOAT=LOW, PULLUP=HIGH

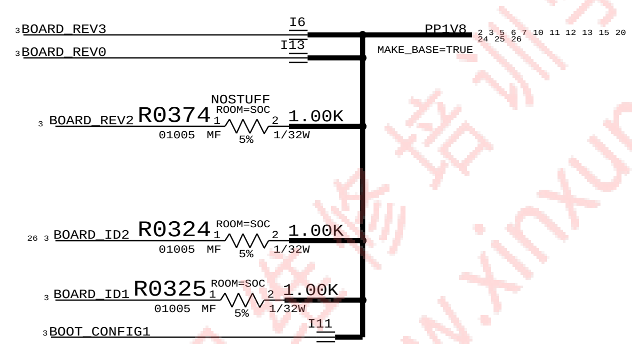
```
1111    PROTOMLB1
1110    PROTOMLB2
1101    PROTO1
1100    PROTO2
1011    EVT
1010    EVT SPLIT CARBON DOE
1001    CARRIER BUILD  <--- SELECTED
1000    DVT
```

BOARD_ID[4:0]={GPIO29, GPIO16, SPIO0_MISO, SPI0_MOSI, SPI0_SCLK}
FLOAT=LOW, PULLUP=HIGH

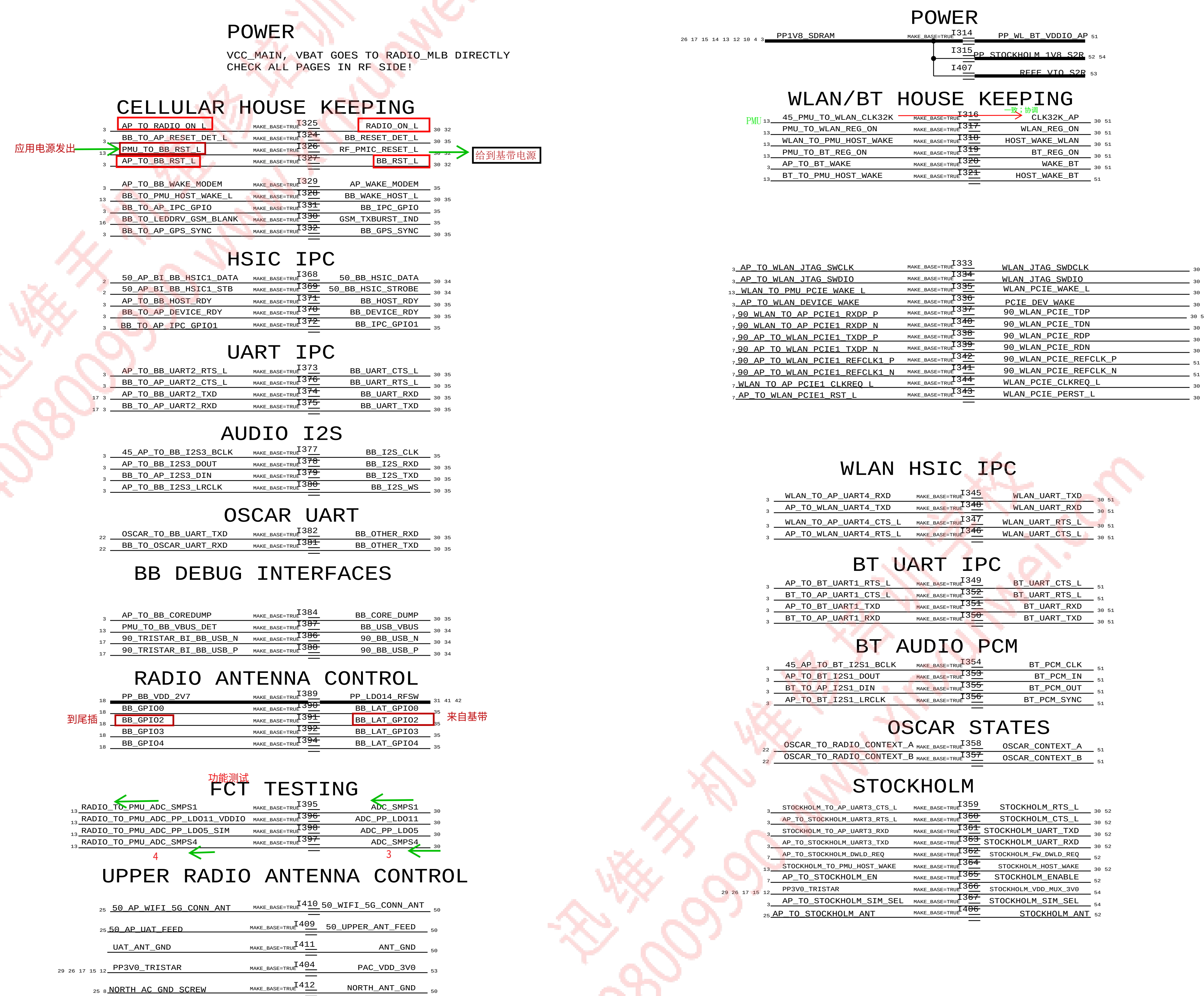
```
00100     N56, T133 MLB
00101     N56 DEV
00110     FIJI N61 MLB  <--- SELECTED
```

BOOT_CONFIG[2:0]={GPIO28, GPIO25, GPIO18}
FLOAT=LOW, PULLUP=HIGH

```
000    SPI0
001    SPI0 TEST MODE
010    NAND                 <--- SELECTED
011    NAND TEST MODE
100    NVME
101    NVME TEST MODE
111    FAST SPI
```

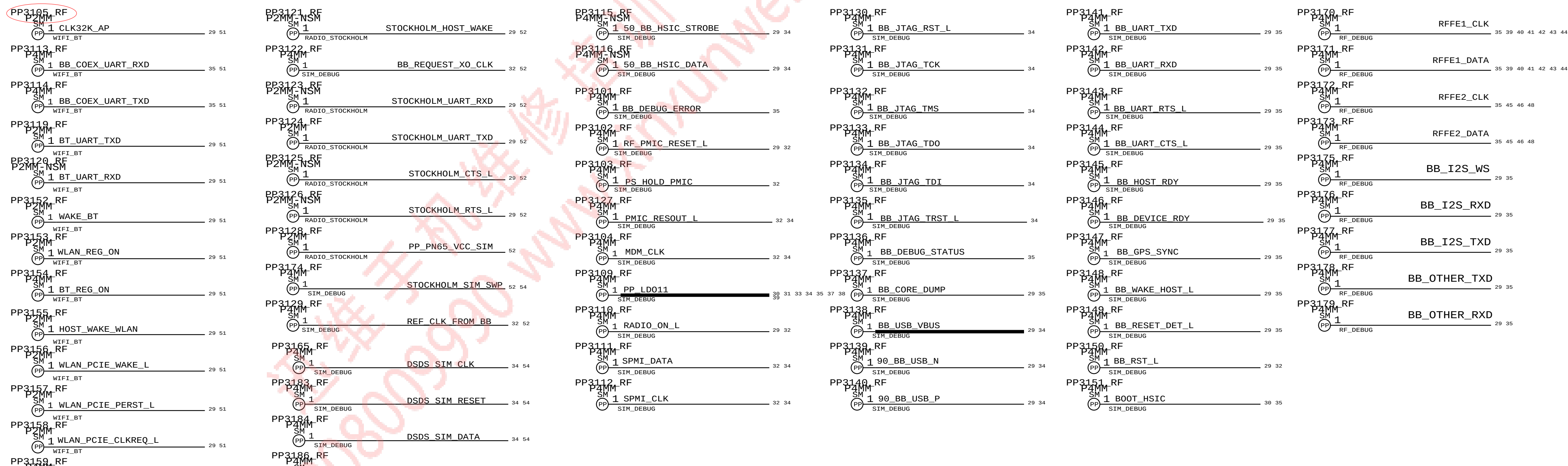


RADIO_MLB HIERARCHICAL SYMBOL

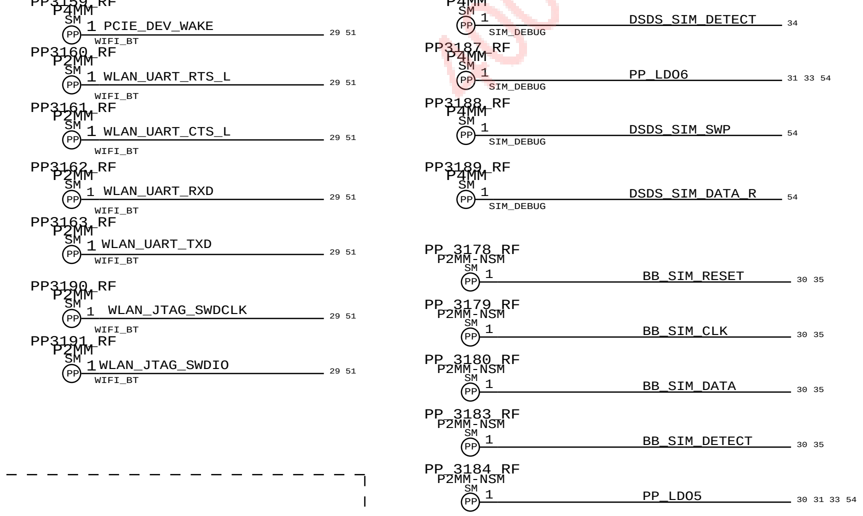


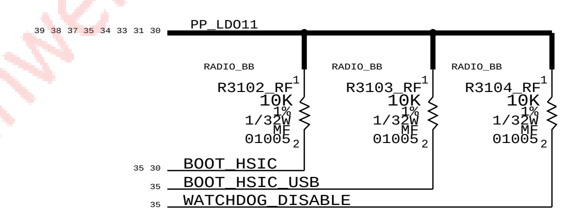
AP INTERFACE & DEBUG CONNECTORS

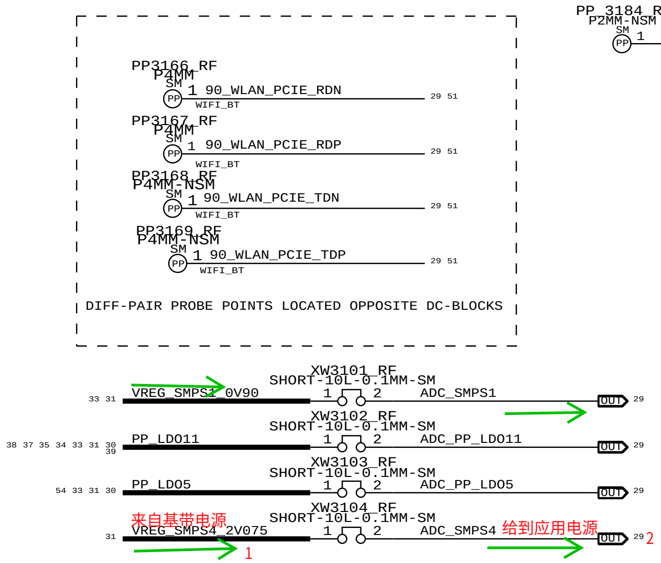
PROBE POINTS



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0565	197S0593	ALTERNATE	Y3301_RF	KDS 19.2MHZ XTAL
197S0598	197S0593	ALTERNATE	Y3301_RF	AVX 19.2MHZ XTAL
138S00005	138S00003	ALTERNATE	C3216_RF	15UF CAPACITOR
138S0739	138S0706	ALTERNATE	C4207_RF	1.0UF CAPACITOR
138S0945	138S0706	ALTERNATE	C4207_RF	1.0UF CAPACITOR
138S1103	138S0719	ALTERNATE	C4007_RF	4.7UF CAPACITOR
339S0231	339S0228	ALTERNATE	U5201_RF	CORONA MODULE USI
339S0242	339S0228	ALTERNATE	U5201_RF	CORONA MODULE TDK
155S00024	155S0950	ALTERNATE	F_TRI_RF	TRIPLEXER BIN2

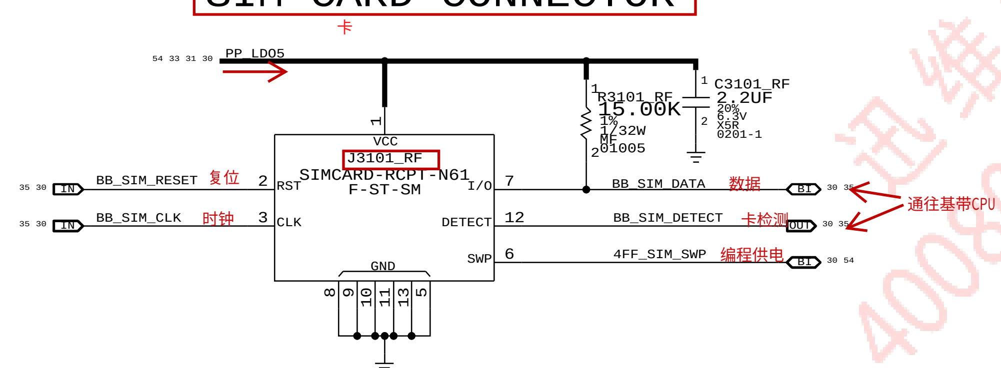






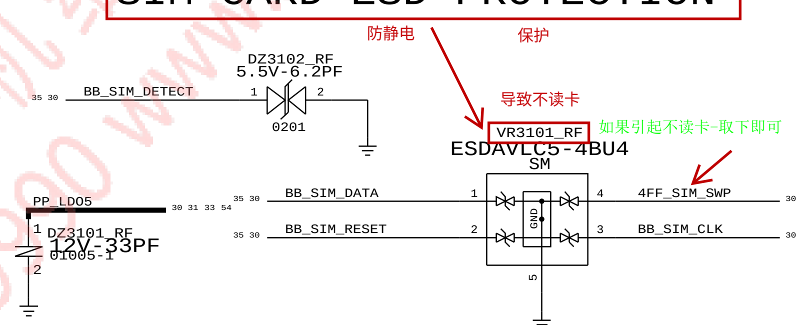
SIM CARD CONNECTOR

连接座
卡



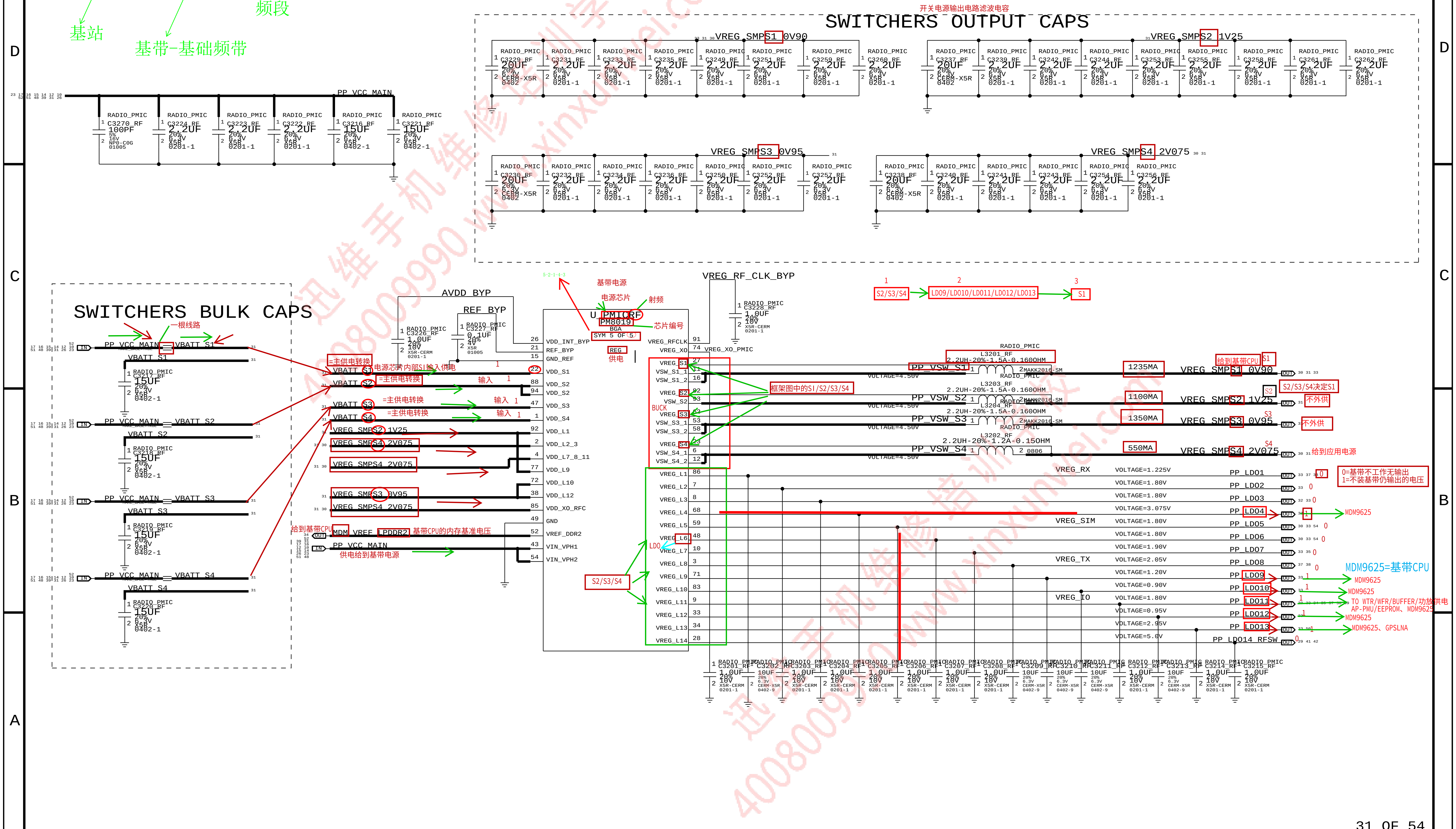
SIM CARD ESD PROTECTION

防静电
保护



BASEBAND PMU (1 OF 2)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

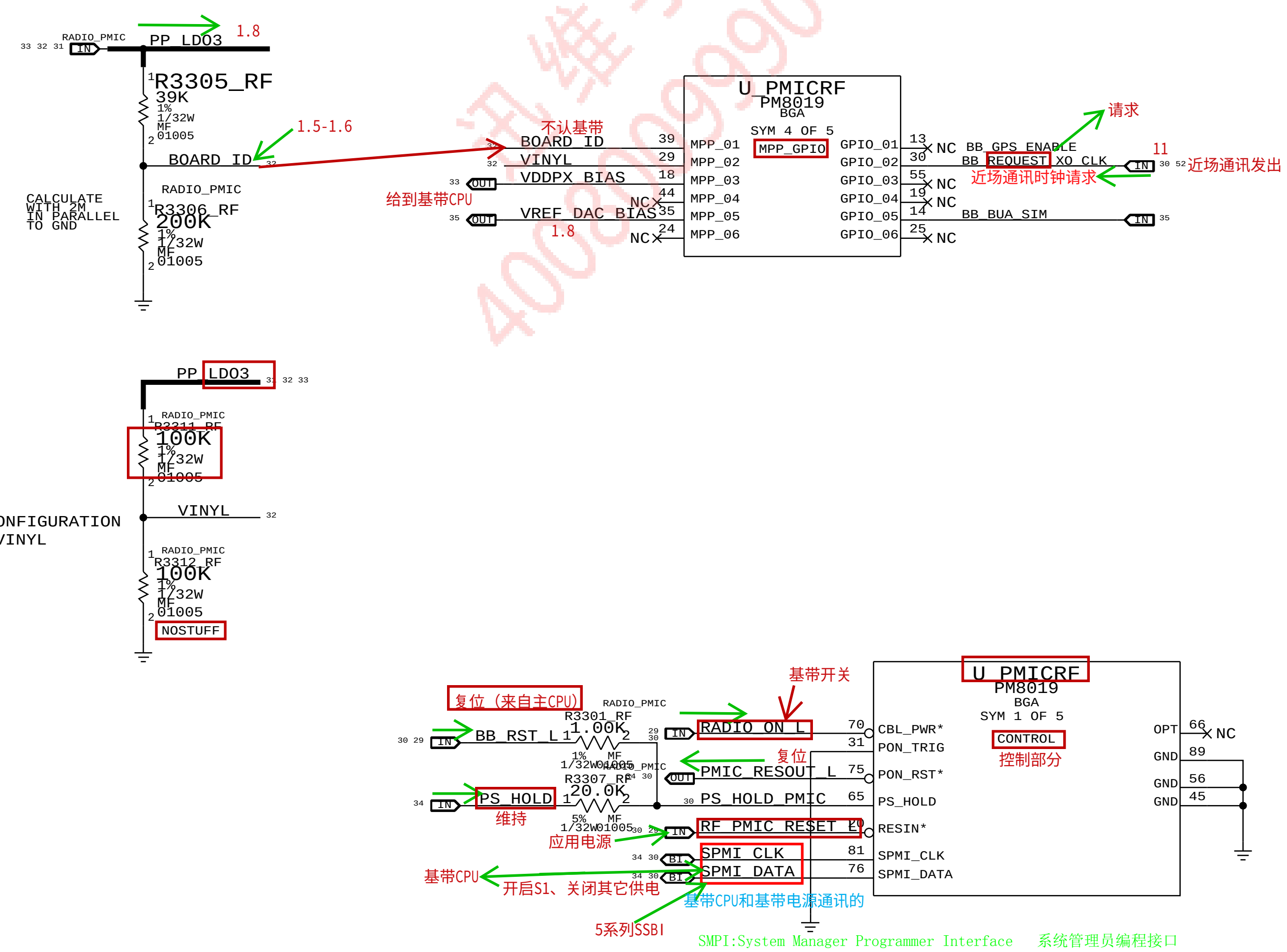


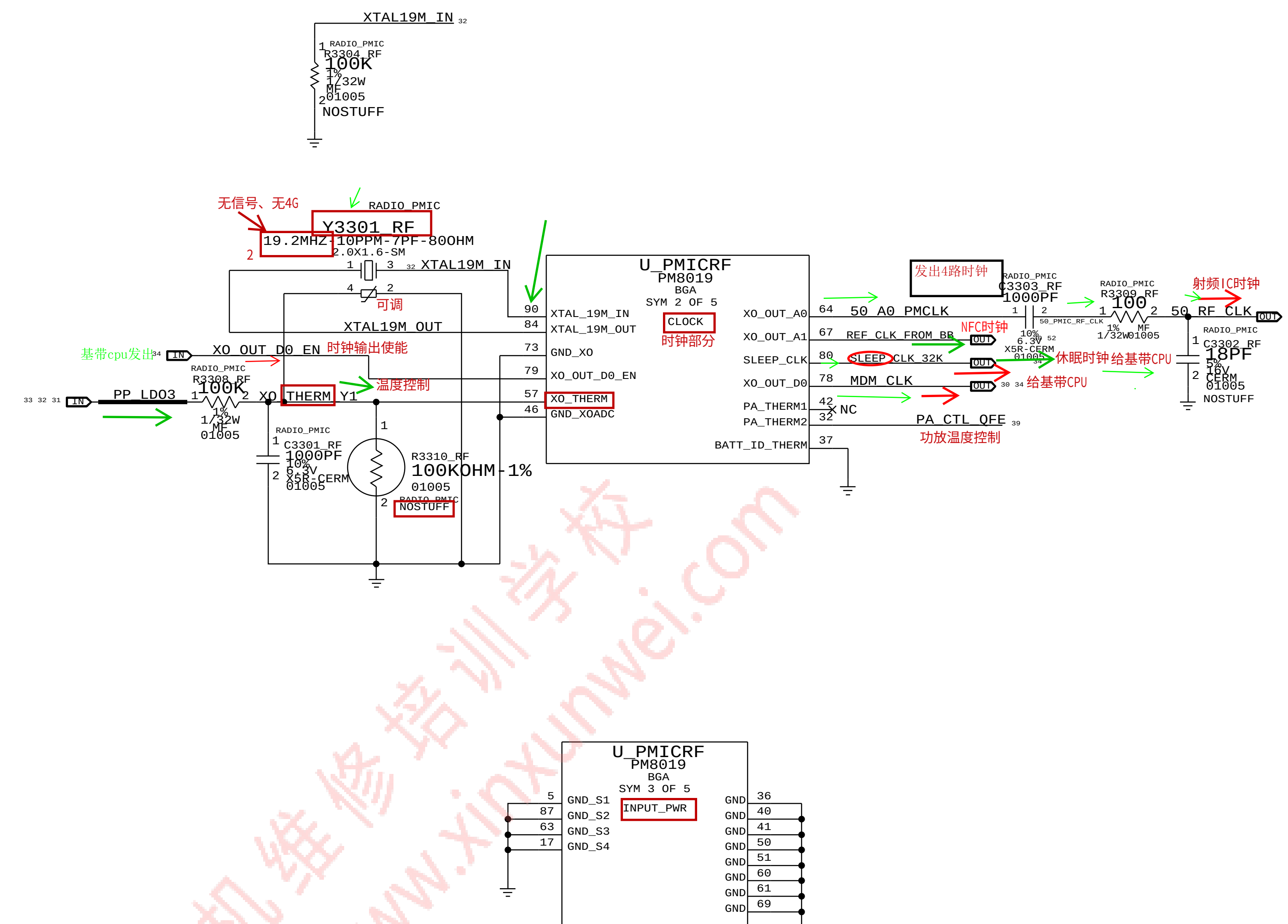
BASEBAND PMU (2 OF 2)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C401
R411
L400
U404

BOARD_ID	REVISION
0.00V	N61 PROTO_MLB1
0.50V	N61 DEV3
0.70V	N61 DEV4
0.90V	N61 PROTO_MLB2
1.10V	N61/N56 PROTO1
1.30V	N61/N56 PROTO2
1.40V	N61/N56 EVT1
1.50V	N61/N56 EVT2 (CARRIER)
1.60V	N61/N56 DVT
1.70V	N61/N56 PVT





BASEBAND (1 OF 3)

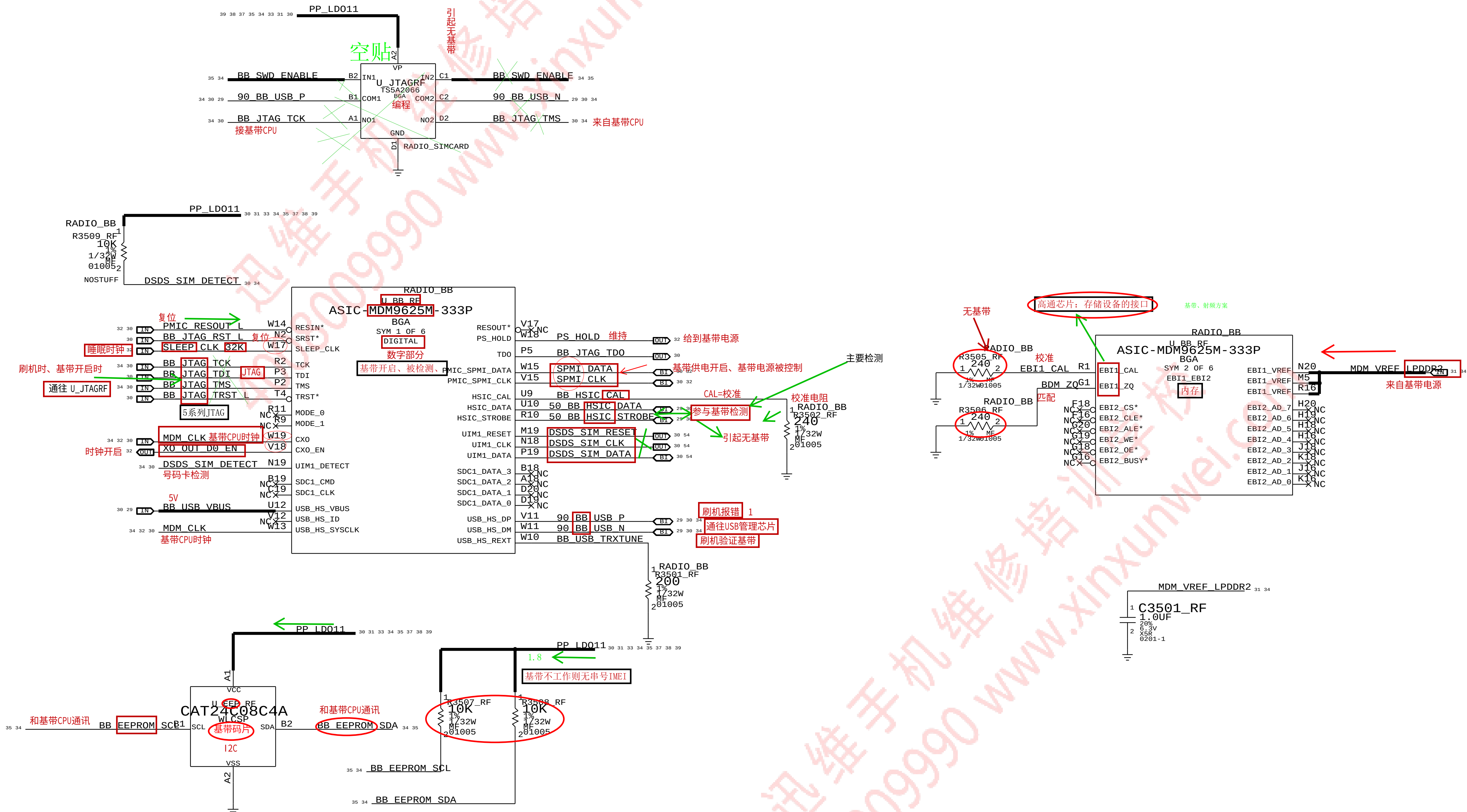
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C538
R500
L500
U502

BASEBAND (2 OF 3)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

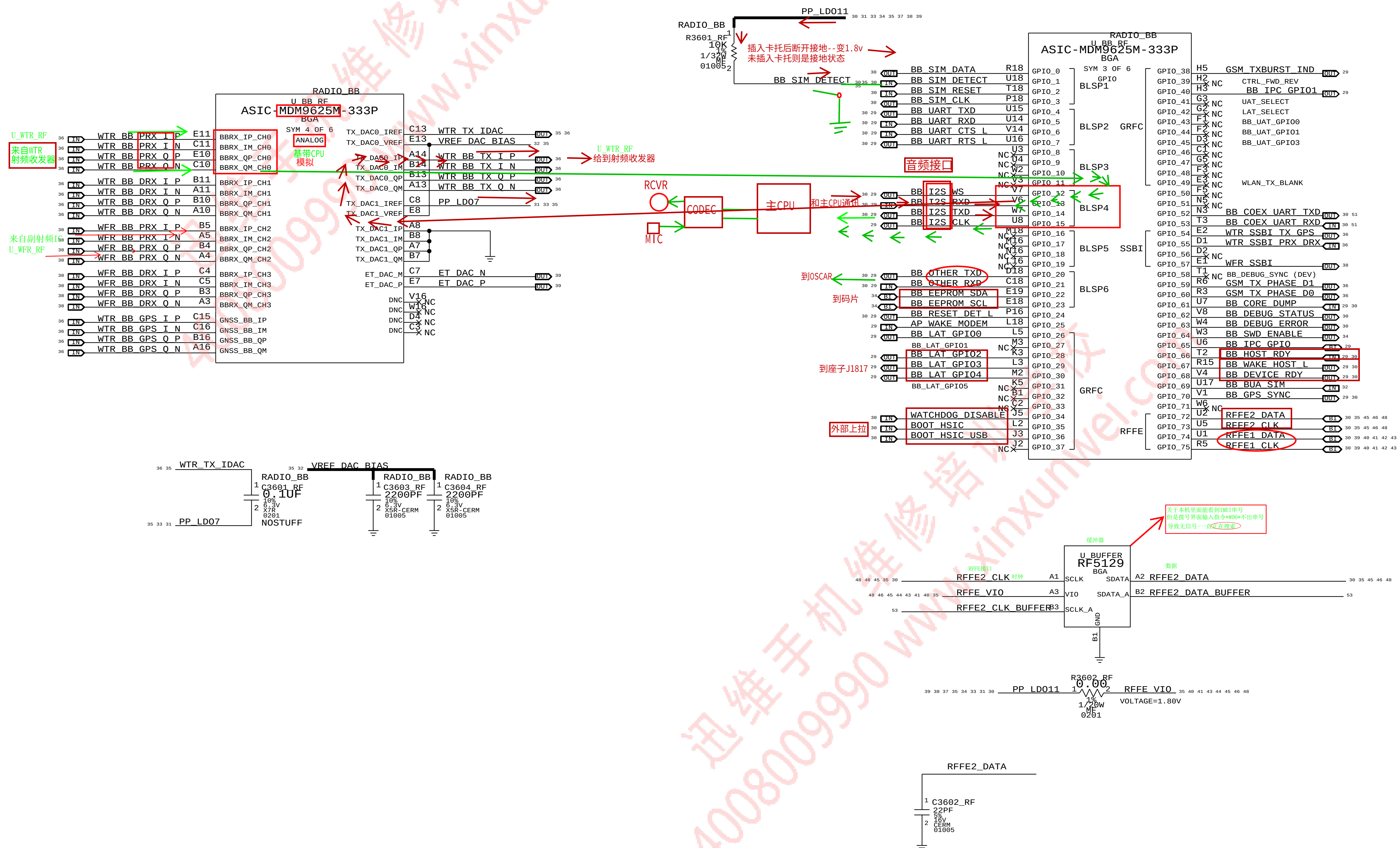
C600
R606
L600
U602



C704
R700
L700
U702

BASEBAND (3 OF 3)

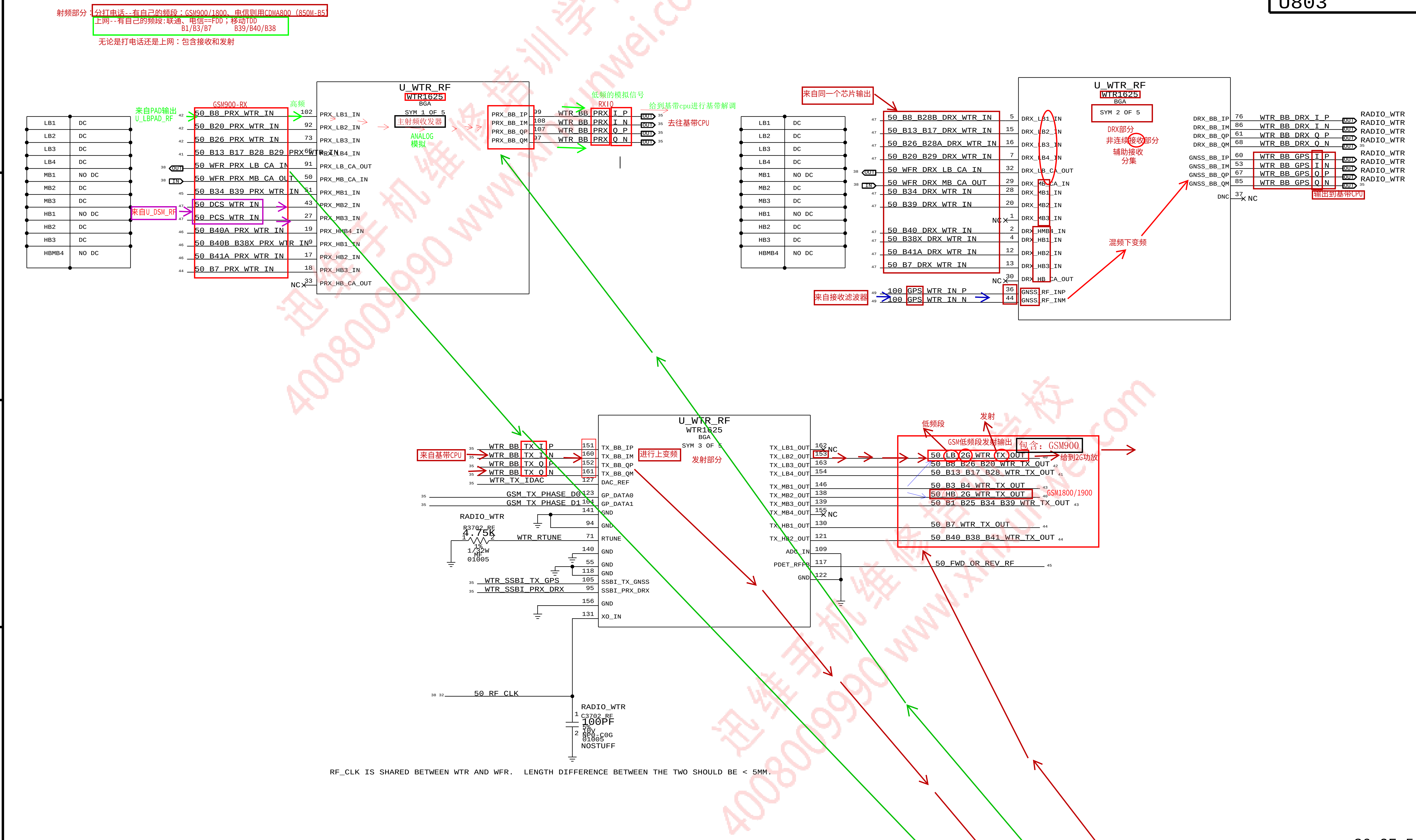
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.



WTR TRANSCEIVER (1 OF 2)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C802
R802
L800
U803

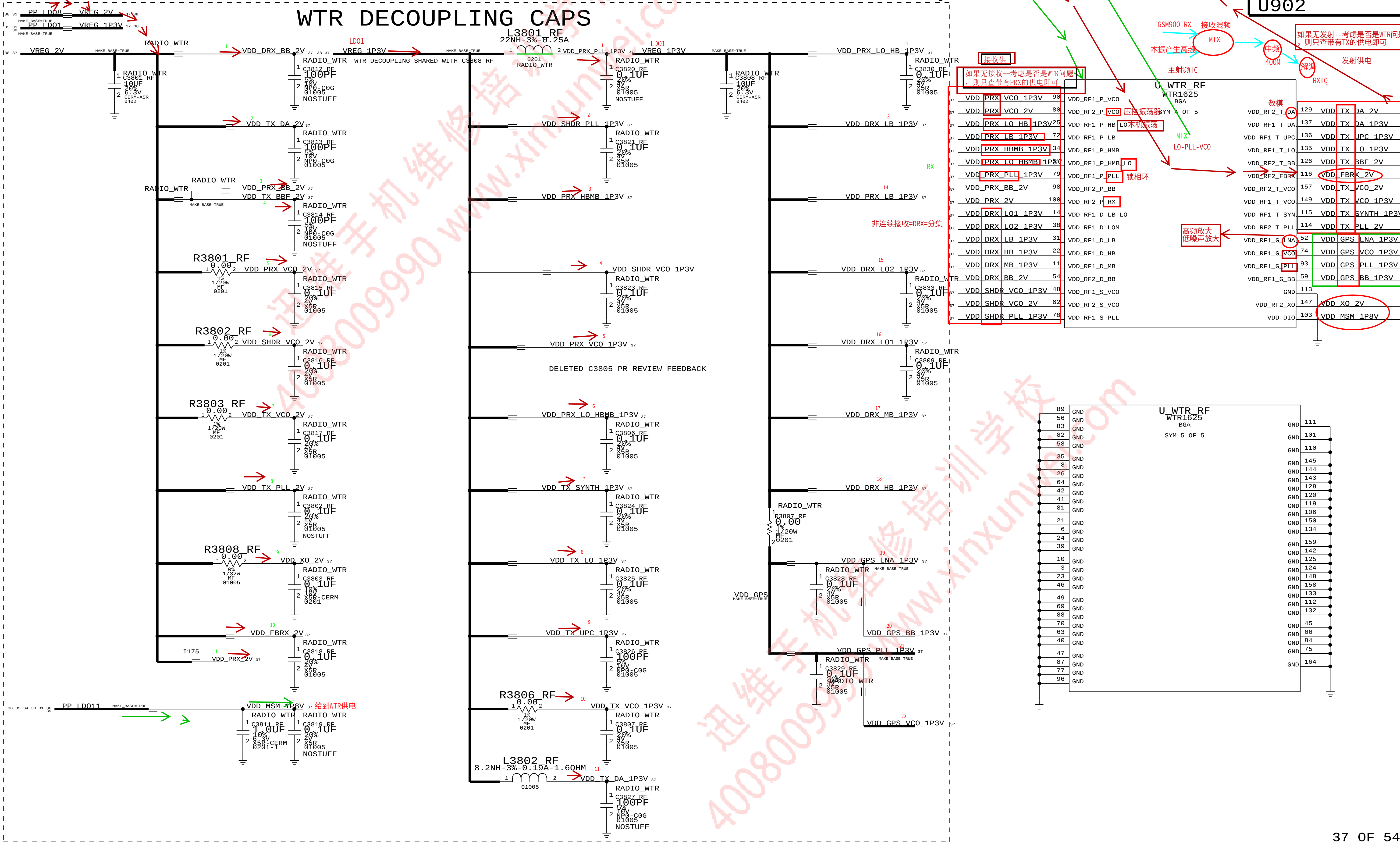


RF_CLK IS SHARED BETWEEN WTR AND WFR. LENGTH DIFFERENCE BETWEEN THE TWO SHOULD BE < 5MM.

WTR TRANSCEIVER (2 OF 2)

C934
R926
L3802_RF
U902

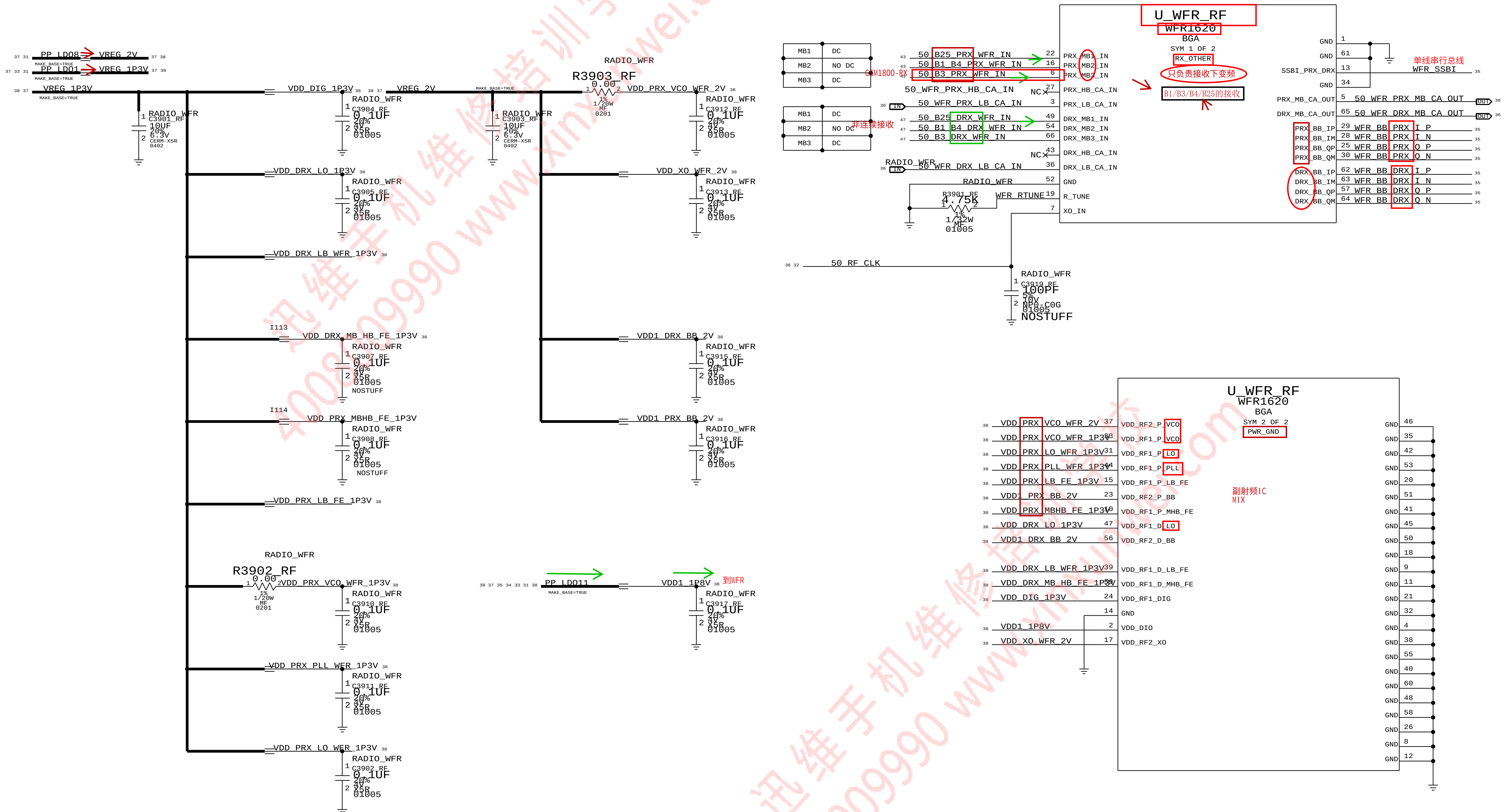
WTR DECOUPLING CAPS



WFR TRANSCEIVER

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

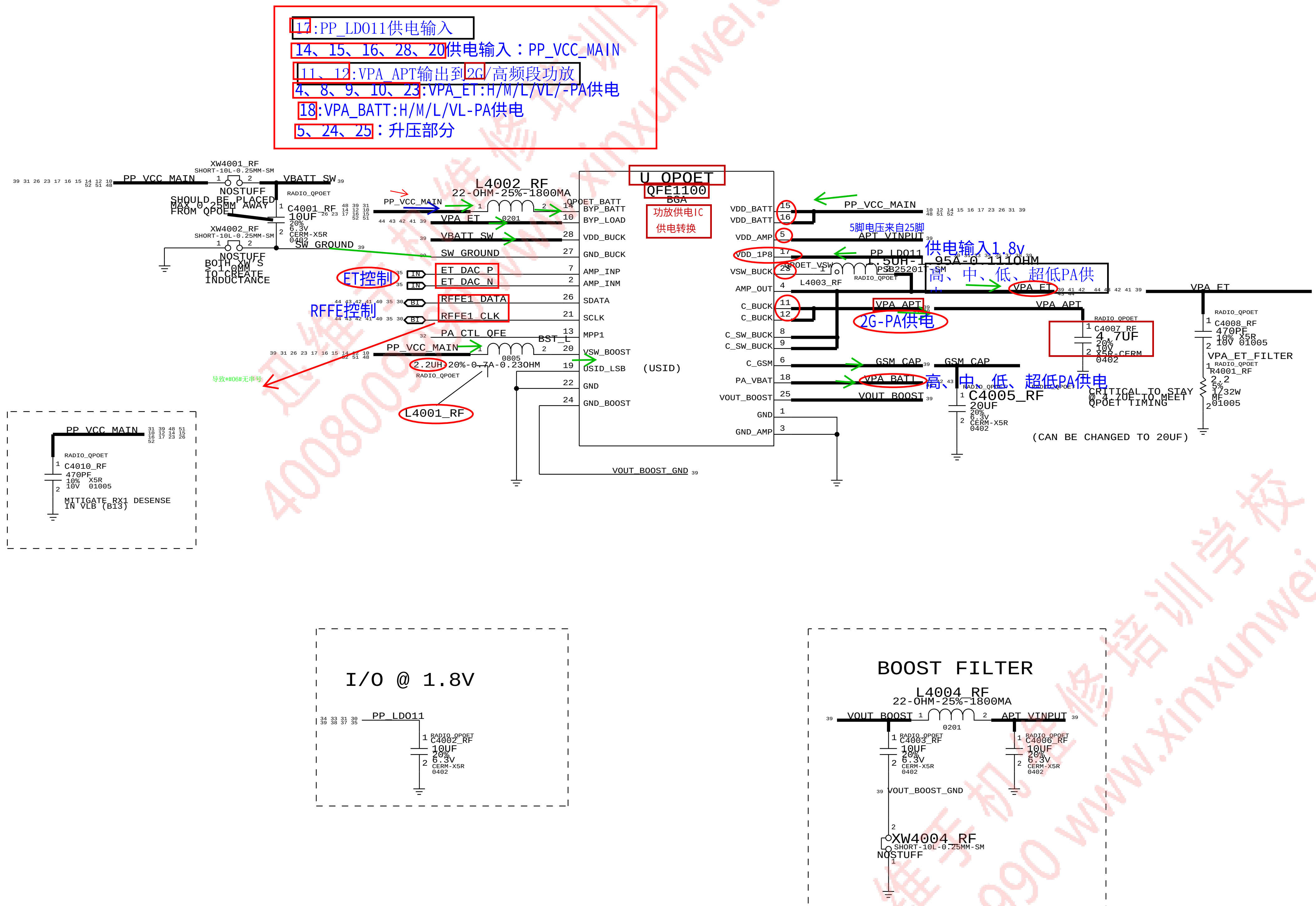
C1019
R1016
L1000
U1002



QFE DCDC

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

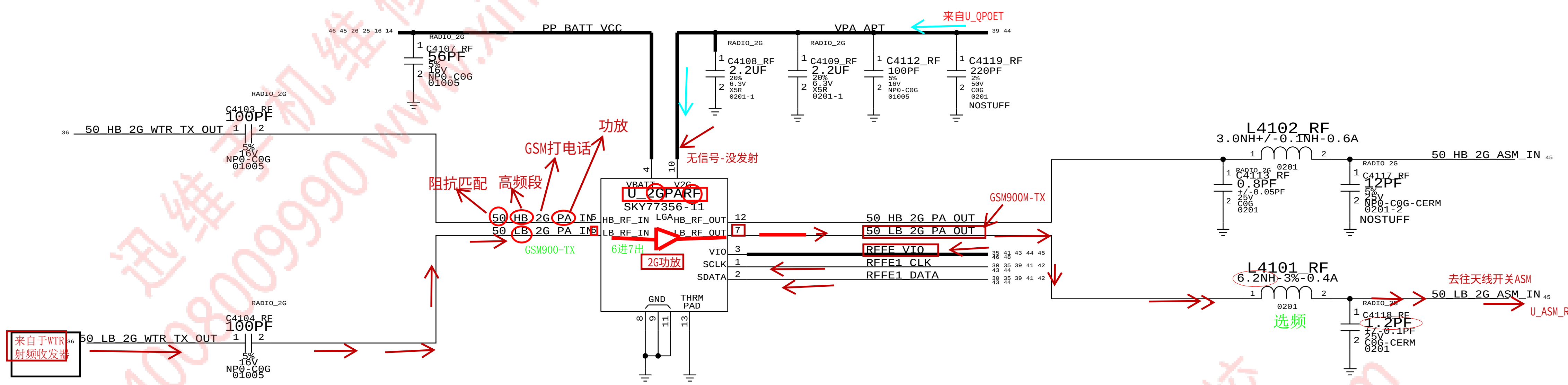
C1110
R1102
L1104
U1101



2G PA

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

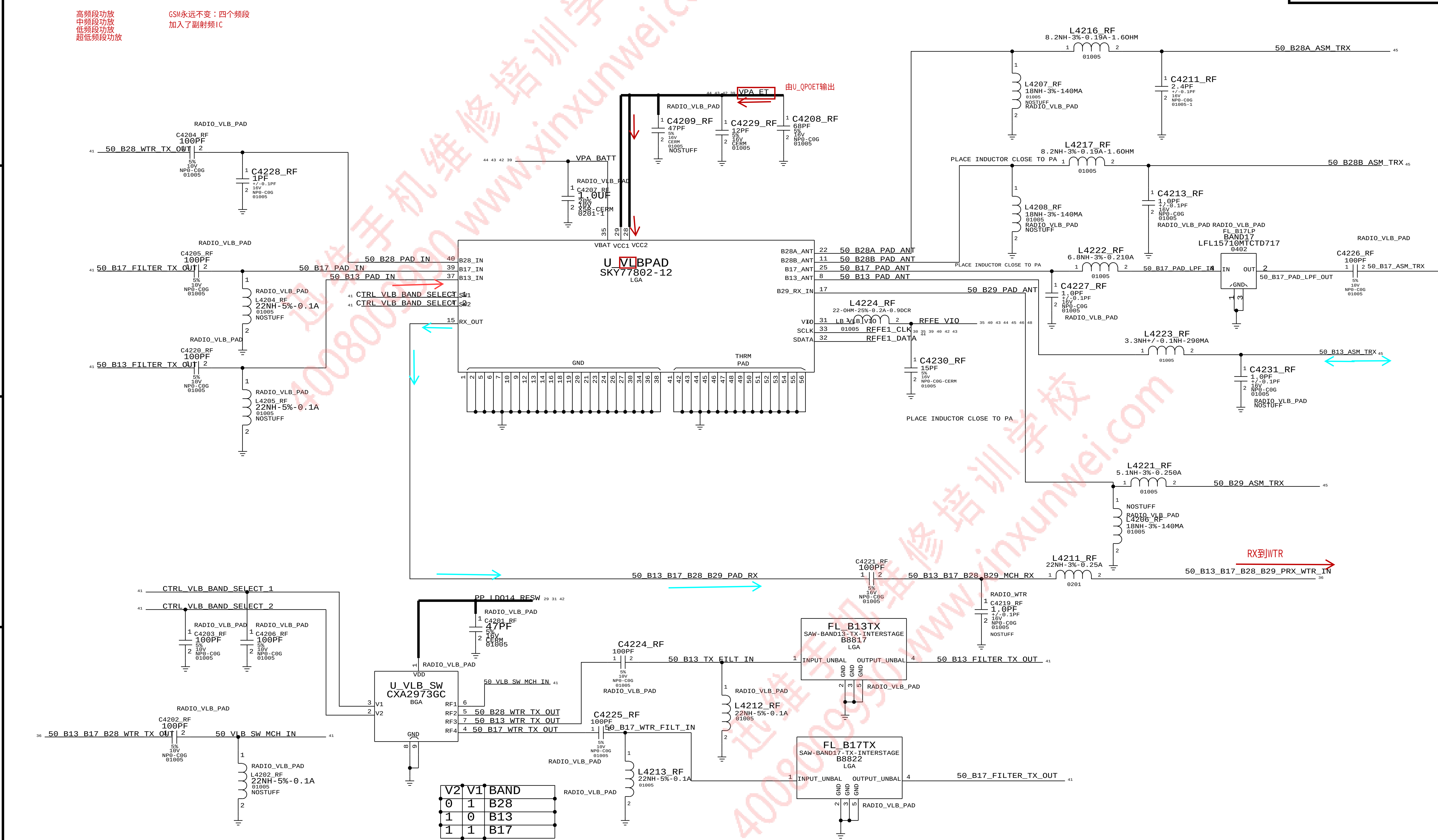
C1208
R1200
L1204
U1201



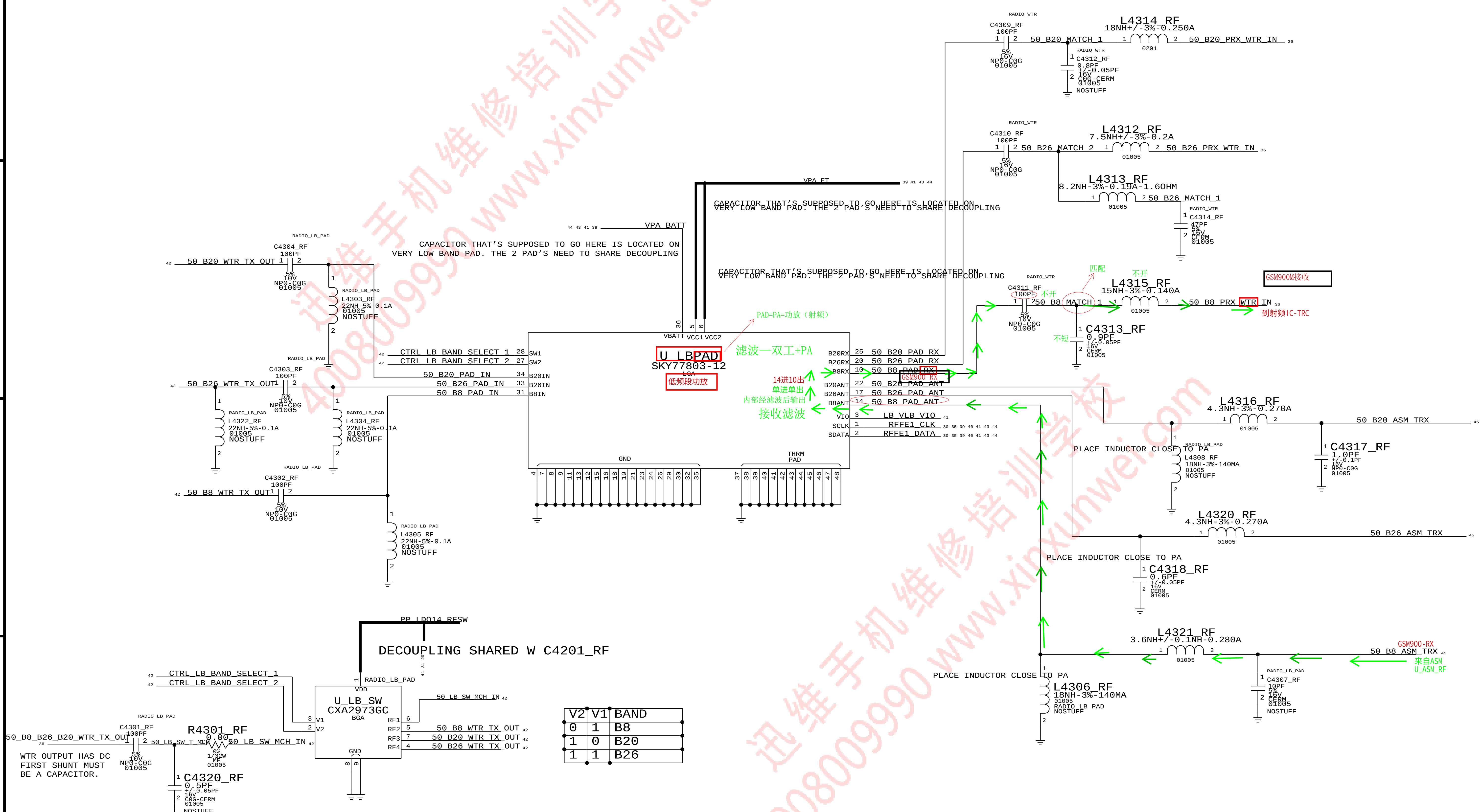
VERY LOW BAND PAD (B13, B17, B28)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C1332
R1300
L4215_RF
U1304



LOW BAND PAD (B8, B26, B20)

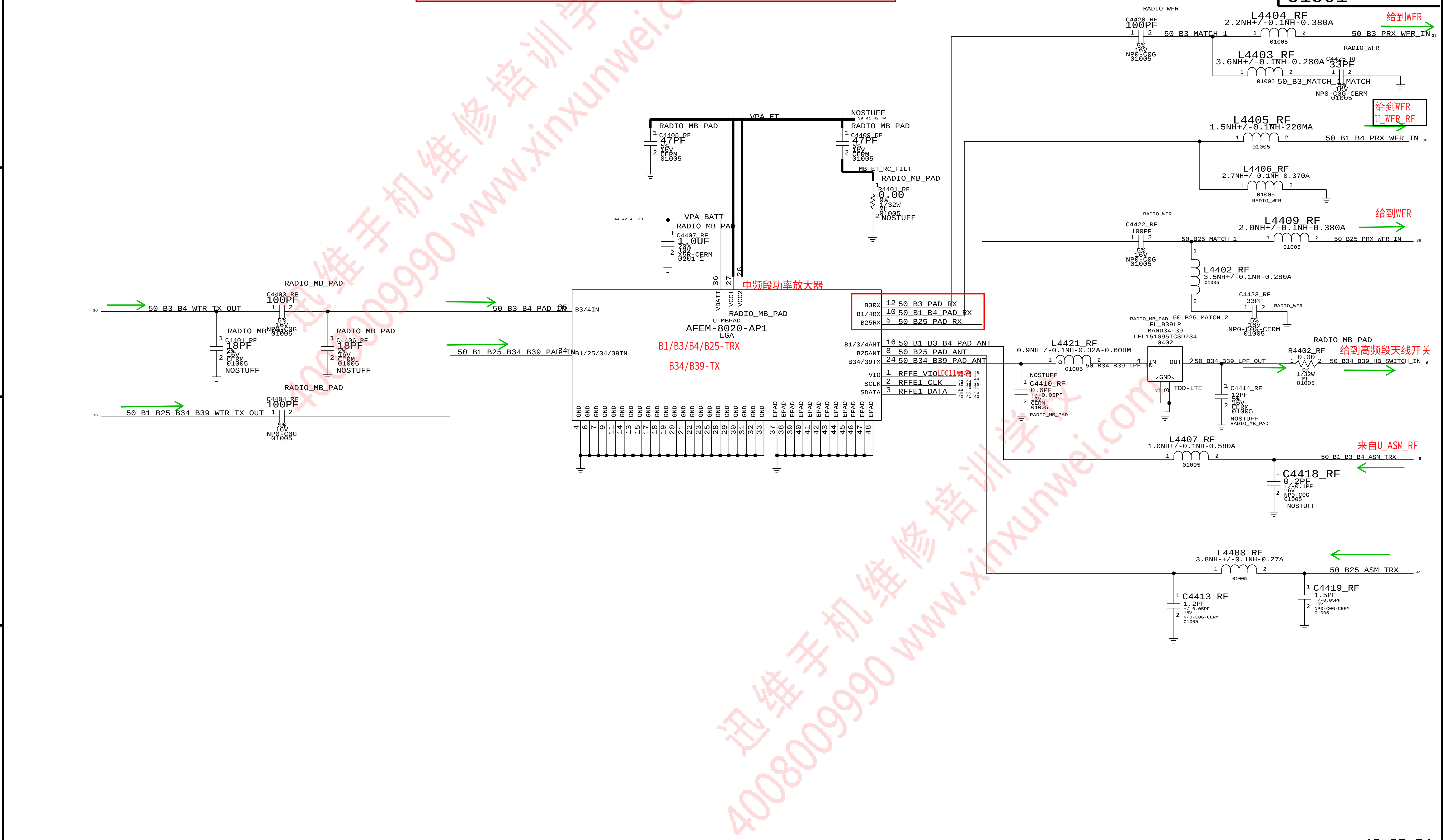


V2	V1	BAND
0	1	B8
1	0	B20
1	1	B26

MID BAND PAD (B1, B25, B3, B4, B34, B39)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

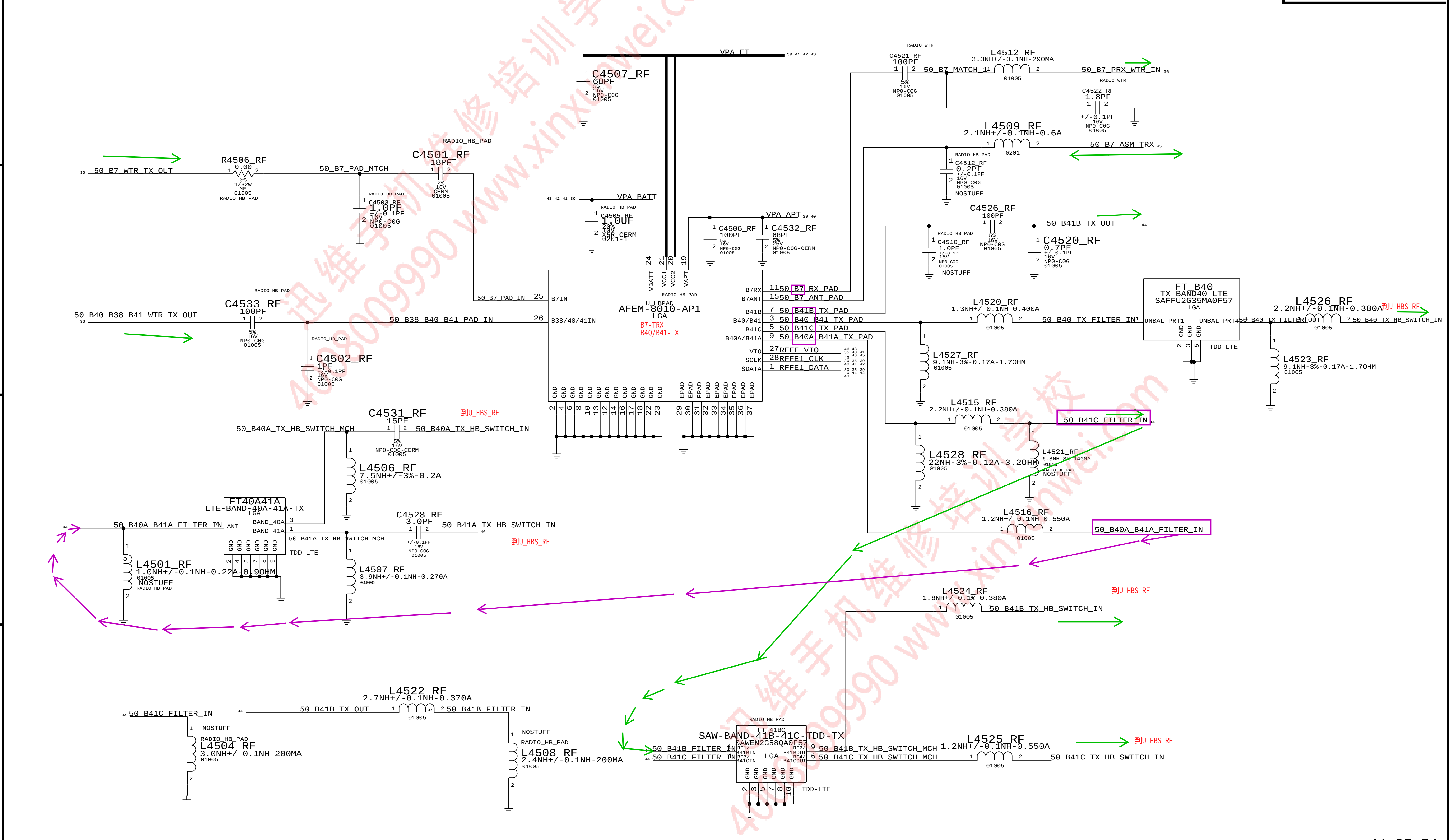
C4426_RF
R1500
L4409_RF
U1501



HIGH BAND PAD (B7, B38, B40, B41, XGP)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

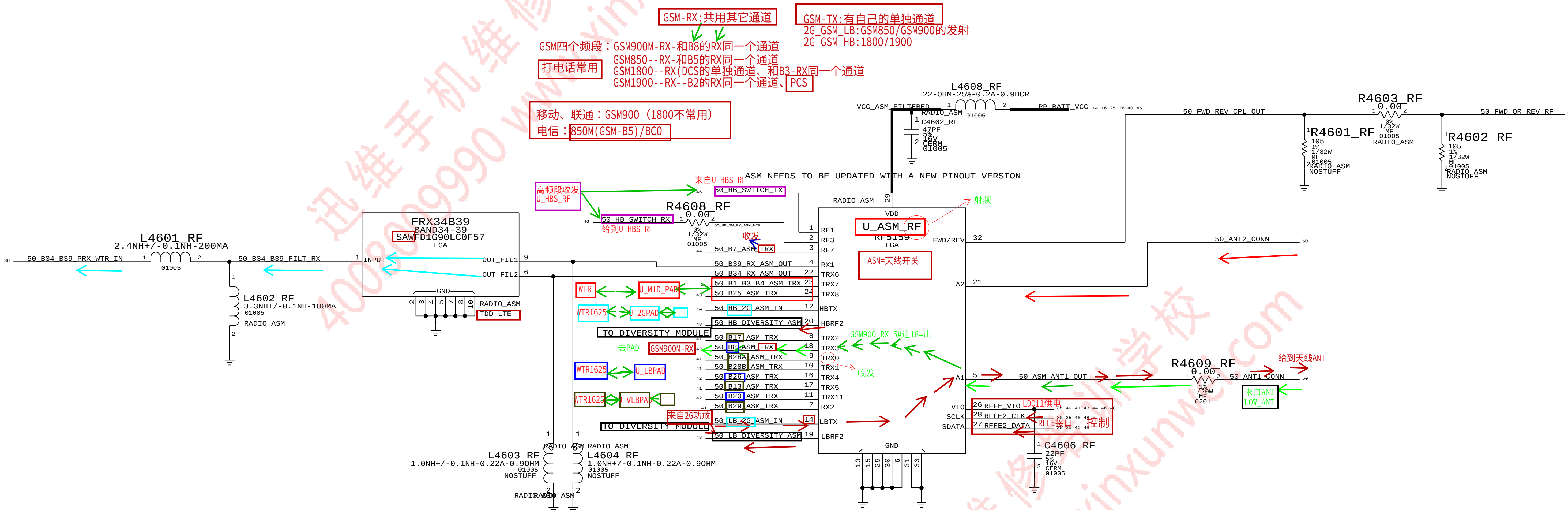
C4533_RF
R1600
L1616
U1601



ANTENNA SWITCH

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

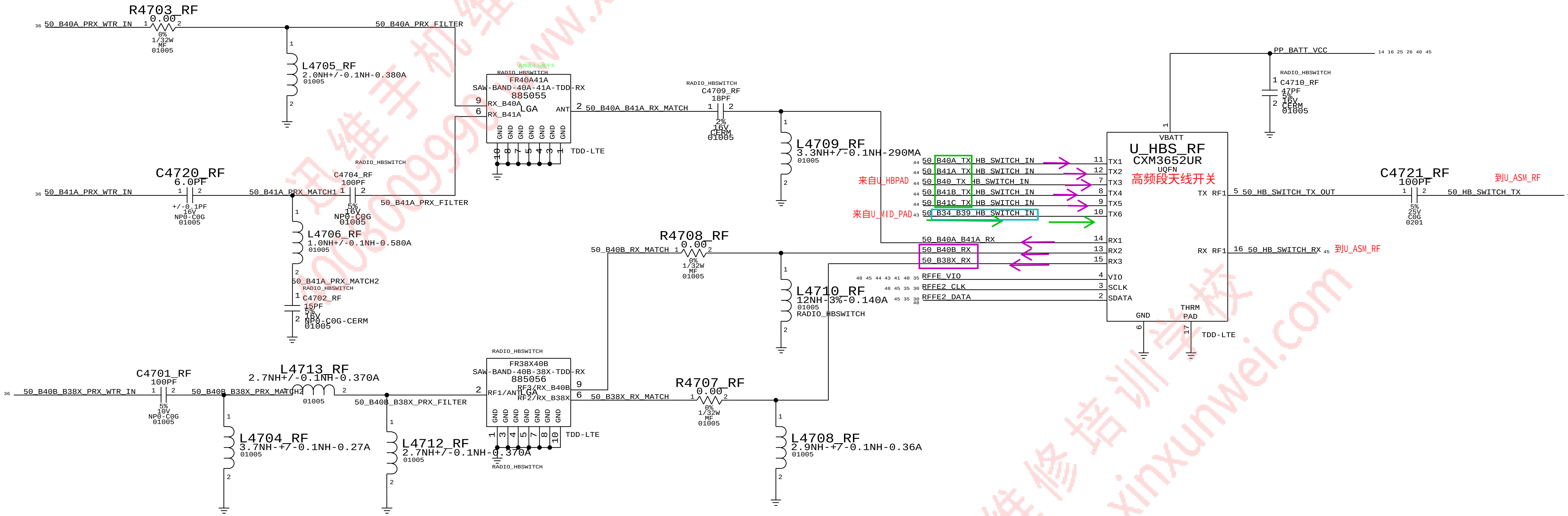
C1702
R1700
L4608_RF
U1702



HIGH BAND SWITCH

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

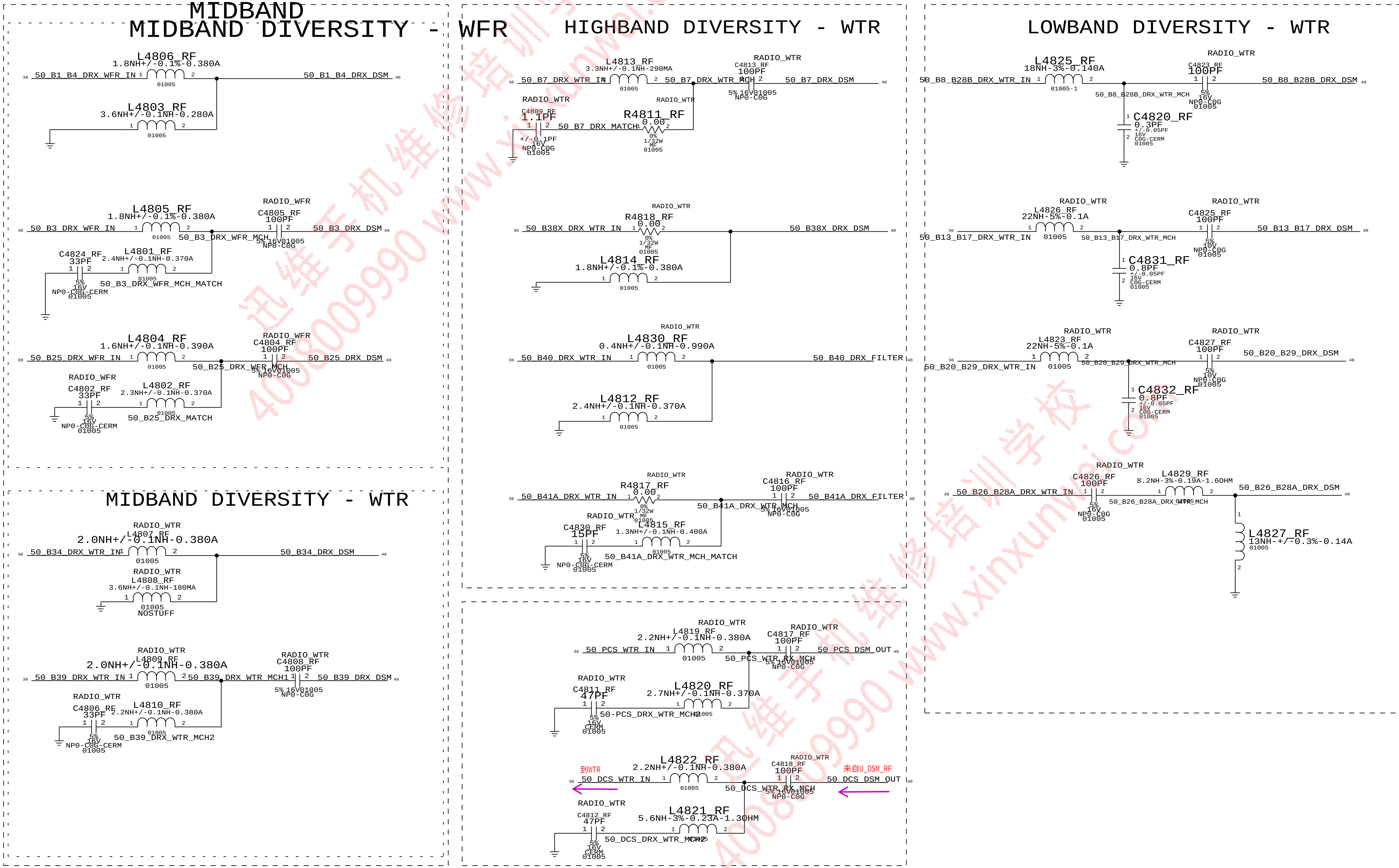
高频段天线开关



RX DIVERSITY (1)

C4826_RF
R1800
L1829
U1801

分集接收-辅助接收

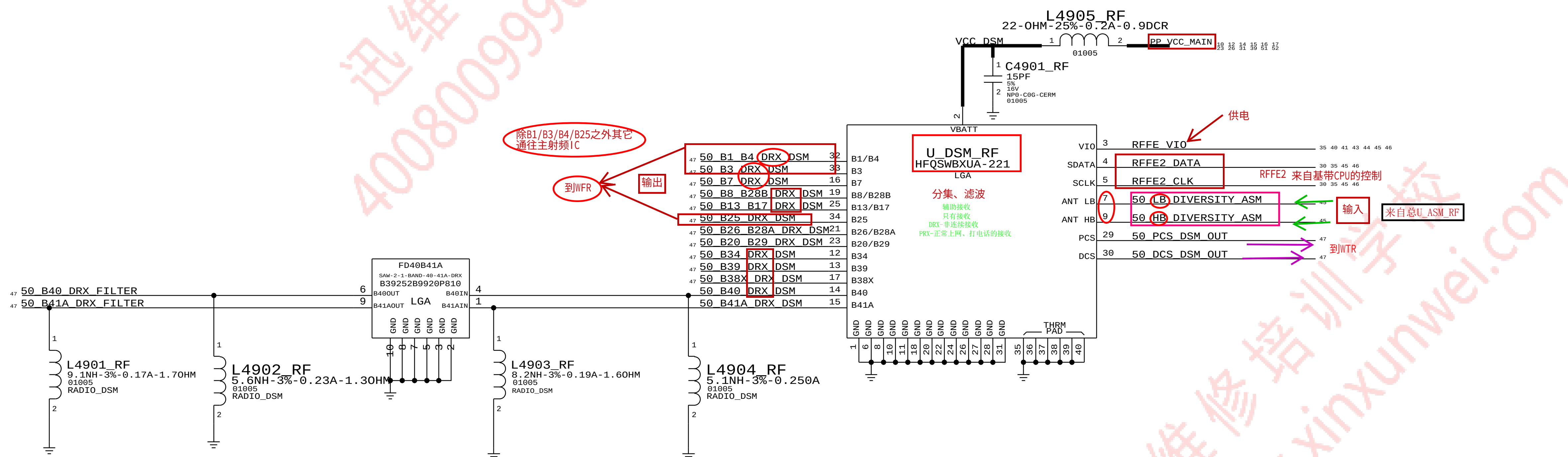


RX DIVERSITY (2)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

分集接收

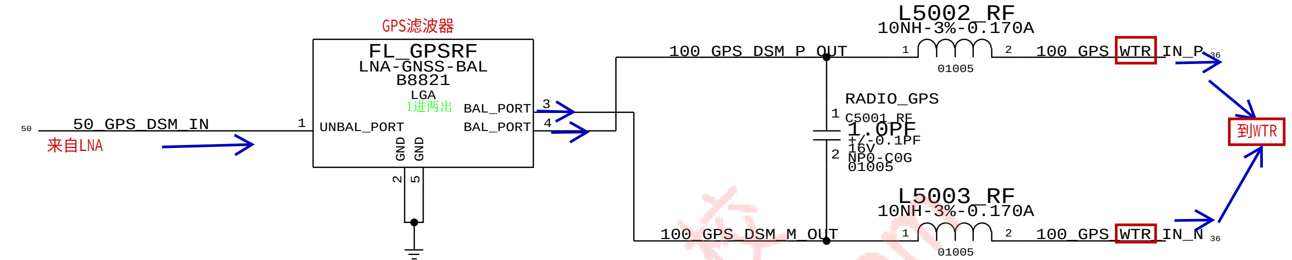
C1900
R1900
L1900
U1901



GPS

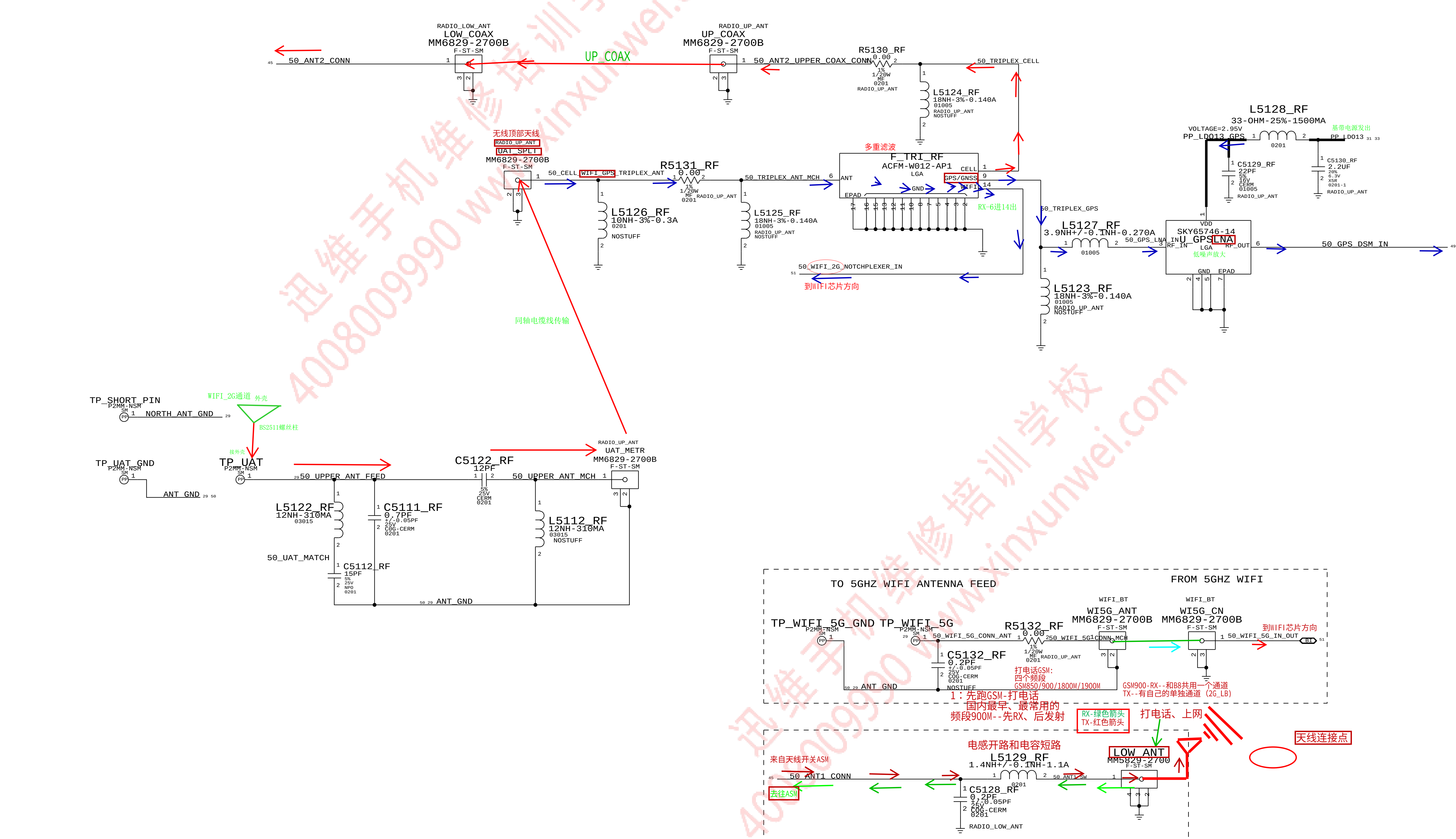
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C1900
R1900
L1900
U1901



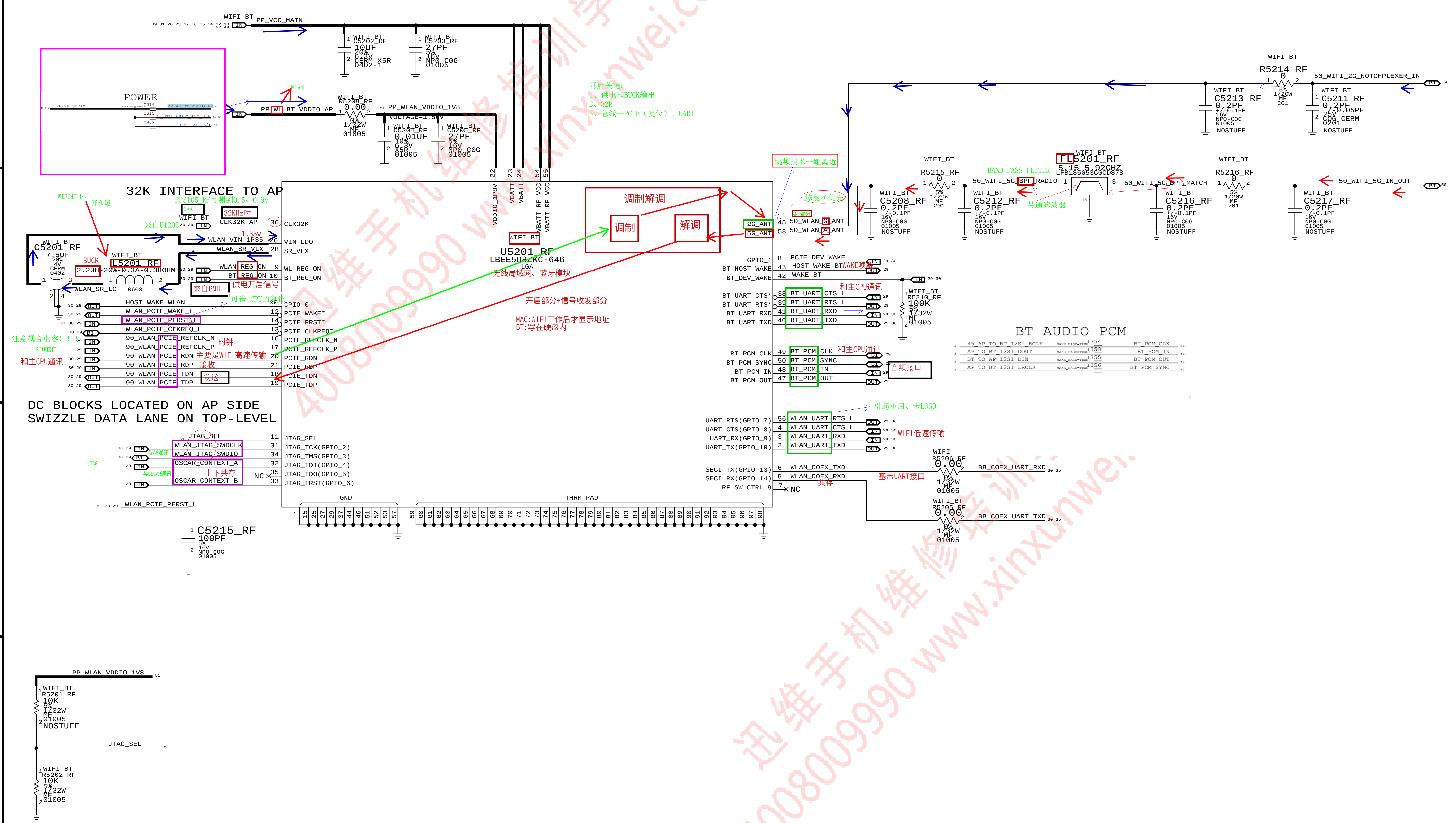
ANTENNA FEED'S

TEST & COAX CONNECTOR FOR LOWER SECTION OF MLB



WLAN/BT

无线局域网--蓝牙



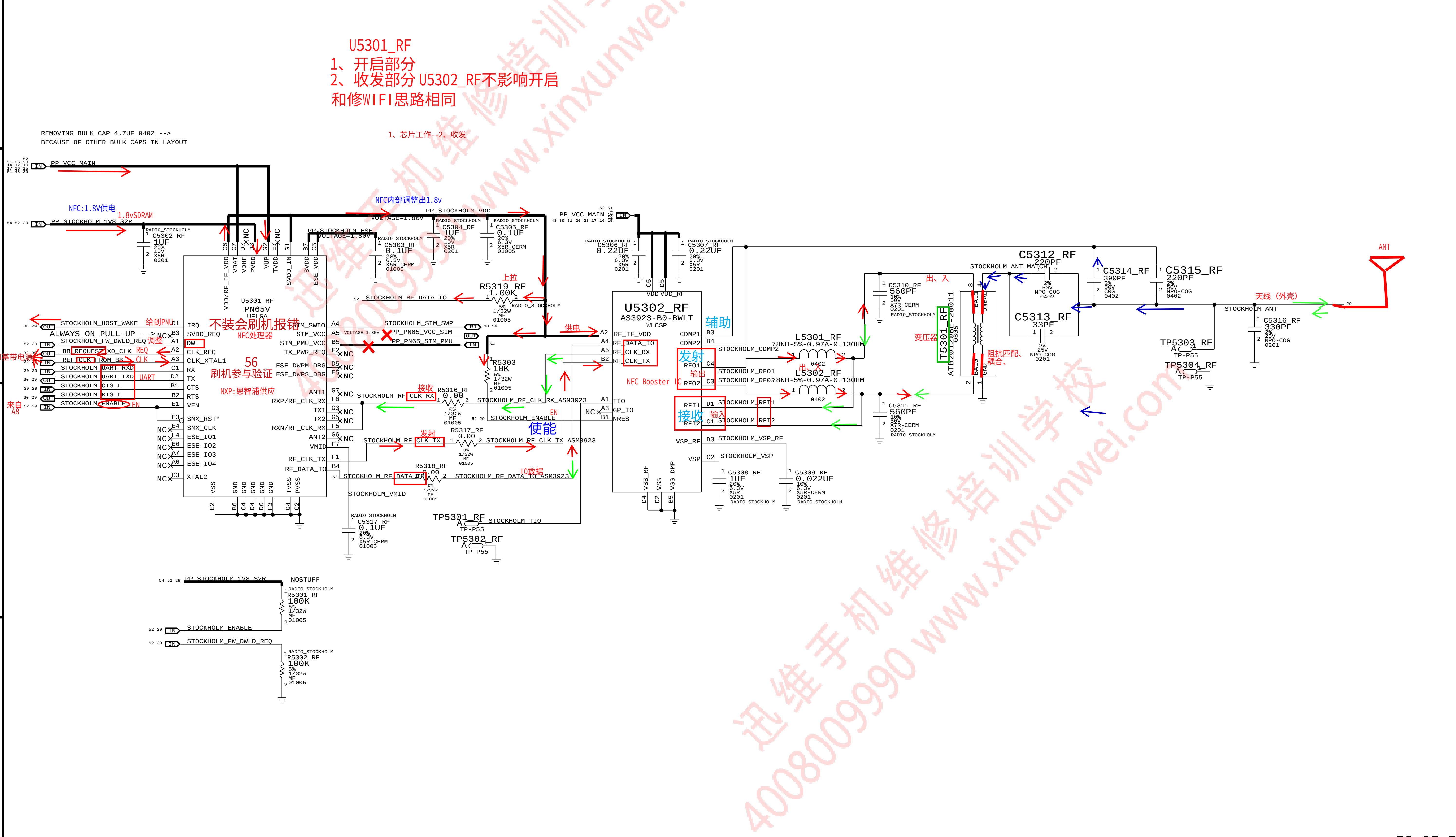
MODULE BOOT-STRAPPED TO PCIE INTERNALLY

STOCKHOLM

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

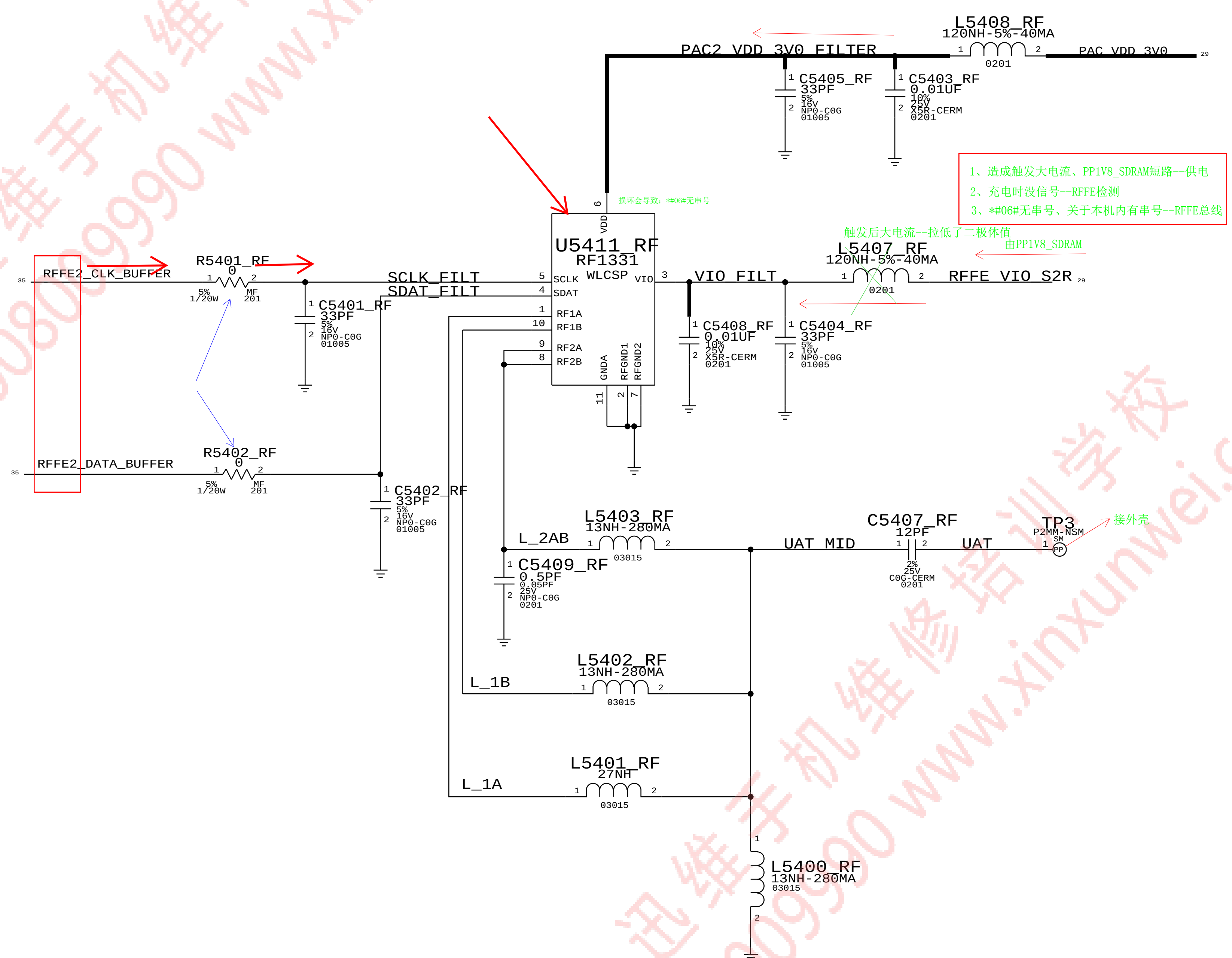
C2101
R2100
L2102
U2100

NFC：接近磁场通讯



ON-BOARD JUMPER FLEX

UAT JUMPER



DSDS

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.